Adaptive Krylov Methods under Hessian Variation

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Abstract

We quantify Hessian-variation sensitivity of secant methods with conjugate-gradient quadratic references. Full conjugacy has a sharp first-order error; one probed correction attains a second-order minimax bound, while one-gradient information leaves a single unresolved Jacobi pivot. The infinite-horizon spectral amplification is asymptotic to $(a/2)\sqrt{\kappa}$ as the condition number $\kappa \to \infty$, where $a \approx 1.3287$ is the value of a unique extremal Krein string. For nonlinear objectives near a nondegenerate minimum in dimension n, convergent full-window symmetric secant iterations with uniformly independent normalized steps have R-order at least $\tau_1(n+1)$ per gradient under a Lipschitz Hessian. Exact endpoint values cancel the leading oldest-pair error and improve this guarantee to $\tau_1(n)$ under a Lipschitz third derivative and a bounded split residual. Here $\tau_1(m) > 1$ solves $t^m = t^{m-1} + 1$; an R-order bound $p > 1$ means $r_k \le C\exp(-cp^k)$ for the local error, with $C, c > 0$. In dimension one the two bounds are the golden ratio and two. The practical method XR combines this model with tested windows, adaptive memory and safeguarded trials. On bounded-below objectives with Lipschitz gradients it has vanishing gradient norms; convergence to a nondegenerate minimum, full memory, uniform independence and adequate trials yield the corresponding local secant rates. CUTEst and full-batch logistic experiments separate gradient savings from model-maintenance cost; on weakly regularized logistic objectives the savings also yield the best observed times.

1. Introduction

A collection of secant pairs records how a function changes along the steps already taken. It does not determine how the Hessian acts on a new correction. This distinction matters even when the algorithms have the same conjugate-gradient (CG) reference on a quadratic, directly or through their virtual projected points. We ask which part of their response to Hessian variation is forced by the observations, and which additional information removes it.

The answer has three levels. Full conjugacy (FC) transports the observed antisymmetric secant defect into a first-order terminal error. One probed latest-model correction (RFC) removes that term and attains a sharp local second-order minimax bound. A method with one gradient per iteration can make the same correction virtually, but it must choose its last trial before observing one remaining Jacobi pivot. The resulting interval of admissible curvatures gives an exact information bound and its optimal trial. Optimizing the first-order coefficient over spectra leads to a continuum string problem with a unique optimizer and a sharp constant.

The nonlinear convergence results give a second reading of this information hierarchy. Let $f : \mathbb{R}^n \to \mathbb{R}$, let x_* be a local minimum with $H_* = \nabla^2 f(x_*) \succ 0$, and measure error by $r_k = \|x_k - x_*\|_{H_*}$. For integers $m \ge 1$, define the order function

$$\tau_1(m) > 1, \qquad \tau_1(m)^m = \tau_1(m)^{m-1} + 1.$$

An R-order bound $p > 1$ means $r_k \le C\exp(-cp^k)$ for some $C, c > 0$ on the local tail [1, 2]. Thus the number of accurate digits grows at least on an exponential scale in k. We count newly evaluated gradients, including probes. For convergent full-window iterations with uniformly independent normalized steps, our local bounds give

	gradients per step	R-order at least
SS1	1	$\tau_1(n+1)$
SS1-F	1	$\tau_1(n)$

$\tau_1(n) > \tau_1(n+1)$.

SS1 needs a Lipschitz Hessian; SS1-F uses exact endpoint values and a Lipschitz third derivative, with the split condition of Theorem 6.1. For $n = 1$ the two orders are $(1+\sqrt{5})/2$ and 2. Both methods are

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Q-superlinear, $r_{k+1}/r_k \to 0$. The gain in R-order comes from cancelling the leading error of the oldest pair: the corrected estimate involves a more recent error and a higher-order remainder. The local error recursions give these orders directly.

The methods apply to nonlinear functions. In the C^3 setting, their leading Hessian increments are slices of the same symmetric third derivative, and the function-level bounds exploit this compatibility. XR globalizes the secant construction: for a bounded-below objective with Lipschitz gradient, Theorem 7.1 gives $\min_{0\le j<K} \|\nabla f(x_j)\| = O(K^{-1/2})$ after K accepted steps. Along iterations converging to a nondegenerate minimum, $m_{\max} = n$, uniform independence, adequate trials and the specified safeguard bounds make XR eventually enter the full-window SS1 regime; XR-F enters SS1-F (Theorem 7.3 and Corollary 7.4).

The experiments compare XR and XR-F with six established methods on controlled objectives, CUTEst and regularized logistic regression. Native implementations and common oracle accounting expose two regimes: inexpensive CUTEst objectives reward low arithmetic overhead, while full-batch logistic objectives reward the reduction in gradient work. Memory and trial ablations separate these effects from the additional information supplied by function values.

Projection identities and quadratic equivalence originate in Hestenes, Dixon and Nazareth [3–9]. The symmetric multipoint secant model and the full-window method SS1 are Burdakov's [10]; compact quasi-Newton representations are classical [11]. Our contribution is the sharp information geometry, its minimax and spectral constants, the function-level obstructions, and the improved R-order guarantee from function values. Section 2 identifies the information; Sections 3–5 determine its price; Sections 6–7 turn it into a method.

Weighted secant fitting and regularized symmetric multisecant updates reconcile inconsistent observations [12, 13]; positive-definite constructions and adaptive quasi-Newton/Anderson guarantees are studied in [14, 15]. Extrapolation and nonlinear Krylov methods share an established algebraic framework [16]. Function values also enter classical modified and multistep secant equations [17, 18]. Here the sharp uncertainty along a fixed CG trajectory identifies which window diagonal is missing and how its reconstruction changes the full-window order.

The companion Chebyshev studies [19, 20] fix a terminal polynomial and compare its ordered realizations. Here the entire reference CG trajectory is fixed. The common insertion kernel and the conversion between query-drift and secant budgets are given in Appendix A.3.

Remarks 3.2–D.23 compare the spectral problem of Section 5 with the isospectral resolvent-diagonal energy of [21]. Both papers read a positive Jacobi chain at the extrapolation point $\lambda = 0$; they price it by the two invariants of its pair of static solutions.

Formal verification. The core secant, perturbation, minimax, spectral-extremum and convergence theorems have formal proofs in Lean. The package is available at `PolynomialRealizationsProgram/krylov`[3].

Source code. The code for the numerical experiments and the generation of figures and tables is available at `PolynomialRealizationsProgram/krylov-code`[4].

2. Secant information and the latest model

For $A \succ 0$, write $\|v\|_A^2 = v^\top A v$ and $\|E\|_A = \|A^{-1/2} E A^{-1/2}\|$, with spectral norm $\|\cdot\|$; $\|\cdot\|_F$ denotes Frobenius norm. For $0 < \mu < L$ with $\operatorname{spec} A \subset [\mu, L]$, write $\kappa = L/\mu$; scalar-spectrum cases are obtained by limits where needed. All operators are real symmetric unless stated otherwise. A full-cycle reference is an n-dimensional quadratic with Hessian A, initial gradient g_0, minimizer $x_* = x_0 - A^{-1} g_0$ and n active spectral atoms. Its CG iterates, gradients, directions and step lengths carry superscript 0 or coefficients α_k, β_k. References before exhaustion may have more active atoms than the horizon.

[3] `https://github.com/dmivilensky/PolynomialRealizationsProgram/tree/main/krylov`
[4] `https://github.com/dmivilensky/PolynomialRealizationsProgram/tree/main/krylov-code`

Definition 2.1 (Observations and variation)**.** A segment $s_k = x_{k+1} - x_k$ produces $g_{k+1} = g_k + A_k s_k$. Probes, correction segments and an unevaluated output have their own operators, denoted B_k or B. For a reference $A \succ 0$ put

$$A_0 = A, \quad A_{l+1} = A_l + A^{1/2}\widehat{E}_l A^{1/2}, \quad \widehat{E}' = A^{-1/2}(B - A_{n-1})A^{-1/2}. \tag{1}$$

Their budgets are $\omega_l = \|\widehat{E}_l\|$, $\omega' = \|\widehat{E}'\|$, $W_0 = \sum_{l=0}^{n-2}\omega_l$, $W_1 = \sum_{l=1}^{n-2}\omega_l$ and $\omega = \max(\max_l \omega_l, \omega')$; empty sums and maxima are zero. Perturbation remainders are uniform near the named reference unless a larger class is explicitly specified. For comparison, the radial error model is $g_s = H_s(x_s - x_*)$, with drift $E_s = H_s - H$ relative to a fixed $H \succ 0$.

For $f \in C^2$, both models are exact: segment averages of $\nabla^2 f$ give A_k, and averages from a stationary x_* to x_s give H_s. Their budgets differ; Proposition A.7 gives the conversion.

Definition 2.2 (Latest model and observed defect)**.** Order pairs from oldest to newest and put $S = [s_0, \dots, s_{p-1}]$, $Y = [y_0, \dots, y_{p-1}]$, $C = S^\top Y$. Define

$$\overline{T}_{ij} = C_{\min(i,j),\max(i,j)}, \qquad \mathsf{D} = C - C^\top, \qquad \delta_{ij} = \frac{\mathsf{D}_{ij}}{\sqrt{C_{ii}C_{jj}}}.$$

The last expression is used when $C_{ii} > 0$. The symmetric model retains the newer observation, $\overline{T}_{ij} = s_{\min(i,j)}^\top y_{\max(i,j)}$.

This is the symmetric multipoint model [10, 22, 23]. Its defect measures disagreements between observations, not their common error.

Let $y_i = A_i s_i$, let $B = A_p$ act on the next correction, and set $E_l = A_{l+1} - A_l$, $X_l = S^\top E_l S$. The coordinate projector $I_{\le l} = \operatorname{diag}(\mathbf{1}_{i \le l})$ selects the old steps. Write tril for the strictly lower triangular part and $[P, X] = PX - XP$.

Lemma 2.3 (Secant flag)**.**

$$S^\top BS - \overline{T} = \sum_{l=0}^{p-1} I_{\le l} X_l I_{\le l}, \qquad \mathsf{D} = \sum_{l=0}^{p-2} [I_{\le l}, X_l], \qquad C - \overline{T} = \operatorname{tril}\mathsf{D}. \tag{2}$$

Proof. For $i \le j$, the first entry is $s_i^\top(B - A_j)s_j = \sum_{l \ge j}(X_l)_{ij}$, and the defect is $s_i^\top(A_j - A_i)s_j = \sum_{i \le l < j}(X_l)_{ij}$. Symmetry gives the remaining entries; the final identity is entrywise. □

The commutator blocks are observed; the old-flag compressions are hidden. To quantify this, fix full-rank S, a base $A \succ 0$, budgets $\omega_l \ge 0$, and set $G_A = S^\top AS$, $W = \sum_{l=0}^{p-1}\omega_l$, $R_{ij} = \sum_{l \ge \max(i,j)}\omega_l$. A sequence is admissible if $A_0 = A$ and $-\omega_l A \preceq E_l \preceq \omega_l A$. For $\overline{T} \succ 0$ put

$$\mathcal{Z} = S^\top BS - \overline{T}, \quad \mathcal{Z}_\star = \overline{T}^{-1/2}\mathcal{Z}\overline{T}^{-1/2}, \quad \Omega_\star(C) = \|\overline{T}^{-1/2}\mathsf{D}\overline{T}^{-1/2}\|.$$

Here $\circ$ denotes entrywise product, and $\mathbf{u}_i$ is the ith coordinate vector, indexed from zero.

Theorem 2.4 (Unresolved curvature)**.** *Every admissible sequence satisfies*

$$-G_A \circ R \preceq \mathcal{Z} \preceq G_A \circ R, \qquad G_A \circ R \succeq 0.$$

For a nonempty window and $W < 1$, there are admissible positive definite sequences with identical quadratic data $C = G_A$ and

$$\Omega_\star = 0, \qquad \mathcal{Z}_\star^\pm = \pm W\hat{u}\hat{u}^\top, \qquad \hat{u} = \frac{G_A^{-1/2}\mathbf{u}_0}{\|G_A^{-1/2}\mathbf{u}_0\|}.$$

Consequently the local radius of information for $\mathcal{Z}_\star$ at $C = G_A$ is at least W for every window, and equals W for an A-conjugate window.

The proof is in Appendix A.2. In particular, a sufficiently large defect refutes a proposed small variation budget, whereas a small defect cannot certify that the hidden model error is small. The oldest diagonal can carry the whole budget.

2.1. Realizations and their information

Definition 2.5 (Information classes)**.** Fixed prior parameters are allowed; unknown operator actions are available only through observations. Evaluated search steps lie in the span of the observed vectors; corrections use the indicated window.

($\mathcal{M}_1$) Given $\mathcal{D}_k = \{g_0, \dots, g_k, s_0, \dots, s_{k-1}\}$, choose s_k and observe $g_{k+1} = g_k + A_k s_k$. The output $\hat{x}_N = x_N + u_N$ is determined by $\mathcal{D}_N$ and is not evaluated; its gradient is $g_N + Bu_N$. CG-consistency means $\hat{x}_k = x_k^0$ on quadratic data, $k \le n$.

($\mathcal{M}$) Given $\mathcal{D}_k$, choose d_k, observe $z_k = A_k d_k$, and take $\alpha_k = -g_k^\top d_k/(d_k^\top z_k)$, $g_k' = g_k + \alpha_k z_k$. Choose corrections $u_k = [S_k, d_k]w_k$ using the enlarged data and evaluate $g_{k+1} = g_k' + B_k u_k$, finitely many times if prescribed. Record the total step and gradient difference. CG-consistency means positive CG direction scaling and zero corrections on quadratic data. Smoothness means that the coefficients in observed vectors are C^2 near the reference data.

($\mathcal{O}$) An oblivious realization has fixed data-independent coefficients: $x_j = x_0 + \sum_{t<j} \gamma_{jt} g_t$. Its relation to the adaptive classes is in Appendix A.3.

The operator model counts an exact search/probe as one gradient observation; every nonfinal correction point is evaluated. Experimental profiles instead count every actual function and gradient evaluation of a numerical search. A cycle in $\mathcal{M}_1$ costs $N+1$ gradients. Classical nonlinear-CG and scalar-initialized quasi-Newton memberships are detailed in Appendix A.1.

Definition 2.6 (FC and RFC)**.** With retained pairs S, Y, FC solves

$$(Y^\top S)\beta = Y^\top g, \qquad d = -g + S\beta, \qquad z = A_k d, \quad \alpha = -\frac{g^\top d}{d^\top z}, \quad g' = g + \alpha z.$$

For an empty window $d = -g$. RFC adds, when the window is nonempty and $\|\delta\| < 1$,

$$H = \begin{pmatrix} \overline{T} & S^\top z \\ z^\top S & d^\top z \end{pmatrix}, \qquad w = -H^{-1}[S,d]^\top g', \qquad g_+ = g' + B_k[S,d]w.$$

It records $s = \alpha d + [S,d]w$, $y = g_+ - g$; without a correction, $s = \alpha d$. At a final output, the correction is not evaluated. All invoked solves must exist. Restart and window rules specify retained pairs.

The gate uses the canonical defect of the retained pairs (S, Y); the correction model is augmented by (d, z). The direction satisfies $Y^\top d = 0$, even when the pairs are not mutually conjugate. On quadratic data these methods are CG (Lemma A.2); the correction identity in Lemma A.1 is the common mechanism behind their different variation orders.

3. The price of a correction

For the reference write

$$e_k = x_k^0 - x_*, \quad \hat{e}_k = A^{1/2} e_k = A^{-1/2} g_k^0, \quad c_k = \|s_k^0\|_A^2 = \|e_k\|_A^2 - \|e_{k+1}\|_A^2, \quad \hat{s}_k = \frac{A^{1/2} s_k^0}{\sqrt{c_k}}. \tag{3}$$

The $\hat{s}_k$ are orthonormal. Put $\mathcal{K}_k = \operatorname{span}\{g_0, Ag_0, \dots, A^{k-1}g_0\}$, let P_l project orthogonally onto $A^{1/2}\mathcal{K}_{l+1}$, and set $v_l = \sum_{j>l} \sqrt{c_j}\hat{s}_j = -\hat{e}_{l+1}$. For the spectral probability measure ν of $\hat{e}_0/\|\hat{e}_0\|$, the CG energy principle [24, 25] gives

$$\frac{\|e_m\|_A^2}{\|e_0\|_A^2} = \lambda_m(0;\nu), \qquad \lambda_m(0;\nu) = \min_{\deg p \le m,\ p(0)=1} \int p^2 \, d\nu.$$

Theorem 3.1 (Full-conjugacy response)**.** *There are positive $C(n,\kappa)$ and $\omega_0(n,\kappa)$ such that, for every reference with n atoms, the defined FC cycle satisfies*

$$A^{-1/2} g_n = F(\widehat{E}) + R, \qquad F(\widehat{E}) = \sum_{l=0}^{n-2} P_l \widehat{E}_l v_l, \qquad \|R\| \le C(n,\kappa) W_0^2 \|e_0\|_A \tag{4}$$

when $W_0 \le \omega_0(n,\kappa)$. The sharp first-order coefficient is

$$\sup_{\|\widehat{E}_l\|\le\omega_l} \frac{\|F(\widehat{E})\|}{\|e_0\|_A} = \sum_{l=0}^{n-2} \omega_l \lambda_{l+1}(0;\nu)^{1/2},$$

attained by $\widehat{E}_l = \omega_l(\hat{s}_0\hat{v}_l^\top + \hat{v}_l\hat{s}_0^\top)$, $\hat{v}_l = v_l/\|v_l\|$. If Γ_ω is the supremum of this sum over n-atom measures on $[\mu, L]$, the worst gradient ratio is at most $\Gamma_\omega + CW_0^2$ and equals $r\Gamma_\omega + o(r)$ for budgets $r\omega_l$, $r \downarrow 0$.

The rank-two attainers couple the unexplored error to the old flag. Before exhaustion the unexplored part of the first variation is common to all smooth CG-consistent exact-search methods; realization-dependent terms lie in the explored space, orthogonal to the reference error. Thus their gradient norms in the inverse-reference metric have the same first variation whenever the reference error is nonzero. The full expansion, its compact-uniform remainder and the virtual-launch terms are Theorems B.11 and B.13 in Appendix B.

Remark 3.2 (Two static solutions). In the Lanczos basis of (A, g_0), put $J = Z^\top AZ$ and $\mathcal{G} = J^{-1}$. Its single-pair Green kernel [26, 27] has factors $h_j^- = \|g_j^0\|^{-1}$ and $h_j^+ = \|e_j\|_A^2/\|g_j^0\|$:

$$|\mathcal{G}_{ij}| = \sqrt{\mathsf{d}_i\mathsf{d}_j}\, e^{-|\mathsf{x}_i - \mathsf{x}_j|}, \qquad \mathsf{d}_j = \frac{\|e_j\|_A^2}{\|g_j^0\|^2}, \quad \mathsf{x}_j = \log\frac{\|e_0\|_A}{\|e_j\|_A}, \quad \lambda_j(0;\nu) = e^{-2\mathsf{x}_j}.$$

The product $h_j^- h_j^+ = \mathsf{d}_j$ is the resolvent diagonal optimized in [21]; their normalized ratio is the CG energy profile priced by Theorem 3.1. Remark D.16 derives the kernel and its discrete string realization.

3.1. The second-order price

A latest-model correction annihilates the measured residual. Its remaining error is the hidden compression acting on that correction. For $W, \omega' \ge 0$ define

$$K(W,\omega') = W \max_{0\le t\le 1} t(W\sqrt{1-t^2} + \omega'), \qquad K(W,0) = \tfrac{1}{2}W^2. \tag{5}$$

To state its information optimality, fix (A, g_0) with $n \ge 2$ atoms. A scenario is a tuple $\Xi = (A_0; E_\bullet; E'; F_\bullet)$, with $A_0 \succ 0$, probe operators $A_k = A_0 + \sum_{l<k} E_l$, correction operators $B_k = A_k + F_k$ for $k \le n-2$, and $B_{n-1} = A_{n-1} + E'$. The class $\mathcal{S}(r;\omega_\bullet,\omega')$ is specified by

$$\|A_0 - A\|_A \le r, \quad \|E_l\|_{A_0} \le \omega_l, \quad \|E'\|_{A_0} \le \omega', \quad \|F_k\|_{A_0} \le \omega.$$

The method does not know A_0. Put $\phi(G) = \|e_{n-1}^G\|_G$ for the penultimate CG error of (G, g_0) and $\rho_M(\Xi) = \|g_n\|_{A_0^{-1}}/\phi(A_0)$. For $A_0 = A$ define

$$\hat{f} = \sum_{l=0}^{n-2} P_l \widehat{E}_l \hat{s}_{n-1}, \qquad \mathcal{L} = \sum_{l=0}^{n-2} P_l \widehat{E}_l P_l + \widehat{E}'. \tag{6}$$

Theorem 3.3 (Second-order minimax bound)**.** *For $A_0 = A$ and $\|F_k\|_A \le \omega$, RFC satisfies*

$$A^{-1/2} g_n = -\|e_{n-1}\|_A \mathcal{L}\hat{f} + O(\omega^3), \qquad \|\mathcal{L}\hat{f}\| \le K(W_0,\omega').$$

There is $r_0 > 0$ such that, for every fixed $0 < r \le r_0$ and every budget profile,

$$\inf_{M\in\mathcal{M}_{\mathrm{CG}}^{(1)}} \sup_{\Xi\in\mathcal{S}(r;\omega_\bullet,\omega')} \rho_M(\Xi) = K(W_0,\omega') + O(\omega^3), \qquad \omega \to 0,$$

where $\mathcal{M}_{\mathrm{CG}}^{(1)}$ consists of CG-consistent methods with at most one correction per iteration. RFC attains the bound. Remainder constants depend only on (A, g_0) and, for the minimax assertion, on r.

The lower bound uses different bases with identical observations, not an adversary that reveals the reference matrix. The information centre and the reflection realizing this indistinguishability are proved in Appendix B.6.

Definition 3.4 (Gated RFC)**.** Fix $\gamma_{\mathrm{g}} \in (0,1)$. Gated RFC makes the correction of iteration $k \geq 1$ only if $(H_k)_{ii} > 0$ for all i, $\|\delta\| < 1$ and $\lambda_{\min}(\widehat{H}_k) \geq \gamma_{\mathrm{g}}$, where $\widehat{H}_k = \operatorname{diag}(H_k)^{-1/2} H_k \operatorname{diag}(H_k)^{-1/2}$; otherwise iteration k is an FC iteration. The tests use only $\mathcal{D}_k \cup \{z_k\}$ and are invariant under $g_0 \mapsto \lambda g_0$, $A_\bullet \mapsto cA_\bullet$.

Here H_k is RFC's augmented model, δ is the canonical defect of the retained pairs, and $\lambda_{\min}$ is the smallest eigenvalue. For a sufficiently large $M_0 = M_0(n, \kappa, \gamma_{\mathrm{g}})$, ordinary RFC has the sharp expansion uniformly under the nondegeneracy condition $\|e_{n-2}\|_A \geq M_0 \omega \|e_0\|_A$. Gated RFC instead satisfies

$$\|g_n\|_{A^{-1}} \leq C(n, \kappa, \gamma_{\mathrm{g}}) \omega^2 \|g_0\|_{A^{-1}}$$

for all defined sufficiently small-variation cycles in the spectral class. Under that nondegeneracy condition its gate never acts. The full statement, including the unrestricted last energy and the low-dimensional cases, is in Theorems B.25 and B.25. A certified singular-model family explains why the defect test alone is insufficient (Proposition B.26).

4. One gradient and the missing curvature

A correction can be made virtually before the next evaluation. With $\overline{T}$ invertible, set

$$w = -\overline{T}^{-1} S^\top g, \quad \hat{x} = x + Sw, \quad \hat{g} = g + Yw, \quad d = -\hat{g} + S\beta, \quad (Y^\top S)\beta = Y^\top \hat{g}. \tag{7}$$

XRFC evaluates $x_+ = \hat{x} + \tau d$ and stores $(x_+ - x, g_+ - g)$; its unevaluated output is the virtual point after this new pair is appended. Empty windows give $w = 0$. On quadratic data, full memory recovers $(\hat{x}_k, \hat{g}_k, d_k) = (x_k^0, g_k^0, d_k^0)$ for arbitrary nonzero trials and reaches $\hat{x}_n = x_*$ after $n+1$ gradients [6–8]; Lemma A.2 gives the precise identities. XRFC is the local model of the practical XR algorithm below.

Fix an n-atom reference. In its Lanczos basis $Z = [z_0, \ldots, z_{n-1}]$, let $J = Z^\top A Z$ have positive couplings $\mathsf{b}_j = J_{j,j+1}$. Let J_0 be its leading $(n-1)$-block, $b_A = J_{n-1,n-1}$, $z = z_{n-1}$, and

$$\sigma_J = \mathsf{b}_{n-2}^2 \mathbf{u}_{n-2}^\top J_0^{-1} \mathbf{u}_{n-2}, \qquad A_b = A + (b - b_A) z z^\top.$$

The information $\mathcal{I} = (g_0, A|_{\mathcal{K}_{n-1}})$ leaves exactly $b > \sigma_J$ free, and the last CG length is $\alpha(b) = (b - \sigma_J)^{-1}$ (Lemma C.6). On an admissible interval $[b_{\mathrm{lo}}, b_{\mathrm{hi}}]$ put $\alpha_{\mathrm{hi}} = (b_{\mathrm{lo}} - \sigma_J)^{-1}$, $\alpha_{\mathrm{lo}} = (b_{\mathrm{hi}} - \sigma_J)^{-1}$ and $r = \alpha_{\mathrm{hi}}/\alpha_{\mathrm{lo}}$. For constant prefixes $A_k = A_b$, define the actual output error

$$E(b, B) = \frac{\|g_n + B u_n\|_{A_b^{-1}}}{\|e_{n-1}(b)\|_{A_b}}, \qquad \omega'(b, B) = \|B - A_b\|_{A_b}.$$

Theorem 4.1 (One-gradient minimax bound)**.** *Let $n \geq 2$, $[b_{\mathrm{lo}}, b_{\mathrm{hi}}] \subset (\sigma_J, \infty)$ be a nondegenerate admissible fibre interval, and $0 \leq \omega' \leq 1$. Then*

$$\min_{M \in \mathcal{M}_1} \sup_{\substack{b \in [b_{\mathrm{lo}}, b_{\mathrm{hi}}] \\ \omega'(b,B) \leq \omega'}} E(b, B) = \frac{r-1}{r+1} \omega'.$$

The lower bound already holds on the two endpoint completions for every method, without CG-consistency. Full-memory XRFC attains it with arbitrary positive earlier trials and

$$\tau_{n-1} = \frac{2\alpha_{\mathrm{lo}} \alpha_{\mathrm{hi}}}{\alpha_{\mathrm{lo}} + \alpha_{\mathrm{hi}}}, \qquad \tau_{n-1}^{-1} = \tfrac{1}{2}(b_{\mathrm{lo}} + b_{\mathrm{hi}}) - \sigma_J.$$

The output point is $\hat{x}_n = x_*(b)$; the residual reflects the unobserved output operator B.

The output coincides with the minimizer of the prefix quadratic; its unevaluated gradient still involves B. An exact search observes the missing moment before choosing its length. The two endpoint completions already force the lower bound, a radius-of-information argument [28]. Over $[\mu, L]$, the uniform supremum is $\rho_\kappa \omega'$, $\rho_\kappa = (\kappa - 1)/(\kappa + 1)$; over a known relative base ball $\mathcal{B}_\rho(A_c) = \{A' : \|A' - A_c\|_{A_c} \leq \rho\}$, $A_c \succ 0$, $0 \leq \rho < 1$, it is $\rho\omega'$ (Corollary C.8). Earlier variation contributes the terms in Theorem C.1.

Remark 4.2 (The unseen boundary value). The information $\mathcal{I}$ fixes the launched static solution h^- and the recurrence for the returned solution h^+; one terminal value determines h^+ throughout (Lemma C.7):

$$\frac{h^+_{n-1}}{\|g^0_{n-1}\|} = \mathsf{d}_{n-1} = \alpha(b) = \frac{1}{b - \sigma_J}.$$

Thus Theorem 4.1 prices one missing boundary compliance. An exact search measures it; the minimax trial uses the two endpoint completions. Remark C.10 gives its terminal-spring realization and the reconstruction of h^+ from this scalar.

4.1. Completion from an arbitrary window

For S of rank $p < n$, let $G = S^\top S$ and $d \notin \operatorname{span} S$. Given $0 < \mu < L$, put $\mathsf{b} = S^\top d$ and

$$\kappa_{\mathrm{lo}} = \mu + \mu^2 \frac{\mathsf{b}^\top (\overline{T} - \mu G)^+ \mathsf{b}}{\|d\|^2}, \quad \kappa_{\mathrm{hi}} = L - L^2 \frac{\mathsf{b}^\top (LG - \overline{T})^+ \mathsf{b}}{\|d\|^2}, \quad \kappa_{\mathrm{mid}} = \tfrac{1}{2}(\kappa_{\mathrm{lo}} + \kappa_{\mathrm{hi}}). \tag{8}$$

Here $+$ denotes the Moore–Penrose inverse; for $p = 0$, $[\kappa_{\mathrm{lo}}, \kappa_{\mathrm{hi}}] = [\mu, L]$. The trial is $\tau = -g^\top d/(\kappa_{\mathrm{mid}}\|d\|^2)$, and clip projects a positive curvature estimate onto $[\kappa_{\mathrm{lo}}, \kappa_{\mathrm{hi}}]\|d\|^2$.

Theorem 4.3 (Window curvature completion)**.** *For the window above, let* $\mathcal{C}_G = \{A' = A'^\top : \mu I \preceq A' \preceq LI,\ S^\top A' S = \overline{T},\ S^\top A' d = 0\}$. *If*

$$\overline{T} - \mu G \succeq 0, \quad LG - \overline{T} \succeq 0, \quad \mathsf{b} \in \operatorname{ran}(\overline{T} - \mu G) \cap \operatorname{ran}(LG - \overline{T}), \quad \kappa_{\mathrm{lo}} \le \kappa_{\mathrm{hi}},$$

then $\{d^\top A' d/\|d\|^2 : A' \in \mathcal{C}_G\} = [\kappa_{\mathrm{lo}}, \kappa_{\mathrm{hi}}]$ *with both endpoints attained; otherwise* $\mathcal{C}_G$ *is empty. If* $Y = AS$, $Y^\top d = 0$ *and* $\mu I \preceq A \preceq LI$, *imposing the full action* $A'S = Y$ *gives the same interval. If* $\mathcal{C}_G \neq \varnothing$ *and* $-g^\top d > 0$, *the midpoint-curvature trial is uniquely minimax:*

$$\min_{\tau} \max_{A' \in \mathcal{C}_G} |1 - \tau/\alpha(A')| = \frac{r-1}{r+1}, \qquad \alpha(A') = \frac{-g^\top d}{d^\top A' d}, \quad r = \frac{\kappa_{\mathrm{hi}}}{\kappa_{\mathrm{lo}}}.$$

At the last quadratic XRFC step it is the harmonic-mean trial of Theorem 4.1. For every consistent $c' = d^\top A' d$ *and positive* $\hat{c}$,

$$|1 - c'/\operatorname{clip}(\hat{c})| \le |1 - c'/\hat{c}|.$$

The proof is a generalized Schur-complement argument [29]; in the Lanczos case its endpoints are the Gauss–Radau rules. The full information fibre and moment interpretation are in Appendix C.

5. Sharp spectral amplification

The response of FC is determined by the CG energy profile. For measures on $[\mu, L]$, define

$$\Gamma^*_{M,\gamma} = \sup_{\nu} \sum_{m=1}^{M} \lambda_m(0;\nu)^\gamma, \quad \Gamma^*_{\infty,\gamma} = \sup_{\nu} \sum_{m \ge 1} \lambda_m(0;\nu)^\gamma, \quad 0 < \gamma \le 1.$$

The finite optimum can be taken on $M + 1$ atoms; general weights, certificates and sharp support bounds are given in Theorem D.1. Thus $\gamma = 1/2$ measures independent spectral increments, and $\gamma = 1$ measures a total Frobenius budget. Let

$$\kappa = L/\mu, \qquad \eta = \frac{\kappa + 1}{\kappa - 1}, \qquad \theta = \operatorname{arccosh} \eta \sim \frac{2}{\sqrt{\kappa}}.$$

The constants are values of an extremal-string problem. A normalized Krein string has length $\ell \in (0, \infty]$, a unit mass at zero and nondecreasing mass function M, with $dM = \varsigma^2 dx + dM_{\mathrm{s}}$. Its state at spectral parameter -1 and its type coordinate satisfy

$$d\varphi = p\, dx, \qquad dp = \varphi\, dM, \qquad dK = \varphi^2\, dM, \qquad (\varphi, p, K)(0) = (1, 1, 1), \qquad dy = \varsigma\, dx.$$

The initial state includes the left atom; subsequent mass increments are on $(0,\ell)$. Define

$$R_\gamma(M)=\int_{[0,\ell)} K^{-\gamma}\,dy, \qquad a_{\infty,\gamma}=\sup_M R_\gamma(M), \qquad a=a_{\infty,1/2}.$$

Definition D.6 gives the integral formulation, including atoms and singular mass. Theorem D.8 identifies its unique optimizer; the planar shooting system (47) computes its value.

Theorem 5.1 (Sharp amplification)**.** *For $\gamma\in\{1/2,1\}$,*

$$\theta\Gamma^*_{\infty,\gamma}\longrightarrow a_{\infty,\gamma}, \qquad \theta\Gamma^*_{n-1,\gamma}\longrightarrow a_{\infty,\gamma} \quad (\kappa\to\infty,\ n\theta\to\infty).$$

For every $\kappa>1$ and every finite n,

$$\Gamma^*_{n-1,1/2}\le\Gamma^*_{\infty,1/2}\le\tfrac{a}{2}\sqrt{\kappa-1}+\tfrac{3}{8}\log\kappa+\tfrac{3}{2}, \qquad a\approx 1.3287, \quad a_{\infty,1}\approx 0.81343.$$

The discrete-to-continuum limit and recovery construction are proved in Appendix D. The optimizer has a positive analytic density. For $\gamma=1/2$, its total mass is a^2 and its initial density is a^{-2}. In the shifted spectral coordinate $\zeta=\lambda/\mu-1$, its principal probability measure satisfies $\nu^*(\{0\})=a^{-2}$ and $\nu^*((\zeta,\infty))\sim 2(\pi a)^{-1}\zeta^{-1/2}$.

Remark 5.2 (Amplification as Green length)*.* The discrete and continuous rewards read the same Green profile:

$$\sum_m \lambda_m(0;\nu)^\gamma=\sum_m e^{-2\gamma \mathsf{X}_m}, \qquad R_\gamma(M)=\int e^{-2\gamma \mathsf{X}(y)}\,dy, \quad \mathsf{X}=\tfrac{1}{2}\log K.$$

In particular, $\theta\sum_m e^{-\mathsf{X}_m}\le a+o(1)$ uniformly over admissible spectra as $\kappa\to\infty$. Every positive Jacobi chain is a Lanczos realization at its first vertex, so the supremum covers all such chains in the interval. The type-to-step correspondence and the distinct isospectral functional of [21] are in Remarks D.10 and D.23.

For $n\ge 2$ and per-increment budgets $\omega_l\le\omega$, the uniform first-order FC contraction threshold is

$$\omega_{\mathrm{crit}}=\frac{1}{\Gamma^*_{n-1,1/2}}, \qquad \omega_{\mathrm{crit}}\sim\frac{2}{a}\kappa^{-1/2}, \qquad 2/a\approx 1.5053. \tag{9}$$

The empty $n=1$ criterion has no first-order obstruction. The limit uses $n\theta\to\infty$; the actual worst cycle ratio has an additional uniform quadratic remainder. Under the total Frobenius budget $\sum_l\|\widehat{E}_l\|_F^2\le\omega^2$, the sharp coefficient is $\omega(\Gamma^*_{n-1,1}/2)^{1/2}\sim\omega\sqrt{a_{\infty,1/4}}\,\kappa^{1/4}$, with $\sqrt{a_{\infty,1/4}}\approx 0.45095$ (Proposition B.9). A rational barrier proves the strict comparison with $\sum_{m\ge1}\operatorname{sech}(m\theta)$ for $\kappa\ge 1600$ (Appendix D).

method	gradients	terminal gradient error, leading term
FC	$n+1$	$\sum_l \omega_l\|e_{l+1}\|_A$
XRFC	$n+1$	$\frac{r-1}{r+1}\omega'\|e_{n-1}\|_A$
RFC	at most $2n-1$	$K(W_0,\omega')\|e_{n-1}\|_A$

Table 1: A full n-atom cycle, $n\ge 2$. Errors are in the inverse-base metric; FC/RFC have quadratic/cubic remainders. XRFC's exact row concerns constant prefixes. Final corrected/virtual outputs are unevaluated. RFC reaches $2n-1$ gradients when every noninitial gate in the evaluated prefix accepts.

The one-gradient lower bound remains positive after normalization by initial error, whereas gated RFC is uniformly second order. On a known base ball of radius r, suitable prior trials instead give $O(r\omega+\omega^2)$. Proposition C.11 specifies these different information regimes.

6. From Hessian variation to function values

Near a stationary x_* with $H_* \succ 0$, use canonical coordinates $H_*^{1/2}(x - x_*)$, scalar errors $r_k = \|x_k - x_*\|_{H_*}$ and normalized canonical steps $\hat{\sigma}_k = H_*^{1/2} s_k / \|s_k\|_{H_*}$. For $f \in C^3$, put $\mathsf{T} = D^3 f(x_*)$ in these coordinates; $\|\mathsf{T}\|_\sigma$ is the supremum of $|\mathsf{T}(u, v, w)|$ over unit vectors.

For a segment $[a_i, a_i + s_i]$, the leading Taylor increment is a slice $\mathsf{T}[h, \cdot, \cdot]$. All segments share this frozen tensor; higher variations enter the remainder. Thus the unrestricted operator minimax is generally too large for a single function. The compatible bounds and their genuine cubic attainers are Theorems E.7–E.9; the one-gradient function obstruction is Theorem E.16. The positive construction uses the information absent from those gradient-only examples.

Let $m_i = a_i + s_i/2$ and define the endpoint trapezoid defect

$$\delta_i^{\mathrm{tr}} = f(a_i + s_i) - f(a_i) - \tfrac{1}{2}\big(\nabla f(a_i) + \nabla f(a_i + s_i)\big)^\top s_i.$$

The corresponding single-pair scalar correction is classical [18]; the point here is its symmetric window assembly. At anchor x choose λ_i, put $w_i = x - m_i - \lambda_i s_i$, and correct the model by

$$\overline{T}_F = \overline{T} - 12\,\mathrm{diag}(\lambda_i \delta_i^{\mathrm{tr}}). \tag{10}$$

The split formulas use the original coordinates; their bounds below use canonical norms. The choices $\lambda_i^Y = y_i^\top (x - m_i)/(y_i^\top s_i)$ and $\lambda_i^E = s_i^\top (x - m_i)/\|s_i\|^2$ make w_i perpendicular to y_i or s_i, respectively. The defect is zero on quadratics and exactly determines the relevant cubic diagonal correction (Lemma E.13). In a superlinear full window it removes the leading contribution of the oldest pair.

For a full window $S_k = [s_{k-n}, \ldots, s_{k-1}]$ and $g_k = \nabla f(x_k)$, the symmetric secant iterations are

$$x_{k+1} = x_k - S_k T_k^{-1} S_k^\top g_k, \qquad T_k = \begin{cases} \overline{T}_k, & \mathrm{SS1}, \\ \overline{T}_{F,k}, & \mathrm{SS1-F}. \end{cases} \tag{11}$$

Each uses one new gradient per step. Their R-orders use the convention and function τ_1 of Section 1. For the splits at x_k, assume

$$\text{(S)} \qquad \|w_j\| \le C_w \|x_k - x_{j+1}\|.$$

The model-metric split has $C_w = 1 + o(1)$ and the Euclidean split has $C_w \le \kappa^{1/2}(1 + o(1))$, where here $\kappa = \mathrm{cond}(H_*)$.

Theorem 6.1 (Trapezoid defects improve the secant R-order bound)**.** *Let $f \in C^3$ near x_*, $H_* \succ 0$, $T_3 = \|\mathsf{T}\|_\sigma$, $\|D^3 f(x) - D^3 f(y)\|_\sigma \le L_3 \|x - y\|$ (canonical coordinates), the splits satisfy (S), and let the SS1-F choice in* (11) *generate $x_k \to x_*$, $x_k \ne x_*$, with $\sigma_{\min}([\hat{\sigma}_{k-n}, \ldots, \hat{\sigma}_{k-1}]) \ge \sigma > 0$. Then $r_{k+1} = o(r_k)$ and*

$$r_{k+1} \le \Big[\frac{(C_w + \frac{1}{2})T_3}{\sigma^2} r_{k-n+1} + \frac{L_3}{12\sigma^2} r_{k-n}^2\Big](1 + o(1))\, r_k.$$

The R-order per gradient is at least $\tau_1(n)$, against $\tau_1(n+1)$ for SS1. If $D^3 f$ is only continuous, still $r_{k+1}/(r_k r_{k-n}) \to 0$.

The theorem uses exact values of f. In dimension one the corrected curvature $y/s - 6\delta^{\mathrm{tr}}/s^2$ is exact on cubics, recovering order two versus the golden ratio of the secant method [1, 30]. Full proofs, restarted-cycle bounds and the complete comparison of sliding methods are in Appendix E.

Let $\tau_{\mathrm{FC}}(n) > 1$ solve $t^n = t + 1$ $(n \ge 2)$.

Classical reset-CG theory gives n-step quadratic convergence [31, 32]. Sliding FC has the sharper product recurrence of Powell's exact-search BFGS estimate, $r_{k+n} = O(r_k r_{k+1})$, and hence the same guaranteed order $\tau_{\mathrm{FC}}(n)$ [2, 33]. The function-value correction improves Burdakov's SS1 bound from $\tau_1(n+1)$ to $\tau_1(n)$ while retaining one new gradient per step. For $n = 2$ its exponent equals $\tau_{\mathrm{FC}}(n)$; for $n \ge 3$ it is larger. This strengthened guarantee uses uniform window independence and a Lipschitz third derivative. Fixed-memory L-BFGS has a global R-linear guarantee on uniformly convex problems under

method	additional hypotheses	guaranteed R-order per gradient
restarted FC	nondegenerate reference cycle, C^3	$2^{1/n}$
restarted RFC	same	$3^{1/(2n-1)}$
sliding FC	defined exact-search iterations	$\tau_{\mathrm{FC}}(n)$
sliding RFC	controlled probes, independent window	$\tau_1(n)^{1/2}$
SS1	uniformly independent window	$\tau_1(n+1)$
SS1-F	independence, (S), Lipschitz $D^3 f$	$\tau_1(n)$

Table 2: Conditional lower guarantees on local R-order, with convergence to $H_* \succ 0$ and a Lipschitz Hessian throughout. The exact-search rows use the operator accounting of Definition 2.5. The C^3 restarted bounds include the sharp tensor coefficients of Theorem E.14. The precise window hypotheses are in Theorems E.18, E.19 and 6.1. With $m_{\max} = n$, XR and XR-F inherit the last two rows under Theorem 7.3 and Corollary 7.4.

its search and initialization conditions [34, Theorem 7.1]; the local full-window results here quantify the benefit of retaining n independent directions.

7. XR: construction and guarantees

XR retains the virtual projection (7), estimates curvature in the remaining direction, and tests the window before using its inverse. The model metric is distinguished by its robust correction: for uncertainty $-\Omega\overline{T} \preceq \mathcal{Z} \preceq \Omega\overline{T}$, full correction is uniquely minimax for $\Omega < 1$ and no correction for $\Omega > 1$ (Theorem F.1). The relative defect has a hereditary test:

$$\overline{T} + i\mathsf{D}/\theta_{\mathrm{w}} \succ 0 \quad \Longleftrightarrow \quad \overline{T} \succ 0, \quad \Omega_\star(C) < \theta_{\mathrm{w}}. \tag{12}$$

Here $i^2 = -1$; positivity is Hermitian positivity. Every ordered subwindow inherits it. Independence of S is checked separately. The subroutine UnrefutedSuffix returns the number of oldest pairs to delete by reverse Cholesky (Algorithm 3, with the contract of Proposition F.2).

7.1. Model step and adaptive iteration

Store $(s_i, y_i, a_i, \delta_i^{\mathrm{tr}}, \varphi_i)$ with $\varphi_i = y_i^\top(x - a_i)$ at the current anchor. Then

$$\lambda_i = \varphi_i / C_{ii} - \tfrac{1}{2} \tag{13}$$

is the model-metric split. For XR-F, limit individual diagonal corrections before testing the selected model:

$$h_i = 12\lambda_i\delta_i^{\mathrm{tr}}\,\mathbf{1}_{\{|12\lambda_i\delta_i^{\mathrm{tr}}| \le C_{ii}/2\}}, \qquad \overline{T}_F^{\mathrm{g}} = \overline{T} - \mathrm{diag}(h_i).$$

Use $\overline{T}_F^{\mathrm{g}}$ if its Hermitian test passes, and $\overline{T}$ otherwise. These size and positivity gates are inactive eventually under the local hypotheses below, recovering the raw correction (10).

Trial policies. The practical policy uses the latest positive secant curvature [35]: initialize $\rho = 1/\tau_0$ and update $\rho \leftarrow s^\top y/(s^\top s)$ whenever a positive pair is observed. Set $\hat{c} = \rho\|d\|^2$ in the missing direction. The spectral policy instead uses the midpoint of (8), with running bounds and an adequacy floor as specified below. Both policies use the same projection, conjugate direction, window and memory controller. The experiments use the practical policy.

Take $1 \le m_0 \le m_{\max} \le n$, integer $q \ge 1$, $\theta_{\mathrm{w}} > 0$, $0 < \sigma_{\mathrm{A}} < 1/2$, $0 < \epsilon < 1$. At a nonstationary initial point set $\tau_0 = 1/\|g_0\|_\infty$, $c_1 = \epsilon\tau_0$, $c_2 = \tau_0/\epsilon$; $m_{\mathrm{cap}} \in [m_0, m_{\max}]$ is the current memory cap. For the spectral policy take $\varpi > 1$, $0 < \eta_0 < 1 - 2\sigma_{\mathrm{A}}$; running bounds $(\hat{\mu}, \hat{L})$ start at optional $0 < \mu_0 \le L_0$, or $(+\infty, 0)$ (bounds unset). Write

$$c_{\mathrm{A}} = 2(1 - \sigma_{\mathrm{A}}) - \eta_0 > 1, \qquad \hat{c} \leftarrow \min\{\max\{\hat{c}, \kappa_{\mathrm{hi}}\|d\|^2/c_{\mathrm{A}}\}, \kappa_{\mathrm{hi}}\|d\|^2\}$$

for its adequacy floor when $\kappa_{\mathrm{hi}} > 0$ (Proposition F.10). Define the fallback curvature $\rho_{\mathrm{b}} = \rho$ for the practical policy, and $(\hat{\mu} + \hat{L})/2$ for the spectral policy when bounds are set, $1/\tau_0$ otherwise.

Algorithm 1 ModelStep at anchor (x, g)

keep the newest $\min(m_{\mathrm{cap}}, n)$ pairs; delete oldest pairs until S has full column rank; $p_{\mathrm{mem}} \leftarrow$ #retained pairs, $p \leftarrow p_{\mathrm{mem}}$
$o \leftarrow \mathrm{UnrefutedSuffix}(\overline{T}, \mathsf{D}, \theta_{\mathrm{w}})$; delete the oldest o pairs; $p \leftarrow p - o$
if $p = 0$ **then**
    $u \leftarrow -g/\rho_{\mathrm{b}}$
else
    $T \leftarrow \overline{T}^{\mathrm{g}}_F$ if F is on and $\overline{T}^{\mathrm{g}}_F + i\mathsf{D}/\theta_{\mathrm{w}} \succ 0$, else $T \leftarrow \overline{T}$
    $w \leftarrow -T^{-1}S^\top g$; $\hat{x} \leftarrow x + Sw$; $\hat{g} \leftarrow g + Yw$
    if spectral policy: compute extreme Ritz values $\lambda_{\mp}$ of $(\overline{T}, S^\top S)$; update $\hat{\mu} \leftarrow \min(\hat{\mu}, \lambda_-/\varpi)$, $\hat{L} \leftarrow \max(\hat{L}, \varpi\lambda_+)$
    if $p = n$ **then**
        $u \leftarrow Sw$
    else
        $d \leftarrow -\hat{g} + S\beta$, $(Y^\top S)\beta = Y^\top \hat{g}$; if singular use $d \leftarrow -\hat{g}$
        if $d = 0$ **then**
            $u \leftarrow Sw$
        else
            if practical policy **then**
                $\hat{c} \leftarrow \rho\|d\|^2$
            else
                compute $\kappa_{\mathrm{lo}}, \kappa_{\mathrm{hi}}, \kappa_{\mathrm{mid}}$ from (8) with $(\hat{\mu}, \hat{L})$
                $\hat{c} \leftarrow \kappa_{\mathrm{mid}}\|d\|^2$ if $\kappa_{\mathrm{lo}} \le \kappa_{\mathrm{hi}}$, else $\min\{\max\{\kappa_{\mathrm{mid}}, \hat{\mu}\}, \hat{L}\}\|d\|^2$; apply adequacy floor if $\kappa_{\mathrm{hi}} > 0$
            end if
            $u \leftarrow Sw - (g^\top d/\hat{c})d$
        end if
    end if
end if
if $g^\top u > -c_1\|g\|^2$ or $\|u\| > c_2\|g\|$, replace u by $-g/\bar{\rho}$, with $\bar{\rho}$ equal to ρ_{b} clipped to $[1/c_2, 1/c_1]$ ▷ (G)
return $u, (p_{\mathrm{mem}}, o)$

A trial $x_t = x + tu$ is accepted when $\chi \le 2(1 - \sigma_{\mathrm{A}})$, where

$$\chi = \frac{2(f(x_t) - f(x) - tg^\top u)}{-tg^\top u}.$$

At $t = 1$, for the raw uncorrected step with a successful nonsingular conjugacy solve (or no added direction), this ratio equals the true averaged curvature divided by the model curvature (Lemma F.6). The first informative rejected trial supplies one additional gradient observation at the same anchor. Subsequent backtracking needs only function values until acceptance. The ordinary virtual model is anchor invariant (Lemma F.7); the F model is recomputed at the anchor.

The memory controller uses the number j of trial evaluations and the first ModelStep's (p_1, o_1). If $o_1 \ge \max(1, \lceil p_1/2 \rceil)$, set $m_{\mathrm{cap}} \leftarrow \max(m_0, \lfloor m_{\mathrm{cap}}/2 \rfloor)$ and reset the clean counter ν. Otherwise increment ν exactly when $j = 1$, $o_1 = 0$ and $p_1 = \min(m_{\mathrm{cap}}, n)$, and reset it in every other case. At $\nu = q$, increase m_{cap} by m_0 up to $m_{\max}$ and reset ν.

Algorithm 2 XR and XR-F (F switch)

evaluate f, g at x_0; return if $g = 0$; $x \leftarrow x_0$, $\tau_0 \leftarrow 1/\|g\|_\infty$, $c_1 \leftarrow \epsilon\tau_0$, $c_2 \leftarrow \tau_0/\epsilon$
start with no pairs, $\rho \leftarrow 1/\tau_0$, $m_{\mathrm{cap}} \leftarrow m_0$, $\nu \leftarrow 0$; initialize optional spectral bounds
while $g \neq 0$ **do**
    call ModelStep; retain $(p_1, o_1) \leftarrow (p_{\mathrm{mem}}, o)$; $t \leftarrow 1$, probed $\leftarrow$ false, $j \leftarrow 0$
    repeat
        $j \leftarrow j + 1$; $x_t \leftarrow x + tu$; evaluate f_t, g_t if not probed, and only f_t otherwise
        compute χ; test finite $f_t \le f + \sigma_{\mathrm{A}} t g^\top u$ with $g^\top u < 0$; if it passes, acquire g_t if needed
        set $\chi \leftarrow +\infty$ for nonfinite acquired data; accept $\leftarrow$ [test passes and g_t is finite]; informative $\leftarrow$ [g_t acquired and f_t, g_t, χ finite]
        if informative, (accept or not probed), and $(tu)^\top(g_t - g) > 0$: update ρ and append $(tu, g_t - g)$ with $a = x$, $\varphi = 0$, $\delta^{\mathrm{tr}} = f_t - f - tg^\top u - \frac{1}{2}(tu)^\top(g_t - g)$
        if not accept **then**
            if informative and not probed: probed $\leftarrow$ true, call ModelStep, $t \leftarrow 1$; else $t \leftarrow t/\min(\max(\chi, 2), 10)$
        end if
    until accept
    $\varphi_i \leftarrow \varphi_i + y_i^\top(tu)$ for every retained pair; $(x, f, g) \leftarrow (x_t, f_t, g_t)$
    update the memory cap and clean counter by the rule above
end while

The nominal parameters are $q = 3$, $\theta_{\mathrm{w}} = 1$, $\sigma_{\mathrm{A}} = 10^{-4}$, $\epsilon = 10^{-10}$; the spectral option uses $\varpi = 2$, $\eta_0 = 10^{-2}$. With maintained Gram data, the standard dense implementation costs $O(n(p+1) + p^3)$ arithmetic work per model step, excluding oracle evaluations.

Theorem 7.1 (Quadratic identity and global convergence)**.** *On a quadratic with n atoms, Algorithm 2 with $m_0 = n$, no safeguard (G), and any nonzero trials retains the full Krylov window. At a model step with $k < n$ pairs,*

$$(\hat{x}, \hat{g}, d) = (x_k^0, g_k^0, d_k^0), \qquad \overline{T}_F = \overline{T}.$$

Before the window is full, each stored pair raises the Krylov dimension. After n independent pairs, the virtual output is x_*, using $n + 1$ gradients including g_0. Unless already evaluated, the next full-window trial evaluates x_*. These identities hold for either trial policy and any acceptance rule with the prescribed rejection update, and for smaller memory until the first deletion.

If ∇f is L-Lipschitz for some $L > 0$ and f is bounded below, the safeguarded algorithm satisfies

$$f(x_{k+1}) \le f(x_k) - \sigma_{\mathrm{A}} t_{\min} c_1 \|\nabla f(x_k)\|^2, \qquad t_{\min} = \min\{1, (1 - \sigma_{\mathrm{A}})c_1/(5Lc_2^2)\} > 0.$$

Thus $\sum_k \|\nabla f(x_k)\|^2 < \infty$ and every limit point is stationary; bounded sublevel sets ensure existence of limit points.

Writing $f_{\inf} = \inf f > -\infty$, summation also gives the explicit nonconvex stationarity rate

$$\min_{0 \le j < K} \|\nabla f(x_j)\| \le \left(\frac{f(x_0) - f_{\inf}}{K \sigma_{\mathrm{A}} t_{\min} c_1}\right)^{1/2}, \qquad K \ge 1.$$

The first assertion is a quadratic identity; the second is the gradient-related Armijo argument [36]. Local fast convergence requires separate hypotheses. The error $r_k = \|x_k - x_*\|_{H_*}$ remains canonical; the norms in the local hypotheses below are Euclidean, with extreme eigenvalues $\lambda_{\min}, \lambda_{\max}$ of H_*.

Assumption 7.2 (Local hypotheses for XR-grow)**.** (G1) $f \in C^2$ near x_*, $\|\nabla^2 f(x) - \nabla^2 f(y)\| \le L_H^{\mathrm{E}} \|x - y\|$, $H_* \succ 0$, and the accepted points satisfy $x_k \to x_*$, $x_k \neq x_*$. (G2) (Uniform independence.) For $k \ge K_0$, $\sigma_{\min}([s_i/\|s_i\|]) \ge \sigma_\flat > 0$ for the incoming chronological window of every model step, after retaining the last $\min(m_{\mathrm{cap}}, n)$ pairs and before rank cleanup or Hermitian deletions; this includes

the model step after a rejected probe. (G3) (Trial adequacy while $p < n$.) There are $0 < \ell \le \Lambda$, $\eta > 0$ with $\ell\|d\|^2 \le \hat{c} \le \Lambda\|d\|^2$ and $d^\top H_* d \le (2(1-\sigma_{\mathrm{A}}) - \eta)\hat{c}$ at every model step with $1 \le p < n$, $d \ne 0$, $k \ge K_0$. (G4) $\sigma_{\mathrm{A}} < \frac{1}{2}$, $c_1 < c_\flat$, $c_2 > C_\flat$, with the constants of Lemma F.8 depending only on $n, \sigma_\flat, \lambda_{\min}, \lambda_{\max}, \ell, \Lambda$. When $m_0 = n$, use its full-window constants, independent of ℓ, Λ.

Theorem 7.3 (Transition to a full secant window)**.** *For XR (F off), assume (G1), (G2), (G4), and (G3) when $m_0 < n$. Eventually the window test deletes no pair, the safeguard is inactive, and every first trial is accepted. The window reaches* $\min(m_{\max}, n)$ *pairs. If $m_{\max} = n$, the iteration is eventually SS1 and*

$$r_{k+1} \le K_k r_k r_{k-n}, \qquad K_k \le \frac{L_H^{\mathrm{E}}}{2\lambda_{\min}^{3/2}\sigma_H^2}(1 + o(1)), \qquad \sigma_H = \sigma_\flat\sqrt{\lambda_{\min}/\lambda_{\max}}.$$

The R-order per gradient is at least $\tau_1(n+1)$.

Corollary 7.4 (XR-F in the superlinear phase)**.** *Under the local hypotheses of Theorem 7.3 with $m_{\max} = n$, a Lipschitz $D^3 f$ and exact function values, XR-F (model-metric or Euclidean split) eventually uses the raw corrected model $\overline{T}_F$, accepts every trial at $t = 1$, has a window consisting of the last n accepted steps, and is SS1-F, of R-order at least $\tau_1(n)$ per gradient.*

A strictly containing spectral interval makes the midpoint trial adequate under the explicit Armijo restriction of Proposition F.9; the floor supplies the more general guarantee in Proposition F.10.

Proposition 7.5 (Trial-policy invariants)**.** *At a fixed observed state, the retained window, selected model and virtual point are independent of the positive missing-direction scale. On faithful quadratic data, suppose the incoming window fits its capacity, its steps are independent,* $\operatorname{range} S = \mathcal{K}_k$, $x - x_0 \in \mathcal{K}_k$, *and CG has not terminated before k. Then the raw trial has the form*

$$x + u = x_k^0 + \tau d_k^0; \qquad p = n \implies u = -ST^{-1}S^\top g,$$

independently of how the scale was obtained. The practical policy above, with safeguard (G), has the global stationarity guarantee of Theorem 7.1 under its global hypotheses; in particular, $\sum_k \|\nabla f(x_k)\|^2 < \infty$.

Proof. Window selection and the virtual projection are independent of the trial scale; Lemma A.2 identifies the quadratic projection and direction. The full-window branch contains no missing direction. The one-probe Armijo proof of Theorem 7.1 uses only the safeguarded slope and norm bounds. The same argument applies to the actual practical recursion, including a recomputation after its single rejected probe. □

The midpoint minimax claim concerns its prescribed spectral information. For the practical policy, the local transition and fast-rate conclusions apply under the stated independence and trial-adequacy hypotheses for that policy.

8. Numerical experiments

We compare the practical XR and XR-F of Section 7.1 with six fixed competitors: compiled L-BFGS-B without bounds [37, 38], L-CG_DESCENT [39], MS-L-BFGS [40], shape-changing MSS [41] and SR1 [42], and O-ACCEL-B [43]. Each retains its published search or trust-region mechanism. XR starts with memory $\min(10, n)$ and grows up to $\min(100, n)$; its parameters are shared across problems. The ablations vary individual components of the same observed model.

The CUTEst selection [44] has 257 instances from 149 source families. Its four nonlinear panels contain 77/43 sum-of-squares and 67/61 general instances, split at $n = 300$; nine classified quadratics are additional controls. The sum-of-squares classification describes the objective construction; both nonlinear groups include objectives that may be nonconvex. The target is $\|g\|_2 \le 10^{-8}\max\{1, \|g_0\|_2\}$, with limits of 10^5 charged points and 120 solver seconds. Every actual gradient is counted, including

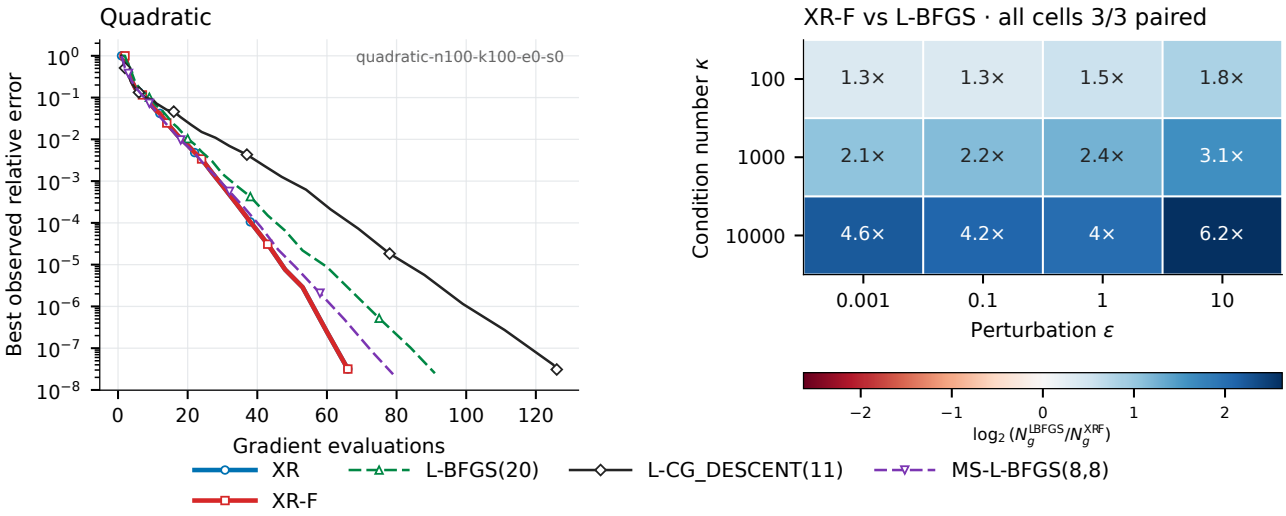


Figure 1: Controlled tests. The left panel shows best observed relative A-energy error, including rejected query points, for $n = 100$, $\kappa = 100$ and seed zero. The right panel includes every condition-number/perturbation cell and seed: color is the median log gradient cost ratio L-BFGS/XR-F on common successes; blue favors XR-F. All three seeds are jointly solved in every cell, as stated above the grid.

at rejected probes. Function-only requests do not acquire a gradient. On nonconvex instances this target measures first-order stationarity. Appendix G gives the experimental settings and additional comparisons.

Controlled tests. The quadratic controls have $n \in \{20, 100\}$, $\kappa \in \{10^2, 10^4\}$ and three seeds. The nonlinear family is

$$f_\varepsilon(x) = \tfrac{1}{2}x^\top Ax + \varepsilon \sum_{j=1}^{n} (e^{a_j^\top x} - 1 - a_j^\top x), \qquad a_j = A^{1/2}u_j, \quad \|u_j\| = 1.$$

Here $A \succ 0$ has a geometric spectrum on $[1, \kappa]$, the minimizer is $x_* = 0$, and $\|x_0\|_A = 1$. The complete grid uses $n = 100$, $\kappa \in \{10^2, 10^3, 10^4\}$, $\varepsilon \in \{10^{-3}, 10^{-1}, 1, 10\}$ and three seeds. The reference-energy curve uses the base matrix A; the perturbed stationary Hessian is $H_* = A + \varepsilon \sum_j a_j a_j^\top$.

Both XR variants reduce paired gradient costs against every comparator in each controlled family. The complete grid isolates a clearer mechanism: the advantage increases with the base condition number and persists across the nonlinear perturbations. It extends beyond the small-perturbation regime, where agreement with CG's virtual reference is most directly visible. The ablations below separate the contributions of window selection and endpoint values when exact quadratic conjugacy is lost.

CUTEst comparisons. Figure 2 gives Dolan–Moré profiles [45]. Failures have infinite ratio, and even problems failed by every method remain in the denominator. Ties all count at ratio one. The profiles separate efficiency near the best observed cost from reliability at larger allowances. XR and XR-F occupy the leading group in gradient cost, with high final coverage on both sum-of-squares and general objectives. Uncorrected XR attains slightly broader coverage. XR-F's paired gradient cost is more favorable on the large general objectives; the value correction's benefit therefore depends on the problem regime. The full and empirical-logit views expose both the initial separation and the closely spaced upper plateaux.

Elapsed-time profiles add the cost of maintaining that information (Figure 5). L-CG_DESCENT has the clearest lead near the best observed time. At larger allowances XR approaches its coverage, matching it on both general-objective cohorts. Its overall coverage is higher than that of the other

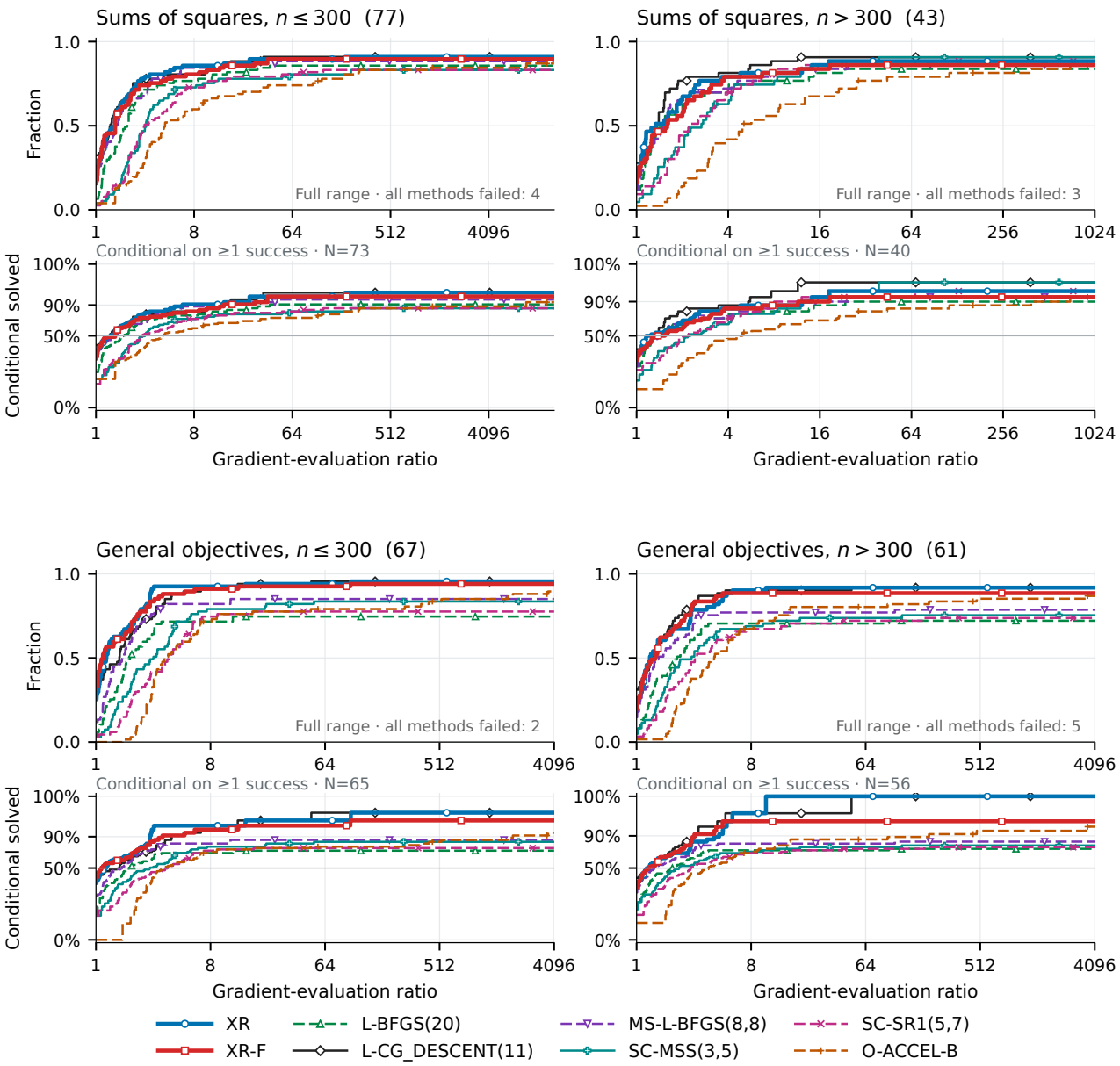


Figure 2: Actual-gradient performance profiles on the four fixed CUTEst cohorts. Ordinary views use every scheduled task. Companion empirical-logit views condition on the tasks solved by at least one method; their cohort sizes are stated separately. Both show the full base-two ratio range. The logit view separates the upper curves while retaining the complete tail.

limited-memory quasi-Newton comparators. Maintaining and testing richer windows carries an arithmetic cost. The full-batch comparison below examines a regime in which reducing oracle work produces a clearer elapsed-time advantage.

Regularized logistic regression. We also solve full-batch logistic objectives on eight datasets from the LIBSVM collection [46], covering dense and sparse designs, large sample counts and high feature dimensions:

$$F_\lambda(w,b) = \frac{1}{N}\sum_{i=1}^{N} \log(1+\exp[-y_i(a_i^\top w + b)]) + \frac{\lambda}{2}(\|w\|^2 + b^2).$$

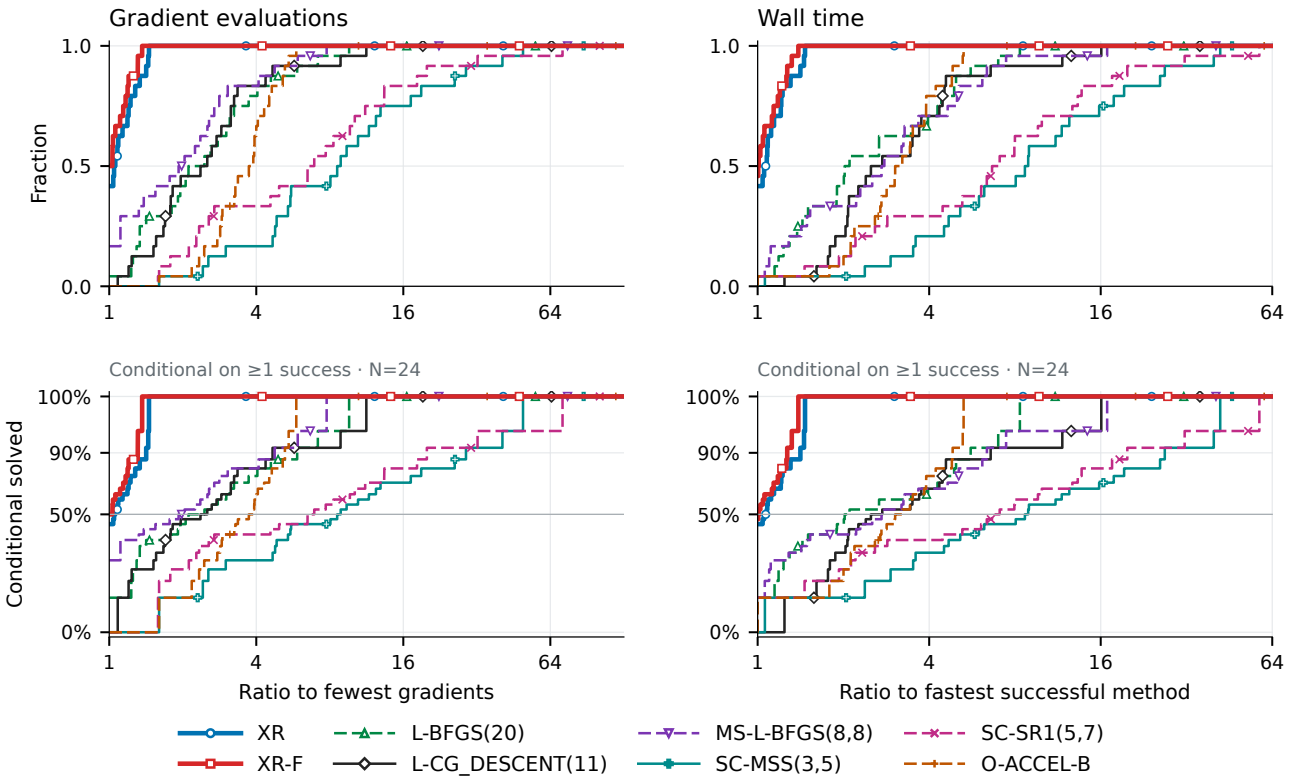


Figure 3: Regularized logistic regression on eight datasets and three penalty levels. Gradient and elapsed-time profiles use all 24 objectives in ordinary panels; empirical-logit panels condition on success by at least one method. The zero start, preprocessing and penalty grid are common to all methods.

Here $y_i \in \{-1, 1\}$ and every nonzero feature vector a_i is normalized to unit length. The intercept is regularized along with the other coefficients. The scalar loss has second derivative at most $1/4$, and row normalization gives $\lambda I \preceq \nabla^2 F_\lambda \preceq (\lambda + 1/2)I$. Thus every objective is globally smooth and strongly convex, with a unique minimizer for each $\lambda \in \{10^{-2}, 10^{-4}, 10^{-6}\}$. All methods start at $(w, b) = 0$ and use exact analytic gradients, the same stationarity target and a 300-second limit. The resulting 24 objectives give 192 method–problem runs. Dataset details are in Table 10; Figure 3 compares costs.

All eight methods reach the target on all 24 objectives. Regularization separates their efficiency regimes. With the strongest penalty, MS-L-BFGS often needs the fewest gradients, while XR and XR-F remain competitive in time. At the two weaker penalties, the fastest run and the run with the fewest gradients both belong to XR or XR-F on every dataset. The advantage appears across dense and sparse designs and grows as the uniform ridge curvature weakens, mirroring the controlled conditioning trend. Lower gradient costs translate into elapsed-time gains on these full-batch objectives. Both XR variants account for the main gain, with a modest additional average benefit from the value correction. Table 11 separates the three regularization regimes.

Full-memory local behavior. A separate diagnostic uses $n \in \{5, 10, 20\}$, $\kappa \in \{10, 100\}$, ten seeds per cell, $\varepsilon = 0.1$ and $\|x_0\|_A = 0.03$. It runs the unchanged XR and XR-F with memory n and records every accepted iterate, to relative error 10^{-12} in the exact stationary norm $H_* = A + \varepsilon \sum_j a_j a_j^\top$. This Hessian is diagnostic information only. Figure 4 shows every accepted iterate for two dimension/condition-number settings, with the first full window identified. The traces locate rapid final contraction relative to the formation of the full window. The conditional asymptotic R-orders of Section 6 describe the limiting regime beyond this finite diagnostic. All 120 runs reach the target after using a full window, with the safeguard inactive throughout. Across the complete cohort, the two methods have similar

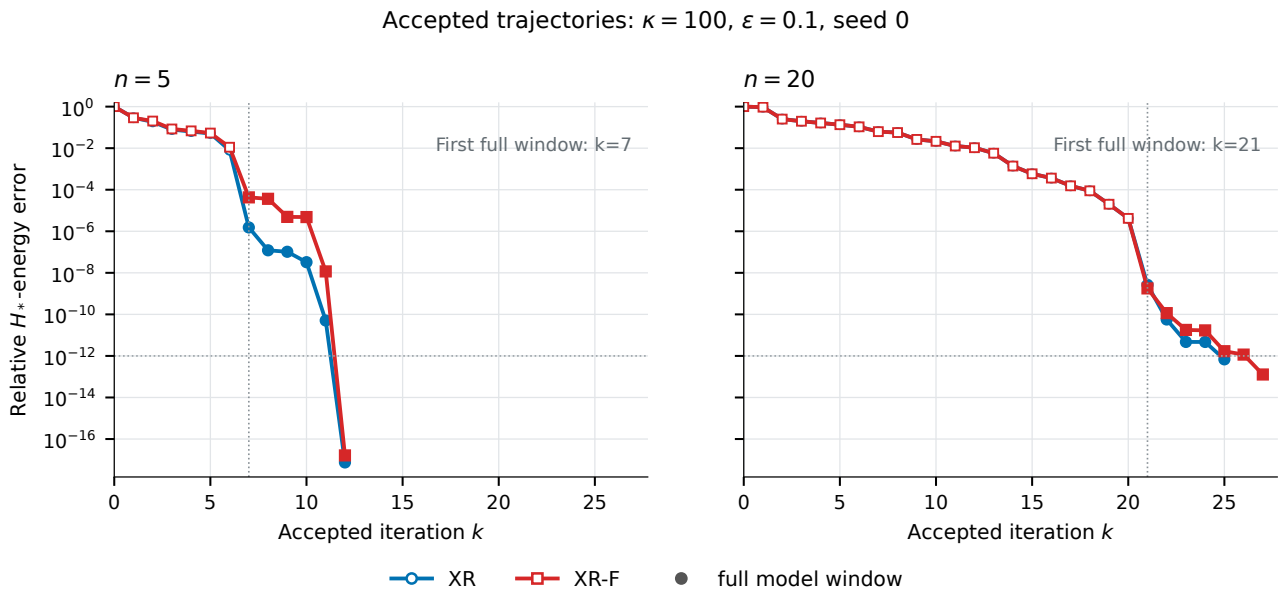


Figure 4: Full-memory local trajectories at $n = 5$ and $n = 20$, $\kappa = 100$, $\varepsilon = 0.1$, seed zero. Every marker is an accepted iterate; filled markers indicate a full retained window. The vertical axis is $\|x_k - x_*\|_{H_*}/\|x_0 - x_*\|_{H_*}$. Both panels use the same axes.

accuracy–cost distributions; the function-value correction's advantage depends on the individual trajectory.

Ablations. Six one-factor variants are compared on the same 64 CUTEst starts, selected by a fixed hash, 16 per panel. F-off equals XR; fixed memory caps storage at ten pairs; the positive-only window tests model positivity alone, keeping the F gate unchanged; no-probe omits the gradient at a rejected trial; midpoint uses the spectral interval and its floor; lagged uses a directional estimate clipped to current individual secant curvatures. Table 3 compares gradient costs on shared successes and the tasks solved exclusively by either variant. Growing memory reduces paired gradient cost and improves coverage relative to the fixed ten-pair window. Larger models increase arithmetic work, and the fixed window has a modest paired elapsed-time advantage. The rejected probe supplies information when the current model fails to predict a satisfactory step. Omitting it saves gradients while shifting work to function-only trials: paired function cost and elapsed time are essentially unchanged, and coverage decreases. F changes the local model's diagonal accuracy; its benefit on the large general objectives accompanies a small overall cost–coverage tradeoff on CUTEst.

The trial comparison illustrates the difference between a minimax rule for a supplied interval and the quality of an interval accumulated online. Historical model bounds can retain a very large curvature estimate long after its generating window has disappeared. The fresh-secant rule follows the recently observed scale and avoids that persistent conservatism. Directional lagged trials also perform well, recovering additional starts and slightly reducing paired gradient cost in this cohort. The main profiles use the fresh-secant policy throughout.

9. Conclusion

The secant flag separates observed disagreement from hidden curvature. It determines the sharp first variation of FC, the second-order price of a probed correction and the missing-moment obstruction for one gradient per iteration. Spectral optimization fixes their scale through a unique extremal string. On actual functions, endpoint values supply additional diagonal information and improve the guaranteed full-window R-order. XR combines this model with tested windows and safeguarded trials, giving global stationarity and conditional local secant rates.

Variant	Solved	$N_g^{\text{variant}}/N_g^{\text{base}}$	Both solved	Base only	Variant only
XR-F	56/64	1	56	0	0
XR-F/F-off	58/64	0.88	56	0	2
XR-F/fixed-memory	54/64	1.16	53	3	1
XR-F/positive-only	56/64	1.06	55	1	1
XR-F/no-probe	54/64	0.87	51	5	3
XR-F/midpoint-trial	30/64	3.24	30	26	0
XR-F/lagged-trial	60/64	0.93	56	0	4

Table 3: One-factor variants of the baseline in the first row. Ratios are geometric means on instances solved by both variants; values above one favor the baseline. Base only and Variant only count exclusive successes.

The experiments show where the richer model pays for itself. On controlled objectives its gradient advantage grows with the base condition number. On CUTEst, XR combines broad coverage with competitive gradient costs, while CG's lighter updates retain a time advantage. On weakly regularized full-batch logistic objectives, the reduction in gradient work yields the best observed times within the comparison. The ablations expose the separate cost and coverage effects of growing memory, rejected probes and the function-value correction.

A common string language also connects the spectral extremum to the isospectral energy of [21]. The product of the two static solutions gives the diagonal Green response; their normalized ratio gives the CG energy profile, $\lambda_m = e^{-2\mathsf{X}_m}$ (Remark 3.2). Here powers of this profile are accumulated over prefixes under a spectral interval constraint. The companion problem fixes the eigenvalues and minimizes the sum of squared diagonal responses. The corresponding strings, change of static frame and type are identified in Remarks D.7 and D.10. For prescribed eigenvalues, the final-prefix energy is maximized by the persymmetric chain (Remark D.3); optimizing the accumulated profile over the interval leads to a.

Use of AI tools. The authors used Claude Opus 5.5 and ChatGPT-6.1 Sol to assist with exposition, formal proof verification and literature checks. The authors verified the statements and take responsibility for the results and accompanying code.

References

[1] J. F. Traub. *Iterative Methods for the Solution of Equations*. Prentice-Hall, Englewood Cliffs, NJ, 1964.

[2] Y. Yuan. On the least Q-order of convergence of variable metric algorithms. *IMA J. Numer. Anal.*, 4(2):233–239, 1984. doi:10.1093/imanum/4.2.233.

[3] M. R. Hestenes. Multiplier and gradient methods. *J. Optim. Theory Appl.*, 4(5):303–320, 1969. doi:10.1007/BF00927673. Also in L. A. Zadeh, L. W. Neustadt, A. V. Balakrishnan (eds.), Computing Methods in Optimization Problems 2, Academic Press, 1969, pp. 143–163.

[4] L. C. W. Dixon. Quasi-Newton algorithms generate identical points. *Math. Program.*, 2(1):383–387, 1972. doi:10.1007/BF01584554.

[5] L. C. W. Dixon. Quasi Newton techniques generate identical points II: the proofs of four new theorems. *Math. Program.*, 3(1):345–358, 1972. doi:10.1007/BF01585007.

[6] L. C. W. Dixon. Conjugate directions without linear searches. *J. Inst. Math. Appl.*, 11(3):317–328, 1973. doi:10.1093/imamat/11.3.317.

[7] L. C. W. Dixon. Conjugate gradient algorithms: quadratic termination without linear searches. *J. Inst. Math. Appl.*, 15(1):9–18, 1975. doi:10.1093/imamat/15.1.9.

[8] L. Nazareth. A conjugate direction algorithm without line searches. *J. Optim. Theory Appl.*, 23(3):373–387, 1977. doi:10.1007/BF00933447.

[9] L. Nazareth. A relationship between the BFGS and conjugate gradient algorithms and its implications for new algorithms. *SIAM J. Numer. Anal.*, 16(5):794–800, 1979. doi:10.1137/0716059.

[10] O. P. Burdakov. Methods of the secant type for systems of equations with symmetric Jacobian matrix. *Numer. Funct. Anal. Optim.*, 6(2):183–195, 1983. doi:10.1080/01630568308816160.

[11] R. H. Byrd, J. Nocedal, and R. B. Schnabel. Representations of quasi-Newton matrices and their use in limited memory methods. *Math. Program.*, 63(1–3):129–156, 1994. doi:10.1007/BF01582063.

[12] S. Gratton, V. Malmedy, and Ph. L. Toint. Quasi-Newton updates with weighted secant equations. *Optim. Methods Softw.*, 30(4):748–755, 2015. doi:10.1080/10556788.2014.971025.

[13] D. Scieur, L. Liu, T. Pumir, and N. Boumal. Generalization of quasi-Newton methods: application to robust symmetric multisecant updates. In *Proc. 24th International Conference on Artificial Intelligence and Statistics*, volume 130 of *Proc. Mach. Learn. Res.*, pages 550–558, 2021. `https://proceedings.mlr.press/v130/scieur21a.html`.

[14] M. Lee and Y. Sun. Advancing multi-secant quasi-Newton methods for general convex functions. arXiv:2504.07330, 2025. doi:10.48550/arXiv.2504.07330. `https://arxiv.org/abs/2504.07330`.

[15] D. Scieur. Adaptive quasi-Newton and Anderson acceleration framework with explicit global (accelerated) convergence rates. In *Proc. 27th International Conference on Artificial Intelligence and Statistics*, volume 238 of *Proc. Mach. Learn. Res.*, pages 883–891, 2024. `https://proceedings.mlr.press/v238/scieur24a.html`.

[16] C. McCoid and M. J. Gander. Extrapolation methods as nonlinear Krylov subspace methods. *Linear Algebra Appl.*, 693:83–97, 2024. doi:10.1016/j.laa.2023.04.004.

[17] J. A. Ford and I. A. Moghrabi. Using function-values in multi-step quasi-Newton methods. *J. Comput. Appl. Math.*, 66(1–2):201–211, 1996. doi:10.1016/0377-0427(95)00178-6.

[18] J. Z. Zhang, N. Y. Deng, and L. H. Chen. New quasi-Newton equation and related methods for unconstrained optimization. *J. Optim. Theory Appl.*, 102(1):147–167, 1999. doi:10.1023/A:1021898630001.

[19] D. Pasechnyuk-Vilensky and M. Takáč. Chebyshev-exact acceleration under Hessian variation, I: Sine–Jacobi method. arXiv:2606.16671v2, 4 October 2026. doi:10.48550/arXiv.2606.16671. `https://arxiv.org/abs/2606.16671v2`.

[20] D. Pasechnyuk-Vilensky and M. Takáč. Chebyshev-exact acceleration under Hessian variation, II: Computer-assisted variational lower bound. arXiv:2610.05164v2, 6 October 2026. doi:10.48550/arXiv.2610.05164. `https://arxiv.org/abs/2610.05164v2`.

[21] D. Pasechnyuk-Vilensky and M. Takáč. Resolvent-diagonal energy of isospectral Jacobi matrices. Preprint, 2026.

[22] O. P. Burdakov, J. M. Martínez, and E. A. Pilotta. A limited-memory multipoint symmetric secant method for bound constrained optimization. *Ann. Oper. Res.*, 117(1–4):51–70, 2002. doi:10.1023/A:1021561204463.

[23] J. B. Erway and M. Rezapour. A new multipoint symmetric secant method with a dense initial matrix. *Optim. Methods Softw.*, 38(4):698–722, 2023. doi:10.1080/10556788.2023.2167993.

[24] G. Meurant. *The Lanczos and Conjugate Gradient Algorithms: From Theory to Finite Precision Computations*. SIAM, Philadelphia, 2006. doi:10.1137/1.9780898718140.

[25] G. H. Golub and G. Meurant. *Matrices, Moments and Quadrature with Applications*. Princeton University Press, Princeton, NJ, 2010. doi:10.1515/9781400833887.

[26] F. Gantmakher and M. Krein. Sur les matrices complètement non négatives et oscillatoires. *Compositio Math.*, 4:445–476, 1937. `https://www.numdam.org/item/CM_1937__4__445_0/`.

[27] G. Meurant. A review on the inverse of symmetric tridiagonal and block tridiagonal matrices. *SIAM J. Matrix Anal. Appl.*, 13(3):707–728, 1992. doi:10.1137/0613045.

[28] J. F. Traub, G. W. Wasilkowski, and H. Woźniakowski. *Information-Based Complexity*. Academic Press, 1988.

[29] A. Albert. Conditions for positive and nonnegative definiteness in terms of pseudoinverses. *SIAM J. Appl. Math.*, 17(2):434–440, 1969. doi:10.1137/0117041.

[30] W. C. Davidon. Variable metric method for minimization. Technical Report ANL-5990 (Rev.), Argonne National Laboratory, Argonne, IL, 1959. doi:10.2172/4222000. Reprinted in SIAM J. Optim. 1 (1991) 1–17, doi:10.1137/0801001.

[31] G. P. McCormick and K. Ritter. Alternative proofs of the convergence properties of the conjugate-gradient method. *J. Optim. Theory Appl.*, 13(5):497–518, 1974. doi:10.1007/BF00933041.

[32] J. Stoer. On the relation between quadratic termination and convergence properties of minimization algorithms. Part I. Theory. *Numer. Math.*, 28(3):343–366, 1977. doi:10.1007/BF01389973.

[33] M. J. D. Powell. On the rate of convergence of variable metric algorithms for unconstrained optimization. In *Proceedings of the International Congress of Mathematicians (Warszawa, 1983)*, volume 2, pages 1525–1539, Warszawa, 1984. PWN. `https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1983.2/ICM1983.2.ocr.pdf`. Also Report DAMTP 1983/NA7, University of Cambridge.

[34] D. C. Liu and J. Nocedal. On the limited memory BFGS method for large scale optimization. *Math. Program.*, 45(1–3):503–528, 1989. doi:10.1007/BF01589116.

[35] J. Barzilai and J. M. Borwein. Two-point step size gradient methods. *IMA Journal of Numerical Analysis*, 8(1):141–148, 1988. doi:10.1093/imanum/8.1.141.

[36] D. P. Bertsekas. *Nonlinear Programming*. Athena Scientific, Belmont, MA, 2nd edition, 1999.

[37] C. Zhu, R. H. Byrd, P. Lu, and J. Nocedal. Algorithm 778: L-BFGS-B: Fortran subroutines for large-scale bound-constrained optimization. *ACM Trans. Math. Software*, 23(4):550–560, 1997. doi:10.1145/279232.279236.

[38] J. L. Morales and J. Nocedal. Remark on "Algorithm 778: L-BFGS-B: Fortran subroutines for large-scale bound constrained optimization". *ACM Trans. Math. Software*, 38(1):7:1–7:4, 2011. doi:10.1145/2049662.2049669.

[39] W. W. Hager and H. Zhang. The limited memory conjugate gradient method. *SIAM J. Optim.*, 23(4):2150–2168, 2013. doi:10.1137/120898097.

[40] F. V. Gubarev. A multi-secant limited-memory BFGS method. Optimization Online, 7 March 2026. `https://optimization-online.org/2026/03/a-multi-secant-limited-memory-bfgs-method/`.

[41] J. J. Brust, J. B. Erway, and R. F. Marcia. Shape-changing trust-region methods using multipoint symmetric secant matrices. *Optim. Methods Softw.*, 39(5):990–1007, 2024. doi:10.1080/10556788.2023.2296441.

[42] J. Brust, O. Burdakov, J. B. Erway, and R. F. Marcia. Algorithm 1030: SC-SR1: MATLAB software for limited-memory SR1 trust-region methods. *ACM Trans. Math. Software*, 48(4):48:1–48:33, 2022. doi:10.1145/3550269.

[43] A. N. Riseth. Objective acceleration for unconstrained optimization. *Numer. Linear Algebra Appl.*, 26(1):e2216, 2019. doi:10.1002/nla.2216.

[44] N. I. M. Gould, D. Orban, and Ph. L. Toint. CUTEst: a constrained and unconstrained testing environment with safe threads for mathematical optimization. *Comput. Optim. Appl.*, 60(3):545–557, 2015. doi:10.1007/s10589-014-9687-3.

[45] E. D. Dolan and J. J. Moré. Benchmarking optimization software with performance profiles. *Math. Program.*, 91(2):201–213, 2002. doi:10.1007/s101070100263.

[46] C.-C. Chang and C.-J. Lin. LIBSVM: A library for support vector machines. *ACM Trans. Intell. Syst. Technol.*, 2(3):27:1–27:27, 2011. doi:10.1145/1961189.1961199. Data collection: `https://www.csie.ntu.edu.tw/~cjlin/libsvmtools/datasets/`.

[47] S. M. Rump. Structured perturbations Part I: normwise distances. *SIAM J. Matrix Anal. Appl.*, 25(1):1–30, 2003. doi:10.1137/S0895479802405732.

[48] S. Chandrasekaran, G. H. Golub, M. Gu, and A. H. Sayed. Parameter estimation in the presence of bounded data uncertainties. *SIAM J. Matrix Anal. Appl.*, 19(1):235–252, 1998. doi:10.1137/S0895479896301674.

[49] L. El Ghaoui and H. Lebret. Robust solutions to least-squares problems with uncertain data. *SIAM J. Matrix Anal. Appl.*, 18(4):1035–1064, 1997. doi:10.1137/S0895479896298130.

[50] M. R. Hestenes and E. Stiefel. Methods of conjugate gradients for solving linear systems. *J. Res. Nat. Bur. Standards*, 49(6):409–436, 1952. doi:10.6028/jres.049.044.

[51] W. W. Hager and H. Zhang. A new conjugate gradient method with guaranteed descent and an efficient line search. *SIAM J. Optim.*, 16(1):170–192, 2005. doi:10.1137/030601880.

[52] T. G. Kolda, D. P. O'Leary, and L. Nazareth. BFGS with update skipping and varying memory. *SIAM J. Optim.*, 8(4):1060–1083, 1998. doi:10.1137/S1052623496306450.

[53] R. B. Schnabel. Quasi-Newton methods using multiple secant equations. Tech. Rep. CU-CS-247-83, Department of Computer Science, University of Colorado at Boulder, 1983. doi:10.21236/ADA131444.

[54] J. Vlček and L. Lukšan. Properties of the block BFGS update and its application to the limited-memory block BNS method for unconstrained minimization. *Numer. Algorithms*, 80(3):957–987, 2019. doi:10.1007/s11075-018-0513-3.

[55] G. Schuller. On the order of convergence of certain quasi-Newton methods. *Numer. Math.*, 23(2):181–192, 1974. doi:10.1007/bf01459951.

[56] Y. Notay. Flexible conjugate gradients. *SIAM J. Sci. Comput.*, 22(4):1444–1460, 2000. doi:10.1137/S1064827599362314.

[57] Ju. L. Daleckii and S. G. Krein. Integration and differentiation of functions of Hermitian operators and applications to the theory of perturbations. In *Thirteen Papers on Functional Analysis and Partial Differential Equations*, volume 47 of *Amer. Math. Soc. Transl. Ser. 2*, pages 1–30. American Mathematical Society, Providence, RI, 1965. doi:10.1090/trans2/047/01. Russian original: Trudy Sem. Funkcional. Anal. Voronezh. Gos. Univ. 1 (1956) 81–105.

[58] G. H. Golub. Some modified matrix eigenvalue problems. *SIAM Rev.*, 15(2):318–334, 1973. doi:10.1137/1015032.

[59] W. Gautschi. The interplay between classical analysis and (numerical) linear algebra—a tribute to Gene H. Golub. *Electron. Trans. Numer. Anal.*, 13:119–147, 2002. `https://etna.ricam.oeaw.ac.at/vol.13.2002/pp119-147.dir/pp119-147.pdf`.

[60] M. G. Krein and A. A. Nudel'man. *The Markov Moment Problem and Extremal Problems*, volume 50 of *Translations of Mathematical Monographs*. American Mathematical Society, Providence, RI, 1977. doi:10.1090/mmono/050.

[61] H. Dette and W. J. Studden. *The Theory of Canonical Moments with Applications in Statistics, Probability, and Analysis*. Wiley Series in Probability and Statistics. Wiley, New York, 1997.

[62] F. R. Gantmacher and M. G. Krein. *Oscillation Matrices and Kernels and Small Vibrations of Mechanical Systems*. Revised edition. AMS Chelsea Publishing, Providence, RI, 2002. doi:10.1090/chel/345.

[63] F. Pukelsheim. *Optimal Design of Experiments*. Wiley, New York, 1993. Unabridged SIAM reprint, 2006. doi:10.1137/1.9780898719109.

[64] J. Kiefer and J. Wolfowitz. The equivalence of two extremum problems. *Canad. J. Math.*, 12:363–366, 1960. doi:10.4153/cjm-1960-030-4.

[65] J. Kiefer. General equivalence theory for optimum designs (approximate theory). *Ann. Statist.*, 2(5):849–879, 1974. doi:10.1214/aos/1176342810.

[66] A. Greenbaum. Comparison of splittings used with the conjugate gradient algorithm. *Numer. Math.*, 33(2):181–193, 1979. doi:10.1007/BF01399553.

[67] P. G. Hoel and A. Levine. Optimal spacing and weighting in polynomial prediction. *Ann. Math. Statist.*, 35(4):1553–1560, 1964. doi:10.1214/aoms/1177700379.

[68] G. Elfving. Optimum allocation in linear regression theory. *Ann. Math. Statist.*, 23(2):255–262, 1952. doi:10.1214/aoms/1177729442.

[69] C. de Boor and E. B. Saff. Finite sequences of orthogonal polynomials connected by a Jacobi matrix. *Linear Algebra Appl.*, 75:43–55, 1986. doi:10.1016/0024-3795(86)90180-1.

[70] *NIST Digital Library of Mathematical Functions*. F. W. J. Olver, A. B. Olde Daalhuis, D. W. Lozier, B. I. Schneider, R. F. Boisvert, C. W. Clark, B. R. Miller, B. V. Saunders, H. S. Cohl, and M. A. McClain, eds. Release 1.2.8 of 2026-09-15. `https://dlmf.nist.gov`.

[71] A. de la Garza. Spacing of information in polynomial regression. *Ann. Math. Statist.*, 25(1):123–130, 1954. doi:10.1214/aoms/1177728851.

[72] E. T. Whittaker and G. N. Watson. *A Course of Modern Analysis*. Cambridge University Press, Cambridge, 4th edition, 1927. Reissued in the Cambridge Mathematical Library, 1996. doi:10.1017/CBO9780511608759.

[73] E. B. Davies and E. M. Harrell II. Conformally flat Riemannian metrics, Schrödinger operators, and semiclassical approximation. *J. Differential Equations*, 66(2):165–188, 1987. doi:10.1016/0022-0396(87)90030-1.

[74] R. Beals, D. H. Sattinger, and J. Szmigielski. Multipeakons and the classical moment problem. *Adv. Math.*, 154(2):229–257, 2000. doi:10.1006/aima.1999.1883.

[75] T. J. Stieltjes. Recherches sur les fractions continues. *Ann. Fac. Sci. Toulouse Math. (1)*, 8(4):J1–J122, 1894. doi:10.5802/afst.108.

[76] I. S. Kac and M. G. Krein. On the spectral functions of the string. *Amer. Math. Soc. Transl. Ser. 2*, 103:19–102, 1974. doi:10.1090/trans2/103/02.

[77] Y. Tourigny. Continued fraction solution of Krein's inverse problem. *Inverse Problems*, 27(8):085008, 2011. doi:10.1088/0266-5611/27/8/085008. `https://arxiv.org/abs/1101.2989`.

[78] M. G. Krein. On a generalization of investigations of Stieltjes. *Dokl. Akad. Nauk SSSR*, 87(6):881–884, 1952. In Russian.

[79] I. S. Kac and V. N. Pivovarchik. On the density of the mass distribution of a string at the origin. *Integral Equations Operator Theory*, 81(4):581–599, 2015. doi:10.1007/s00020-014-2209-y.

[80] O. Lafitte, M. Williams, and K. Zumbrun. Block-diagonalization of ODEs in the semiclassical limit and C^ω versus C^∞ stationary phase. *SIAM J. Math. Anal.*, 48(3):1773–1797, 2016. doi:10.1137/15M103042X.

[81] L. Ambrosio, N. Fusco, and D. Pallara. *Functions of Bounded Variation and Free Discontinuity Problems*. Oxford Mathematical Monographs. Clarendon Press, Oxford, 2000. doi:10.1093/oso/9780198502456.001.0001.

[82] J. Liesen and Z. Strakoš. *Krylov Subspace Methods: Principles and Analysis*. Oxford University Press, Oxford, 2013. doi:10.1093/acprof:oso/9780199655410.001.0001.

[83] E. Hairer, S. P. Nørsett, and G. Wanner. *Solving Ordinary Differential Equations I: Nonstiff Problems*, volume 8 of *Springer Series in Computational Mathematics*. Springer, Berlin, 2nd edition, 1993. ISBN 9783540566700. doi:10.1007/978-3-540-78862-1.

[84] N. H. Bingham, C. M. Goldie, and J. L. Teugels. *Regular Variation*, volume 27 of *Encyclopedia of Mathematics and its Applications*. Cambridge University Press, Cambridge, 1987. doi:10.1017/CBO9780511721434.

[85] Yu. Nesterov. *Lectures on Convex Optimization*, volume 137 of *Springer Optimization and Its Applications*. Springer, Cham, 2nd edition, 2018. doi:10.1007/978-3-319-91578-4.

[86] S. Banach. Über homogene Polynome in (L^2). *Studia Math.*, 7(1):36–44, 1938. doi:10.4064/sm-7-1-36-44.

[87] S. Friedland and L.-H. Lim. Nuclear norm of higher-order tensors. *Math. Comp.*, 87(311):1255–1281, 2018. doi:10.1090/mcom/3239.

[88] J. M. Ortega and W. C. Rheinboldt. *Iterative Solution of Nonlinear Equations in Several Variables*. Academic Press, New York, 1970. Unabridged SIAM reprint, 2000. doi:10.1137/1.9780898719468.

[89] G. Szegő. Über einen Satz des Herrn Serge Bernstein. *Schriften der Königsberger Gelehrten Gesellschaft, Naturwissenschaftliche Klasse*, 5(4):59–70, 1928.

[90] J. G. van der Corput and G. Schaake. Berichtigung zu: Ungleichungen für Polynome und trigonometrische Polynome. *Compositio Mathematica*, 3:128, 1936. `https://www.numdam.org/item/CM_1936__3__128_0/`.

[91] K. Ritter. On the rate of superlinear convergence of a class of variable metric methods. *Numer. Math.*, 35(3):293–313, 1980. doi:10.1007/bf01396414.

[92] J. Stoer. On the convergence rate of imperfect minimization algorithms in Broyden's β-class. *Math. Program.*, 9(1):313–335, 1975. doi:10.1007/BF01681353.

[93] H. Crowder and P. Wolfe. Linear convergence of the conjugate gradient method. *IBM J. Res. Develop.*, 16(4):431–433, 1972. doi:10.1147/rd.164.0431. Earlier report: IBM Thomas J. Watson Research Center, DTIC AD0742126.

[94] M. J. D. Powell. Some convergence properties of the conjugate gradient method. *Math. Program.*, 11(1):42–49, 1976. doi:10.1007/BF01580369.

[95] J. Vlček and L. Lukšan. Two limited-memory optimization methods with minimum violation of the previous secant conditions. *Comput. Optim. Appl.*, 80(3):755–780, 2021. doi:10.1007/s10589-021-00318-y.

[96] P. T. Boggs and R. H. Byrd. Adaptive, limited-memory BFGS algorithms for unconstrained optimization. *SIAM J. Optim.*, 29(2):1282–1299, 2019. doi:10.1137/16M1065100.

[97] D.-H. Li and M. Fukushima. On the global convergence of the BFGS method for nonconvex unconstrained optimization problems. *SIAM J. Optim.*, 11(4):1054–1064, 2001. doi:10.1137/S1052623499354242.

[98] F. Mannel. A globalization of L-BFGS and the Barzilai–Borwein method for nonconvex unconstrained optimization. *J. Optim. Theory Appl.*, 204(3), Article 47, 2025. doi:10.1007/s10957-024-02565-5.

[99] D. B. Szyld. The many proofs of an identity on the norm of oblique projections. *Numer. Algorithms*, 42(3–4):309–323, 2006. doi:10.1007/s11075-006-9046-2.

[100] J. Brust, J. B. Erway, and R. F. Marcia. On solving L-SR1 trust-region subproblems. *Comput. Optim. Appl.*, 66(2):245–266, 2017. doi:10.1007/s10589-016-9868-3.

[101] D. F. Shanno and K.-H. Phua. Matrix conditioning and nonlinear optimization. *Math. Program.*, 14(1):149–160, 1978. doi:10.1007/bf01588962.

[102] J. J. Moré and D. J. Thuente. Line search algorithms with guaranteed sufficient decrease. *ACM Trans. Math. Software*, 20(3):286–307, 1994. doi:10.1145/192115.192132.

[103] J. C. Platt. Fast training of support vector machines using sequential minimal optimization. In B. Schölkopf, C. J. C. Burges, and A. J. Smola, editors, *Advances in Kernel Methods—Support Vector Learning*. MIT Press, Cambridge, MA, 1998. doi:10.7551/mitpress/1130.003.0016.

[104] R. Collobert, S. Bengio, and Y. Bengio. A parallel mixture of SVMs for very large scale problems. *Neural Computation*, 14(5):1105–1114, 2002. doi:10.1162/089976602753633402.

[105] I. Guyon, S. Gunn, A. Ben-Hur, and G. Dror. Result analysis of the NIPS 2003 feature selection challenge. In *Advances in Neural Information Processing Systems 17*, pages 545–552. MIT Press, 2005. `https://proceedings.neurips.cc/paper_files/paper/2004/hash/5e751896e527c862bf67251a474b3819-Abstract.html`.

[106] G.-X. Yuan, C.-H. Ho, and C.-J. Lin. An improved GLMNET for ℓ_1-regularized logistic regression. *J. Mach. Learn. Res.*, 13(64):1999–2030, 2012. `https://www.jmlr.org/papers/v13/yuan12a.html`.

[107] D. D. Lewis, Y. Yang, T. G. Rose, and F. Li. RCV1: A new benchmark collection for text categorization research. *J. Mach. Learn. Res.*, 5:361–397, 2004. `https://www.jmlr.org/papers/v5/lewis04a.html`.

A. Secant geometry and quadratic references

A.1. Correction identities and quadratic realizations

Use the CG notation of Section 3. For $0 < \mu < L$, put $\eta = (L+\mu)/(L-\mu) = \cosh\theta$, $q = e^{-\theta}$, $z(\lambda) = (L+\mu-2\lambda)/(L-\mu)$ and $p_N^*(\lambda) = T_N(z(\lambda))/T_N(\eta)$, $\varepsilon_N^* = \operatorname{sech}(N\theta)$. Here T_m, U_m are Chebyshev polynomials and $T_m(\eta) = \cosh(m\theta)$. We use $\operatorname{sym} X = (X + X^\top)/2$, tensor symmetrization Sym, the all-ones vector $\mathbf{1}$ and the strict upper part triu. The CG coefficient is $\beta_k = \|g_{k+1}^0\|^2/\|g_k^0\|^2$. A reference with N active atoms has $\dim \mathcal{K}_k = \min(k, N)$ and $g_k^0 \perp \mathcal{K}_k$ [24, 25]. Relative balls obey $(1-\rho)A_c \preceq A' \preceq (1+\rho)A_c$; congruence preserves Loewner order. FC/RFC commute with $A_k \mapsto cQA_kQ^\top$, $g_0 \mapsto c'Qg_0$, for orthogonal Q and $c, c' > 0$.

The following calculus combines symmetric backward error [47], robust least squares [48, 49] and generalized Schur complements [29].

Lemma A.1 (Window calculus)**.***(a) For nonsingular H, $w = -H^{-1}U^\top g$, $g_+ = g + BUw$,*

$$U^\top g_+ = (U^\top BU - H)w, \qquad r_{p+1} = -(U^\top B_p U - H)H^{-1}r_p,$$

for successive corrections with fixed U, H, $r_p = U^\top g_p$. If $Hw = -U^\top(g + \alpha A^{\mathrm{p}} d)$ and $g_+ = g + \alpha A^{\mathrm{s}} d + BUw$, then

$$U^\top g_+ = \alpha U^\top (A^{\mathrm{s}} - A^{\mathrm{p}})d + (U^\top BU - H)w. \tag{14}$$

If U is square and $J = U^{-\top}HU^{-1}$, then $Uw = -J^{-1}g$ and $g_+ = (B-J)Uw$. In canonical coordinates, with $\widehat{J} = A^{-1/2}JA^{-1/2}$ and $\mathcal{E} = A^{-1/2}(B-J)A^{-1/2}$,

$$\hat{g}_+ = -\mathcal{E}\widehat{J}^{-1}\hat{g}. \tag{15}$$

If normalized models are $I + O(\omega)$ and their errors $L_j + O(\omega^2)$, $L_j = O(\omega)$, iteration gives

$$\hat{g}^{(t)} = (-1)^t L_t \cdots L_1 r + O(\omega^{p+t+1}), \qquad \hat{g}^{(0)} = r + O(\omega^{p+1}), \quad r = O(\omega^p), \tag{16}$$

for fixed t (including $t = 1$). If the last column of U is d and $H\mathbf{u}_{\mathrm{last}} = U^\top z$, then $\alpha d - UH^{-1}U^\top(g + \alpha z) = -UH^{-1}U^\top g$.

(b) For $\Omega \geq 0$, $\{Ev : E = E^\top, \|E\| \leq \Omega\}$ is the ball of radius $\Omega\|v\|$. Hence

$$\max_{E=E^\top, \|E\|\leq\Omega} \|\rho + (N+E)b\| = \|\rho + Nb\| + \Omega\|b\|.$$

For $N = I$, the minimum is $\min(1, \Omega)\|\rho\|$; its minimizer is uniquely $-\rho$ for $\Omega < 1$, uniquely 0 for $\Omega > 1$, and $[-\rho, 0]$ for $\Omega = 1$.

(c) If $E = E^\top$, $\|E\| \le \omega$, P is orthogonal, $Pe = 0$, $\|e\| = 1$, and $Y = PEP$, $x = PEe$, then

$$Y^2 + xx^\top \preceq \omega^2 P, \qquad \|YX\|^2 \le \omega^2\|X\|^2 - (x^\top X)^2.$$

(d) Equal data force equal decisions, and

$$\max_{\pm} \|a \pm b\|^2 \ge \|a\|^2 + \|b\|^2, \qquad \max_{\pm} \|\rho_\pm + Nu\| \ge \tfrac{1}{2}\|\rho_+ - \rho_-\|.$$

If $\mathcal{F} \supset \mathcal{F}_0 \cup (2J - \mathcal{F}_0)$, every seminorm ν satisfies

$$\sup_{B\in\mathcal{F}} \nu(B - K_0) \ge \sup_{B\in\mathcal{F}_0} \nu(B - J).$$

For invertible G and $K = \Delta G^{-1}$,

$$\frac{\|K\rho\|}{\sqrt{1 + \|K\|^2}} \le \inf_v \max_{\pm} \|\rho + (G \pm \Delta)v\| \le \|K\rho\|,$$

where the upper bound uses $v = -G^{-1}\rho$.

(e) Let $T(E) = \sum_l L_l E_l v_l$, $1 \le p \le \infty$, $1/p + 1/q = 1$ and $\omega_l \ge 0$, with $E_l = E_l^\top$ constrained by $\|(\|E_l\|/\omega_l)_l\|_p \le 1$, with zero budgets interpreted as $E_l = 0$. Then

$$\sup_E \|T(E)\| = \max_{\|u\|=1} \big\|(\omega_l \|L_l^\top u\| \|v_l\|)_l\big\|_q. \tag{17}$$

For Frobenius budgets replace $\|L_l^\top u\|\|v_l\|$ by $\|\operatorname{sym}(L_l^\top u \otimes v_l)\|_F$. Also, for orthogonal P and symmetric Z, $\max_{E=PEP,\|E\|\le\omega}\langle E, Z\rangle = \omega\|PZP\|_*$, attained at $E = \omega \operatorname{sign}(PZP)$.

(f) Let $\omega > 0$, $x \in \mathbb{R}^q$ and a symmetric $K \in \mathbb{R}^{q\times q}$ with $\|K\| < \omega$. The values y satisfying $\left\|\left(\begin{smallmatrix} y & x^\top \\ x & K \end{smallmatrix}\right)\right\| \le \omega$ exist iff $x^\top(\omega^2 I - K^2)^{-1}x \le 1$, and form

$$\big[-\omega + x^\top(\omega I + K)^{-1}x, \;\; \omega - x^\top(\omega I - K)^{-1}x\big],$$

with centre $-x^\top K(\omega^2 I - K^2)^{-1}x$ and half-length $\omega - \omega\, x^\top(\omega^2 I - K^2)^{-1}x$. For $\|K\| \le \omega$, symmetric Y satisfies $\left\|\left(\begin{smallmatrix} Y & X^\top \\ X & K \end{smallmatrix}\right)\right\| \le \omega$ iff

$$-\omega I + X^\top(\omega I + K)^+ X \preceq Y \preceq \omega I - X^\top(\omega I - K)^+ X, \tag{18}$$

with $\operatorname{ran} X \subset \operatorname{ran}(\omega I + K) \cap \operatorname{ran}(\omega I - K)$.

Proof. (a) Substitute the model solve; for square U use $g_+ = -(B - J)J^{-1}g$ and expand the inverse. The anchor identity uses $UH^{-1}U^\top z = d$. (b) The symmetric image-ball identity is [47, Lem. 5.1]: on $\operatorname{span}\{v, z\}$ the block $\left(\begin{smallmatrix} a & b \\ b & -a \end{smallmatrix}\right)$ realizes $v \mapsto av + bz$ with exactly the target norm. Alignment and the triangle inequality give the maximum, minimum and equality cases. (c) Compress $E^2 \preceq \omega^2 I$ and take quadratic forms. (d) Use the parallelogram identity and pair reflected instances. Setting $v' = Gv$, the minimum average squared residual is $\rho^\top K^\top K(I + K^\top K)^{-1}\rho$; spectral calculus gives the bounds. (e) Apply (b), Hölder duality and the adjoint of the symmetric matrix action; diagonalization gives nuclear duality. (f) Apply the generalized Schur criterion [29] to $\omega I \pm \left(\begin{smallmatrix} Y & X^\top \\ X & K \end{smallmatrix}\right)$. It gives both range conditions and interval bounds; their sum and difference give the scalar centre, half-length and feasibility. □

With exact line search, HS and HZ nonlinear CG [50, 51], the convex Broyden class initialized by cI, $c > 0$ [4, 5, 9], and L-BFGS with positive scalar initialization and any positive memory retaining the newest pair [52, Cor. 2.5] are correction-free CG-consistent members of $\mathcal{M}$. FC uses the block-BFGS projector $P = I - Y(S^\top Y)^{-1}S^\top$, $d = -P^\top g$ [53–55]; unlike flexible-CG Gram–Schmidt [56], it enforces $Y^\top d = 0$ for nonconjugate pairs.

Lemma A.2 (CG-consistency)**.** *Let ν have n atoms, and let all operators equal A.*

(a) FC and RFC, restarted every n iterations or with a sliding window of any length $m \geq 1$, are well defined for $k < n$, generate the CG iterates and gradients, make only zero corrections, and terminate at $k = n$.

(b) XRFC with full memory is CG on quadratics for every sequence of nonzero trial steps, whatever rule chooses them: for $k < n$, $\hat{x}_k = x_k^0$, $\hat{g}_k = g_k^0$, $d_k = d_k^0$, $x_{k+1} = \hat{x}_k + \tau_k d_k$ and $\hat{x}_{k+1} - x_{k+1} = (\alpha_k - \tau_k) d_k^0$; $\hat{x}_n = x_*$ after $n+1$ gradients, without any line search.

Proof of Lemma A.2. On quadratic data $Y = AS$. Older conjugacy coefficients vanish because $As_i^0 \in \mathcal{K}_k \perp g_k^0$ for $i \leq k-2$; retaining the newest pair therefore gives the CG recurrence [52, Thm. 2.2, Cor. 2.3]. Exact search makes $g' \perp \operatorname{span}[S, d]$, so RFC's positive Gram model gives zero correction. This proves (a) by induction. For (b), $\operatorname{span} S_k = \mathcal{K}_k$ makes the virtual solve the CG Galerkin minimum. A nonzero trial adds the next Krylov direction, and $\hat{x}_{k+1} - x_{k+1} = (\alpha_k - \tau_k) d_k^0$ closes the induction. □

A.2. Secant observations and unresolved curvature

Lemma A.3 (Canonical flag identities)**.** *In Lemma 2.3, if $p = n$ and S is invertible, put $\Sigma = A^{1/2} S$, $\Pi_l = \Sigma I_{\leq l} \Sigma^{-1}$ and $J = S^{-\top} \overline{T} S^{-1}$. Then*

$$A^{-1/2}(B - J)A^{-1/2} = \sum_{l<n} \Pi_l^\top \widehat{E}_l \Pi_l, \qquad \Sigma^{-\top} \mathsf{D} \Sigma^{-1} = \sum_{l<n-1} (\Pi_l^\top \widehat{E}_l - \widehat{E}_l \Pi_l).$$

For A-conjugate steps, Π_l is orthogonal and the latter summands are $[\Pi_l, \widehat{E}_l]$.

For arbitrary $p \leq n$, a reference with at least p atoms and $s_i = s_i^0 + O(\|\widehat{E}\|)$, put $Q = [\hat{s}_0, \ldots, \hat{s}_{p-1}]$, $D_c = \operatorname{diag}(c_i)$. Under (1)*,*

$$\delta = Q^\top \sum_{l<p-1} [P_l, \widehat{E}_l] Q + O(\|\widehat{E}\|^2), \qquad D_c^{-1/2} \mathcal{Z} D_c^{-1/2} = Q^\top \sum_{l<p} P_l \widehat{E}_l P_l Q + O(\|\widehat{E}\|^2).$$

In particular $\|\delta\| \leq \sum_{l<p-1} \omega_l + O(\|\widehat{E}\|^2)$, since $\|[P_l, \widehat{E}_l]\| = \|P_l \widehat{E}_l (I - P_l)\| \leq \omega_l$.

Lemma A.4 (Defect transfer and reconstruction)**.** *For a correction-free exact-search method,*

$$s_i^\top g_N = \sum_{j=i+1}^{N-1} (\mathsf{D}_{ij} + s_j^\top y_i), \qquad \mathsf{D}_{ij} = s_i^\top (A_j - A_i) s_j.$$

If pair i is retained by FC at every later iteration, the second terms vanish. Thus in a full cycle $S^\top g_N = (\operatorname{triu} \mathsf{D})\mathbf{1}$, and in a sliding window $r_i^{(k)} = \sum_{j=i+1}^{k} \mathsf{D}_{ij}$ for $k - n + 1 \leq i < k$. The last search gives $d^\top g_+ = 0$. For any old-step residual vector r, whenever $\mathbf{S} = [\text{old steps}, d]$ is square and invertible,

$$g_+ = \mathbf{S}^{-\top} (r, 0)^\top,$$

with $r_i = \sum_{j>i} \mathsf{D}_{ij}$ in a cycle and $r = r^{(k)}$ for the last $n-1$ pairs of a sliding window without restart.

Proof of Lemma A.3. Congruence of (2) by $\Sigma^{-\top}(\cdot)\Sigma^{-1}$ gives the exact identities. At the reference $\Sigma = Q \operatorname{diag}(\sqrt{c_i})$, so $\Pi_l = P_l$. Insert $S = S^0 + O(\omega)$ into the exact sums; every summand already contains one increment, so the error is $O(\omega^2)$. Normalizing by $C_{ii} = c_i(1 + O(\omega))$ preserves this order. The norm of $[P, E]$ is the norm of its off-diagonal block $PE(I - P)$. □

Proof of Lemma A.4. Exact search gives $s_i^\top g_{i+1} = 0$. Telescope the gradient updates and write $s_i^\top y_j = \mathsf{D}_{ij} + s_j^\top y_i$. FC conjugacy annihilates the latter term whenever pair i is retained. The last exact search supplies the zero coordinate; inversion proves reconstruction in either window. □

We use the entry formulas of Lemma 2.3 together with $C_{ij} = s_i^\top A_j s_j$ for $i \leq j$ and $C_{ji} = s_j^\top A_i s_i$: with $A_0 = A$ and $G_A = S^\top A S$, for $i \leq j$,

$$\mathcal{Z}_{ij} = \sum_{l \geq j} (X_l)_{ij}, \qquad C_{ij} = \overline{T}_{ij} = (G_A)_{ij} + \sum_{l<j} (X_l)_{ij}, \qquad C_{ji} = (G_A)_{ij} + \sum_{l<i} (X_l)_{ij}. \tag{19}$$

Every entry $(X_l)_{ij}$, $i \le j$, enters the data if $l < \max(i,j) = j$ and the model error if $l \ge j$, never both.

Proof of Theorem 2.4. Compress each $-\omega_l A \preceq E_l \preceq \omega_l A$ by $SI_{\le l}$ and sum: $G_A \circ R = \sum_l \omega_l I_{\le l} G_A I_{\le l} \succeq 0$. Equality on the positive side is equivalent to $\widehat{E}_l = \omega_l I$ on $A^{1/2}\,\mathrm{span}\{s_0, \dots, s_l\}$, and is attained by that subspace projector. It gives $\mathsf{D}_{ij} = (G_A)_{ij} \sum_{i \le l < j} \omega_l$; with positive internal budgets this vanishes exactly for diagonal G_A. The negative case is identical.

For the indistinguishable pair take $h = ASG_A^{-1}\mathbf{u}_0, \quad \gamma = ((G_A^{-1})_{00})^{-1}, \quad A_i^{\pm} = A \pm \gamma \sum_{l<i} \omega_l h h^\top$. Then $S^\top h = \mathbf{u}_0$, $h^\top A^{-1} h = \gamma^{-1}$ and $\gamma h h^\top \preceq A$. Hence all operators are at least $(1-W)A$, the increment norms are ω_l, and every observed pair remains $y_i = As_i$. Their errors are $\pm\gamma W \mathbf{u}_0 \mathbf{u}_0^\top$, of normalized norm W. Their distance forces radius at least W. Conversely the flag bound gives $\|G_A^{-1/2} \mathcal{Z} G_A^{-1/2}\| \le \left\| \sum_l \omega_l \Pi_l^{G_A} (\Pi_l^{G_A})^\top \right\|, \quad \Pi_l^{G_A} = G_A^{-1/2} I_{\le l} G_A^{1/2}$. For diagonal G_A the right side is W, so the zero estimator attains the radius. □

A.3. Insertion kernels and perturbation budgets

For $R \in \mathcal{O}$, fixed affine recursions give polynomials r_s, R_s such that, for $g_s = Ae_s + \iota_s$,

$$e_N = p_R(A)e_0 + \sum_{s<N} R_s(A)\iota_s, \qquad K_s(\lambda,\nu) = R_s(\lambda) r_s(\nu). \tag{20}$$

An error-model drift E_s injects $\iota_s = E_s e_s$; at first order it acts on $r_s(A)e_0$. In [19] the spectral-pair ℓ_2 gain controls a total Hilbert–Schmidt budget. Our independent relative budgets use a different norm of the same linear response (Lemma A.1(e)). Secant increments instead give cumulative injections; Proposition A.7 makes the conversion explicit. These budgets cannot be interchanged without that conversion.

Proposition A.5 (The realization dictionary and the static direction)**.** *(i) Let T be a positive definite tridiagonal matrix with negative off-diagonal entries, $T = LDL^\top$, $D_{ss} = 1/\alpha_s$, $L_{s+1,s} = -\sqrt{\beta_s}$. Its Jacobi realization is the frozen CG recurrence*

$$x_{s+1} = x_s - \alpha_s g_s + \frac{\alpha_s \beta_{s-1}}{\alpha_{s-1}}(x_s - x_{s-1}). \tag{21}$$

With empty determinants equal to one,

$$r_s(\lambda) = \frac{\det(T_{0:s-1} - \lambda I)}{\det T_{0:s-1}}, \qquad R_s(\lambda) = -\frac{\det T_{0:s-1}\,\det(T_{s+1:N-1} - \lambda I)}{\det T}.$$

These are CG polynomials at the spectral measure of T; adaptive CG uses its own measured Lanczos matrix. Final Chebyshev exactness fixes the eigenvalues of T at the zeros of p_N^*, leaving the spectral weights free [19, Thms. 4.1–4.2].

(ii) Every fixed realization, of any memory, obeys

$$\sum_{s<N} K_s(\lambda,\lambda) = p_R'(\lambda).$$

For $p_R = p_N^*$ and relative query drifts $\|A^{-1/2} E_s A^{-1/2}\| \le \omega$, static drift along a top eigenvector already gives the exact error

$$\varepsilon_N^* T_N \left(1 + \frac{2\kappa\omega}{\kappa - 1}\right) \|e_0\|_A, \qquad \varepsilon_N^* = \mathrm{sech}(N\theta),$$

and first-order relative coefficient at least $2\kappa N^2 \varepsilon_N^*/(\kappa-1)$. This coefficient is the worst one for the prefix Jacobi chain and, more generally, chains whose leading and tail polynomials attain their uniform norms at the same endpoint.

(iii) Every CG-consistent realization terminates on a static shifted quadratic after its number of active atoms: static drift produces zero terminal error. Its nonzero variation price therefore comes from changing operators or missing observations, as quantified below.

Proof of Proposition A.5. Unrolling the affine recurrence gives (20); the leading-minor recurrence for $LDL^\top$ gives both determinant formulas and (21), as in [19, Thms. 4.1–4.2]. The same three-term recurrence identifies its spectral measure with frozen CG. Cofactor telescoping gives the useful constant-injection identity

$$\varphi_{N,j}(\lambda) = 1 + \lambda \sum_{s \geq j} R_s(\lambda) = \frac{\det(T - \lambda I_{\geq j})}{\det T}. \tag{22}$$

Expansion across the preceding edge gives the two-tail form. Inserting $\iota_s = \epsilon e_s$ and differentiating the recurrence for $A + \epsilon I$ proves the diagonal sum rule. Static drift on $Av = Lv$ gives $p_N^*(L(1+\omega))e_0$; $|T_N'(\pm 1)| = N^2$ gives its coefficient.

By symmetric-ball duality the sharp independent-budget gain is

$$\sigma_R = \max_{\|u\|=1} \sum_s \|r_s(A)\hat{e}_0\| \|A^{1/2} R_s(A) A^{1/2} u\|. \tag{23}$$

Shared endpoint maxima bound this by $L \sum_s |K_s(\mu,\mu)| \|\hat{e}_0\|$. The Chebyshev terms have one sign and sum to $2N^2\varepsilon_N^*/(L-\mu)$. The prefix leading/tail polynomials $2^{1-s}T_s$, $2^{-k}U_k$ have this property. CG-consistency proves (iii). □

Remark A.6 (Frozen realizations carry the product). Take T in (i) to be the Lanczos chain of Remark 3.2, up to the signs of the links, and put $\tilde{R}_s(\lambda) = \det(T_{s+1:N-1} - \lambda I)/\det T_{s+1:N-1}$. The cofactor formula $\mathsf{d}_s = \det T_{0:s-1} \det T_{s+1:N-1}/\det T$ factors the frozen kernel through the bead weight:

$$K_s(\lambda,\lambda') = -\mathsf{d}_s \, \tilde{R}_s(\lambda) \, r_s(\lambda'), \qquad \tilde{R}_s(0) = r_s(0) = 1.$$

At $\lambda = 0$ the sum rule (ii) reads $\sum_s \mathsf{d}_s = \operatorname{Tr} T^{-1}$. When 0 is the lower end of the spectral interval, as in [21, Prop. 1.2], the amplitude d_s is the endpoint channel, and the energy $\sum_s \mathsf{d}_s^2$ is its squared ℓ_2 norm. By (iii), CG-consistency removes this amplitude, and the adaptive price of Theorem 3.1 depends on the ratio coordinate alone. Fixed and adaptive realizations of one chain are thus priced by the two invariants of its static pair.

Proposition A.7 (Converting the two perturbation budgets)**.** *Error-model drifts give independent canonical injections $\hat{\iota}_j = \hat{E}_j \hat{e}_j$, ranging over balls of radii $\omega_j \|\hat{e}_j\|$. Secant increments give, at first order,*

$$\hat{\iota}_j = \sum_{l \leq j-2} \widehat{E}_l (\hat{e}_j - \hat{e}_{l+1}) + O(\omega^2).$$

Here the common reference trajectory is CG and $\hat{e}_j = \hat{e}_j$. Conversely an error-model pair has secant perturbation of minimum canonical norm $\|\hat{\iota}_{i+1} - \hat{\iota}_i\|/\sqrt{c_i}$ at first order. For another reference trajectory replace $\sqrt{c_i}$ by $\|A^{1/2} s_i^0\|$. Thus the conversion can be large near a small error or a short step.

Proof. The first assertion is the symmetric-ball identity of Lemma A.1(b). For the second, telescope $\hat{\iota}_j = \sum_{k<j} (\sum_{l<k} \widehat{E}_l) A^{1/2} s_k$ and replace the steps by their reference values; exchanging sums gives $\sum_{k=l+1}^{j-1} A^{1/2} s_k^0 = \hat{e}_j - \hat{e}_{l+1}$. Conversely $y_i = A s_i + \iota_{i+1} - \iota_i$, so the same ball identity gives the minimum norm of a symmetric operator with the required image. These are local expansions on compact nondegenerate reference sets. □

Lemma A.8 (Model error and the oldest pair)**.** *Let $U = [u_0, \ldots, u_{n-1}]$ be invertible, $v_i = A_i u_i$ with symmetric A_i, $H_{ij} = u_{\min(i,j)}^\top v_{\max(i,j)}$, and $J = U^{-\top} H U^{-1}$. Relative to a base $A \succ 0$, put $\widehat{A}_i = A^{-1/2} A_i A^{-1/2}$, $\widehat{J} = A^{-1/2} J A^{-1/2}$ and $\hat{u}_i = A^{1/2} u_i / \|u_i\|_A$, $\hat{U} = [\hat{u}_i]$, $\sigma = \sigma_{\min}(\hat{U})$, and*

$$X_{ij} = \hat{u}_i^\top (\widehat{A}_{\max(i,j)} - I) \hat{u}_j.$$

Then $\widehat{J} - I = \hat{U}^{-\top} X \hat{U}^{-1}$ *and* $\|\widehat{J} - I\| \le n \max_i \|\widehat{A}_i - I\|/\sigma^2$. *More precisely, if* $X = X_{00}\mathbf{u}_0\mathbf{u}_0^\top + X'$, *then*

$$\widehat{J} - I = X_{00} bb^\top + \hat{U}^{-\top} X' \hat{U}^{-1}, \qquad b = \hat{U}^{-\top}\mathbf{u}_0,$$
$$\|b\| = \Delta^{-1}, \qquad \Delta = \operatorname{dist}(\hat{u}_0, \operatorname{span}\{\hat{u}_1, \dots, \hat{u}_{n-1}\}) \ge \sigma.$$

For a model step $x_+ = x - J^{-1}g$, $g = A^*x$, *write* $\hat{x} = A^{1/2}x$ *and* $\widehat{A}^* = A^{-1/2}A^*A^{-1/2}$. *Then*

$$\hat{x}_+ = \widehat{J}^{-1}(\widehat{J} - \widehat{A}^*)\hat{x},$$
$$\|\hat{x}_+\| \le \|\widehat{J}^{-1}\|\Big(\frac{|X_{00}|}{\Delta^2} + \frac{\|X'\|}{\sigma^2} + \|\widehat{A}^* - I\|\Big)\|\hat{x}\|. \tag{24}$$

For any symmetric correction D, *these identities hold for* $J_D = U^{-\top}(H + D)U^{-1}$ *with* $X_D = X + N^{-1}DN^{-1}$, $N = \operatorname{diag}(\|u_i\|_A)$. *In a convergent sequence, uniform* $\sigma > 0$, $X_D \to 0$ *and* $\widehat{A}^* \to I$ *imply* $\widehat{J}_D \to I$, $\|\widehat{J}_D^{-1}\| = 1 + o(1)$ *and* $r_{k+1} = o(r_k)$, *where* $r_k = \|\hat{x}_k\|$. *If* $(X_D)_{00} = O(q_k)$ *and* $\|X_D - (X_D)_{00}\mathbf{u}_0\mathbf{u}_0^\top\| + \|\widehat{A}^* - I\| = o(q_k)$, *then*

$$r_{k+1} \le \left(\frac{|(X_D)_{00}|}{\Delta_k^2} + o(q_k)\right) r_k.$$

Proof. With $N = \operatorname{diag}(\|u_i\|_A)$, symmetry gives $H - U^\top AU = NXN$. Congruence proves the model identities and $\|X\| \le n \max_i \|\widehat{A}_i - I\|$ gives the bound. The dual vector b annihilates the newer columns and satisfies $b^\top \hat{u}_0 = 1$, so $\|b\| = \Delta^{-1} \le \sigma^{-1}$. Transforming $x_+ = x - J^{-1}A^*x$ proves (24). Adding D replaces X by X_D; continuity of inversion proves both asymptotic conclusions. □

B. Perturbation responses and correction minimax

Use Definition 2.1 and $B_k = A_k + A^{1/2}F_kA^{1/2}$; ω is the largest canonical perturbation norm. Unless stated otherwise, the reference (A, g_0) is fixed. All remainders are local in these perturbations. For a full n-atom cycle write $Q = [\hat{s}_0, \dots, \hat{s}_{n-1}]$.

B.1. A common perturbation calculus

Let $\mathcal{U}_N$ be the open set of positive definite references with at least N active atoms. Their first N CG steps are analytic. Write $\Phi(A, g_0, \Theta)$ for a finite run, with point outputs expressed as displacements from x_0 and Θ containing all canonical perturbations.

Lemma B.1 (Uniform remainders)**.** *Suppose* Φ *is* C^r, $r \ge 2$, *near* $\mathcal{U} \times \{0\}$, $\mathcal{U} \subset \mathcal{U}_N$ *open. For each compact* $K \subset \mathcal{U}$ *there are* $\delta_K, C_K > 0$ *such that every vector output has*

$$\|R_k y\| \le C_K \|\Theta\|^k \|e_0\|_A \quad (\|\Theta\| \le \delta_K,\ 1 \le k \le r),$$

where R_k *is the Taylor remainder after degree* $k-1$. *The degree-*k *coefficient is* C^{r-k} *in the reference. A relative base change* η *changes it by* $O_K(\eta)\|e_0\|_A$ *for* $k < r$, *and by* $o_K(1)\|e_0\|_A$ *for* $k = r$; *the latter is Lipschitz if* $\Phi \in C^{r,1}$.

For outputs of degree one in g_0, *with degree-zero decision coefficients, the same constants hold on the cone over* K. *These hypotheses hold on* $\mathcal{U}_n$ *with* $r = \infty$ *for FC/RFC, and* $r = 2$ *for positive* C^2 *degree-zero XRFC trial rules and smooth degree-zero CG-consistent methods of* $\mathcal{M}$. *Thus the remainders in Theorems 3.1, 3.3, C.1 are uniform on such cones.*

Proof. Positive reference CG energies, secant determinants and trial lengths give an open domain on which the stated rules have their asserted regularity; gates pass strictly. Taylor's integral formula on a compact tube gives the bounds. The coefficient is $D_\Theta^k \Phi/k!$, so bounded derivatives or uniform continuity give the reference estimates. Homogeneity gives the cone statement. □

Lemma B.2 (Aggregated compressions)**.** *Let P_l be orthogonal projectors, $P_l h = 0$, $\|h\| = 1$, $E_l = E_l^\top$, $\|E_l\| \le \omega_l$, and put*

$$W = \sum_l \omega_l, \quad X = \sum_l P_l E_l h, \quad Y = \sum_l P_l E_l P_l, \quad \gamma = \sum_l h^\top E_l h.$$

Then

$$\|X\| \le W, \qquad \|YX\| \le \|X\|\sqrt{W^2 - \|X\|^2}, \qquad |\gamma| \le \sqrt{W^2 - \|X\|^2}. \tag{25}$$

For independent Frobenius budgets $\|E_l\|_F \le \omega_l$, the first two bounds improve to

$$\sqrt{2}\|X\| \le W, \qquad \|YX\| \le \|X\|\sqrt{W^2 - 2\|X\|^2}. \tag{26}$$

These bounds are simultaneously sharp when a unit vector e belongs to every $\operatorname{ran} P_l$.

Proof. For $v = X/\|X\|$, set $a_l = v^\top P_l E_l h/\omega_l$, discarding zero budgets. Compression gives $\|P_l E_l P_l v\|, |h^\top E_l h| \le \omega_l\sqrt{1-a_l^2}$. Since $\sum_l \omega_l a_l = \|X\|$, concavity proves (25). In the Frobenius case use $\|E_l\|_F^2 \ge \|P_l E_l P_l\|_F^2 + 2\|P_l E_l h\|^2$ and replace a_l by $\sqrt{2}a_l$. Simultaneous attainment uses, with $s = \sqrt{1-a^2}$, $E_l = \omega_l[-see^\top + a(eh^\top + he^\top) + shh^\top]$ in the spectral case, and $\omega_l[\pm see^\top + (a/\sqrt{2})(eh^\top + he^\top)]$ in the Frobenius case. Zero cases follow by continuity. □

Lemma B.3 (Projected residual after a correction)**.** *In canonical coordinates suppose $y_i = A_i s_i + \xi_i$, $\|A_i - I\|, \|A - I\|, \|B - I\| \le \varepsilon$ and $\|\xi_i\| \le K\varepsilon\|s_i\|$. Take an RFC exact-search step with $d \ne 0$, $Y^\top d = 0$, $g' = g + \alpha Ad$, $U = [S, d]$ of p columns, and latest model H. Let D contain the column lengths of U, $T = UD^{-1} = [V, d/\|d\|]$ and*

$$\rho = D^{-1}U^\top g', \quad N = D^{-1}(U^\top BU - H)D^{-1}, \quad \chi = -UH^{-1}U^\top g', \quad R = \|V^\top g\|, \quad t = \|\alpha d\|.$$

If $D^{-1}HD^{-1}$ and $T^\top T$ have eigenvalues in $[a, b] \subset (0, \infty)$, then

$$\|N\| \le Cp\varepsilon, \quad \|\rho\| \le R + C\sqrt{p}\varepsilon t, \quad \|\chi\| \le C\|\rho\|, \quad R_+ \le Cp\varepsilon(R + C\sqrt{p}\varepsilon t). \tag{27}$$

Here $C = C(a, b, K)$ is independent of lengths and p; the last bound concerns the normalized recorded-step residual and assumes $\|\chi\| \le K\|\alpha d + \chi\|$. On finite nondegenerate RFC prefixes, $w_k = O(\omega)$ and both projected residuals $U_k^\top g_{k+1}, S_{k+1}^\top g_{k+1}$ are $O(\omega^2)$, with

$$y_i = A_i s_i + (B_i - A_i)U_i w_i = A_i s_i + O(\omega^2). \tag{28}$$

Proof. Conjugacy gives $s_i^\top Ad = s_i^\top(A - A_i)d - \xi_i^\top d$, and exact search gives $d^\top g' = 0$. These bound ρ; entrywise bounds control N, and the normalized model inverse controls χ. The identity

$$U^\top g_+ = (U^\top BU - H)w \tag{29}$$

then bounds the new residual. The assumption on χ bounds the new recorded unit column's coordinates in T. On a nondegenerate prefix Lemma B.1 supplies these bounds uniformly; finite induction gives the stated orders. Adding the two segments gives (28). □

Lemma B.4 (A two-dimensional data fibre)**.** *Let $n \ge 2$, $e = \hat{s}_0$, $h = \hat{s}_{n-1}$ and $c = c_{n-1}$. In canonical coordinates take*

$$\widehat{A}_0 = I - zhh^\top, \qquad \widehat{E}_l = x_l(eh^\top + he^\top) + y_l ee^\top + z_l hh^\top, \qquad z = \sum_l z_l, \quad x = \sum_l x_l, \quad b = \sum_l y_l + \zeta,$$

with $\widehat{E}' = \zeta ee^\top$ and no intermediate correction perturbations. Whenever the operators are positive definite and $|x| < 1$, every CG-consistent method of $\mathcal{M}$ has the reference data through iteration $n - 2$, the common normalized last probe $A_{n-1}A^{-1/2}h = A^{1/2}(h + xe)$, the reference last line step, and post-search gradient $\rho = x\sqrt{c}\,e$. The final correction operator is

$$\widehat{B} = M + bee^\top, \qquad M = I + x(eh^\top + he^\top).$$

Thus all data available before the final correction depend on x alone. For two admissible choices $b = \pm\beta$, every common correction has worst residual at least

$$\frac{\|\beta e e^\top M^{-1}\rho\|}{\sqrt{1+\|\beta e e^\top M^{-1}\|^2}},$$

and the centre correction $-M^{-1}\rho$ leaves $\mp\beta e e^\top M^{-1}\rho$.

Proof. The base change annihilates e, every increment annihilates the middle reference directions, and the accumulated $hh^\top$ coefficient vanishes at the last probe. Induction using CG-consistency fixes all preceding probes and zero corrections. The last curvature is unchanged, giving the stated gradient; summing the operators gives $\widehat{B}$. The radius and centre are Lemma A.1(d). □

Lemma B.5 (No stagnation)**.** *If p_k is the CG residual polynomial of step k and $k < N$ (the number of atoms of ν), then $\|\hat{e}_{k+1}\| \le \frac{\kappa-1}{\kappa+1}\|\hat{e}_k\|$ and*

$$c_k = \|\hat{e}_k\|^2 - \|\hat{e}_{k+1}\|^2 \;\ge\; \frac{4\kappa}{(\kappa+1)^2}\|\hat{e}_k\|^2 \;\ge\; \frac{\|\hat{e}_k\|^2}{\kappa}.$$

Proof. $p_k(t)(1-\frac{2t}{L+\mu})$ is admissible at step $k+1$, and $|1-\frac{2t}{L+\mu}| \le \frac{\kappa-1}{\kappa+1}$ on $[\mu, L]$. The identity for c_k is Lemma B.6 applied from step k, and $1-(\frac{\kappa-1}{\kappa+1})^2 = \frac{4\kappa}{(\kappa+1)^2} \ge \frac{1}{\kappa}$. □

B.2. Full-cycle variation

Lemma B.6 (Energy and Christoffel functions)**.** *If ν has n atoms, then $\|e_l\|_A^2 = \sum_{j\ge l} c_j$, and $v_l = A^{1/2}(x_n^0 - x_{l+1}^0)$ has $\|v_l\| = \|e_{l+1}\|_A$. $\|e_l\|_A = \|e_0\|_A\,\lambda_l(0;\nu)^{1/2}$. If ν has at least m atoms, $\lambda_m(0;\nu) = \det H_{m+1}^{(0)}/\det H_m^{(2)}$, $H_k^{(s)} = (\int t^{i+j+s}d\nu)_{i,j<k}$. $c_k = \alpha_k\|g_k^0\|^2$.*

Proof of Lemma B.6. CG's energy minimum [24, 25] gives the Christoffel formula and orthogonal steps give its energy differences. For the determinant identity, write $p = 1 + tq$: the minimum is the Schur complement of the positive definite moment Gram matrix $H_m^{(2)}$, hence $\det H_{m+1}^{(0)}/\det H_m^{(2)}$. Exact search gives $c_k = \alpha_k\|g_k^0\|^2$. □

Pointwise response in Theorem 3.1. Let $\Sigma = A^{1/2}S$ and $\Pi_l = \Sigma I_{\le l}\Sigma^{-1}$. Combining the staircase and flag identities gives exactly

$$A^{-1/2}g_n = \sum_l \Pi_l^\top \widehat{E}_l A^{1/2}(x_n - x_{l+1}). \tag{30}$$

At the reference, $\Pi_l = P_l$ and the final factor is v_l; Lemma B.1 therefore gives the first variation. Its norm is at most $\sum_l \omega_l\|v_l\|$. The stated rank-two increments have norm ω_l and send v_l to $\omega_l\|v_l\|\hat{s}_0$, aligning every term. Lemma B.6 gives the coefficient. □

Corollary B.7 (Correction-free methods are first order)**.** *Let a method of $\mathcal{M}$ be CG-consistent and correction-free, $n \ge 2$, and $W_0 < 1$. For $\widehat{E}_l = \omega_l(\hat{s}_0\hat{s}_{n-1}^\top + \hat{s}_{n-1}\hat{s}_0^\top)$ all operators are positive definite and, exactly, $\|g_n\|_{A^{-1}} = W_0\|e_{n-1}\|_A$. Hence the first-order minimax error of correction-free CG-consistent methods of $\mathcal{M}$ over $\|\widehat{E}_l\| \le \omega_l$ lies in $[W_0\|e_{n-1}\|_A,\ \sum_l \omega_l\|e_{l+1}\|_A]$, and FC attains the upper end.*

Proof. Apply Lemma B.4 with $x_l = \omega_l$, $y_l = z_l = \zeta = 0$. The operators have eigenvalues $1 \pm \sum_{l<k}\omega_l$ and 1, hence are positive definite. The exact residual is $W_0\sqrt{c_{n-1}}e$; Theorem 3.1 supplies the upper bound. □

The following step bound controls references arbitrarily close to early termination.

Lemma B.8 (Perturbed conjugacy and line steps)**.** *Normalize $\mu = \|\hat{e}_0\| = 1$. At a defined step $k < n$, let σ_j be preceding canonical steps, $\tau_j = \sigma_j/\|\sigma_j\|$, $\hat{y}_j$ their images and $\hat{g}_k$ the current gradient. Suppose*

$$\|\hat{g}_k - \hat{e}_k\|, \|\sigma_j - \sqrt{c_j}\hat{s}_j\| \le C_a\delta, \quad |\tau_i^\top \tau_j| \le C_a\delta \ (i \ne j), \quad \|\hat{y}_j - \sigma_j\| \le C_a\delta\|\sigma_j\|, \quad \|\widehat{A}_k - I\| \le C_a\delta.$$

For $d_k = -g_k + S_k\beta_k$, $Y_k^\top d_k = 0$, put $u_k = A^{1/2}d_k/\|A^{1/2}d_k\|$ and $\sigma_k^\ell = \alpha_k A^{1/2}d_k$, with exact search α_k. Then $|\tau_j^\top u_k| \le C\delta$ and $\|\sigma_k^\ell - \sqrt{c_k}\hat{s}_k\| \le C\delta$ for sufficiently small δ, with constants depending only on (n, κ, C_a). Consequently every defined FC cycle with small $W = \sum_l \|\widehat{E}_l\|$ satisfies $|\tau_i^\top \tau_j| \le W$ and $\|\sigma_k - \sqrt{c_k}\hat{s}_k\| \le CW\|e_0\|_A$, uniformly over the spectral class without a smallest-energy bound.

Proof. Put $T = [\tau_j]$, $G = T^\top T$, and let P, P^0 project onto the actual and reference old spans. Conjugacy bounds $PD_k = O(\delta\|\hat{g}_k\|)$ for $D_k = A^{1/2}d_k$. Moreover

$$P - P^0 = T(G^{-1} - I)T^\top + \sum_{j<k}(\tau_j\tau_j^\top - \hat{s}_j\hat{s}_j^\top). \tag{31}$$

Use $\|\tau_j - \hat{s}_j\| \le \min(2, 2C_a\delta/\sqrt{c_j})$ and nonstagnation $\|\hat{e}_k\|/\sqrt{c_j} \le \sqrt{\kappa}$ to obtain $D_k - D_k^0 = O(\delta)$. If $\|\hat{e}_k\| \gtrsim \delta$, substitute its unit-direction bound into the exact search; otherwise both steps are $O(\delta)$. FC conjugacy gives the angle bound exactly, and the telescoped secant identity closes finite induction. Scaling restores $\|e_0\|_A$. □

For a square unit-column matrix T, the same identity with $I_{\le l}$ inserted gives

$$TI_{\le l}T^{-1} - P_l = TI_{\le l}(G^{-1} - I)T^\top + \sum_{j\le l}(\tau_j\tau_j^\top - \hat{s}_j\hat{s}_j^\top).$$

Hence, if $G = I + O(\delta)$ and the step errors are $O(\delta)$,

$$\|\hat{e}_{l+1}\|\,\|TI_{\le l}T^{-1} - P_l\| \le C\delta, \qquad \sqrt{c_{n-1}}\,\|TI_{\le l}T^{-1} - P_l\| \le C\delta. \tag{32}$$

The bounds follow termwise from nonstagnation, so no smallest CG energy enters C.

Uniformity in Theorem 3.1. The step lemma makes $T = [\tau_j]$ uniformly invertible. In (30) put $\rho_l = \sum_{j>l}\sigma_j$ and subtract the linear term: $\sum_l \Pi_l^\top \widehat{E}_l(\rho_l - v_l) + \sum_l(\Pi_l - P_l)^\top \widehat{E}_l v_l$. The step errors bound the first sum by $CW_0^2\|e_0\|_A$; (32) and $\|v_l\| = \|\hat{e}_{l+1}\|$ give the same bound for the second. The upper spectral bound follows. For the limit, choose an n-atom measure within ϵ of Γ_ω and use the aligned rank-two increments. The pointwise lower bound and uniform upper bound, followed by $r \downarrow 0$ and $\epsilon \downarrow 0$, give the claim. □

Proposition B.9 (Hilbert–Schmidt budgets)**.** *For $\sum_l \|\widehat{E}_l\|_F^2 \le \omega^2$, Theorem 3.1 has sharp linear response*

$$\frac{\omega}{\sqrt{2}}\Big(\sum_{l\le n-2}\|e_{l+1}\|_A^2\Big)^{1/2} = \frac{\omega}{\sqrt{2}}\|e_0\|_A\Big(\sum_{m=1}^{n-1}\lambda_m(0;\nu)\Big)^{1/2},$$

attained by $\widehat{E}_l \propto \mathrm{sym}(\hat{s}_0 \otimes v_l)$.

Proof of Proposition B.9. In (17) use Frobenius balls and $p = q = 2$. Since $P_l u \perp v_l$, the squared adjoint norm is $\frac{1}{2}\sum_l \|P_l u\|^2\|v_l\|^2$, maximized at $u = \hat{s}_0$. The aligned adjoints attain it. □

With $\gamma = 1$, Lemma D.4 and Theorem D.9, the worst Hilbert–Schmidt response over all references with spectrum in $[\mu, L]$ and n atoms is $\frac{\omega}{\sqrt{2}}\Gamma^*_{n-1,1}(\kappa)^{1/2}\|e_0\|_A = (\frac{a_{\infty,1}}{4})^{1/2}\kappa^{1/4}\omega\|e_0\|_A(1 + o(1)) = 0.45095\,\kappa^{1/4}\omega\|e_0\|_A(1 + o(1))$ as $\kappa \to \infty$, $n\theta \to \infty$.

B.3. Variation before exhaustion

Definition B.10 (Continuation polynomials and common tilt)**.** Let $\varphi_{j,j} = 1$, $\psi_{j,j} = -1$, and, for $j \le k < m$, define

$$\varphi_{k+1,j} = \varphi_{k,j} + \alpha_k \lambda \psi_{k,j}, \qquad \psi_{k+1,j} = -\varphi_{k+1,j} + \beta_k \psi_{k,j}.$$

Put $\zeta_j = (A_{j-1} - A)s_{j-1}^0$, $\chi_{k,j} = (\varphi_{k,j} - 1)/\lambda$, $\xi_k = \sum_{j \le k} \chi_{k,j}(A)\zeta_j$, and

$$\varkappa_m = \sum_{i<m} c_i^{-1/2} \langle g_m^0, \xi_{i+1} - \xi_i \rangle \hat{s}_i. \tag{33}$$

The polynomial $\varphi_{m,j}$ propagates a gradient injection at step j through the frozen CG recurrence. Assume:

(H1) the reference has $N \ge m$ active atoms;

(H2) all canonical increment and correction perturbations have norm at most ω;

(H3) the method belongs to $\mathcal{M}$, is CG-consistent on all quadratic data, makes finitely many corrections per iteration, and has C^2 coefficients near the first m reference iterations.

Theorem B.11 (Variation before Krylov exhaustion)**.** *Under (H1)–(H3), with $P = P_{m-1}$,*

$$A^{-1/2} g_m = \hat{e}_m + \iota_m + \nu_m + \varrho_m, \qquad \iota_m = (I - P) \sum_{l \le m-2} \sum_{i=l+1}^{m-1} \sqrt{c_i} \varphi_{m,i+1}(A) \widehat{E}_l \hat{s}_i,$$

where $\nu_m \in \operatorname{ran} P$ depends on the method, $\hat{e}_m + \iota_m \perp \nu_m$, and $\|\varrho_m\| \le C_{\mathcal{K}} \omega^2 \|e_0\|_A$ uniformly on compact $\mathcal{K} \subset \mathcal{U}_m$. In every regime,

$$\left| \|g_m\|_{A^{-1}} - \sqrt{\|\hat{e}_m + \iota_m\|^2 + \|\nu_m\|^2} \right| \le \|\varrho_m\|.$$

If $a_m = \|\hat{e}_m\| > 0$ and $a_m \ge 2\|\iota_m\|$, the common first variation in the inverse-reference norm is

$$\|g_m\|_{A^{-1}} = a_m + a_m^{-1} \sum_l \langle \widehat{E}_l, Z_l \rangle + O\big(\omega^2 \|e_0\|_A (1 + \|e_0\|_A / a_m)\big),$$

$$Z_l = \sum_{i=l+1}^{m-1} \sqrt{c_i}\, \operatorname{sym}\big(\varphi_{m,i+1}(A)\hat{e}_m \otimes \hat{s}_i\big).$$

Write δ for differentiation along a fixed perturbation direction. Let $\mathcal{V}_k$ contain every vector observed before choosing d_k, including intermediate correction gradients and preceding steps. On the reference these vectors lie in $\mathcal{K}_{k+1}$, whose orthogonal basis is $(g_0^0, \dots, g_k^0)$. Let $\rho_{v,i}$ be the coordinates of v^0 in this basis. The probes are combinations of the observed gradients; the exact-search denominators are positive near the reference.

Lemma B.12 (Filtration)**.** *For every $k \le m-1$ and every $v \in \mathcal{V}_k \cup \{s_k\}$,*

$$\delta v \equiv \sum_{i \le k} \rho_{v,i}\, \delta g_i \pmod{\mathcal{K}_{k+1}},$$

and $\delta x_k - \xi_k \in \mathcal{K}_k$ for $k \le m$, where ξ_k are the frozen displacements above. Moreover $\delta g_k = A\,\delta x_k + J_k$ with $J_k = \sum_{j<k}(A_j - A)s_j^0 = \sum_{j \le k} \zeta_j$.

Proof. Differentiated gradient telescoping gives $\delta g_k = A\delta x_k + J_k$. Derivatives of scalar decision coefficients multiply reference vectors in $\mathcal{K}_{k+1}$; correction gradients differ by $AU_j^0 \delta w_j \in \mathcal{K}_{j+2}$. Induction over observations and steps proves the congruence. Modulo $\mathcal{K}_k$, the displacement derivative therefore obeys the frozen CG recurrence with the constant gradient injections ζ_j from time j. Its impulse response is $\chi_{k,j} = (\varphi_{k,j} - 1)/\lambda$ by (22), so $\delta x_k \equiv \xi_k \pmod{\mathcal{K}_k}$. □

Proof of Theorem B.11. The filtration lemma gives $\delta g_m = \sum_j \varphi_{m,j}(A)\zeta_j + A\Delta_m$ with $\Delta_m \in \mathcal{K}_m$. Projection gives ι_m and ν_m; Lemma B.1, Pythagoras and the norm Taylor formula give all remainder estimates. Differentiating $s_i^\top g_m$ also gives

$$\nu_m = \varpi_m - \varkappa_m, \qquad \varpi_m = \sum_{i<m} c_i^{-1/2}\delta(s_i^\top g_m)\hat{s}_i. \tag{34}$$

With $u_l = \sum_{j=l+1}^{m-1}\sqrt{c_j}\hat{s}_j$, Lemma A.4 gives $\varpi_m^{\mathrm{FC}} = \sum_l P_l\widehat{E}_l u_l$, while Lemma B.3 gives $\varpi_m^{\mathrm{RFC}} = 0$. At exhaustion $P = I$, $\hat{e}_m = \iota_m = \varkappa_m = 0$ and $u_l = v_l$. □

Let τ_k be the trial steps of XRFC, α_k the CG step lengths of the reference, $\rho_k = \alpha_k - \tau_k$, $\epsilon_k = \rho_k/\alpha_k$ and $\epsilon'_k = \rho_k/\tau_k = \epsilon_k/(1-\epsilon_k)$. At the reference the stored steps of XRFC are $\bar{s}_k = \rho_{k-1}d^0_{k-1} + \tau_k d^0_k$ ($\rho_{-1} = 0$), and its virtual corrections have coefficients $w^0_{k,i}$. Let $\widehat{E}_{m-1} = A^{-1/2}(B - A_{m-1})A^{-1/2}$ for the unevaluated output. The virtual injections and output coefficients are

$$\hat{J}_k = \sum_{j<k}(A_j - A)\bar{s}_j + \sum_{i<k} w^0_{k,i}(A_i - A)\bar{s}_i, \qquad \hat{\zeta}_k = \hat{J}_k - \hat{J}_{k-1}, \qquad a^{(m)}_l = (-1)^{m-1-l}\epsilon_l\Big(\prod_{i=l+1}^{m-1}\epsilon'_i\Big)\sqrt{c_l}, \tag{35}$$

and $J_k = \sum_{i<k}(A_i - A)s^0_i$, so $\hat{J}_k = J_k + \Delta^{\mathrm{tr}}_k$ defines Δ^{tr}_k. The regularity follows from Lemma B.1: the reference Gram matrices are positive definite and $\tau_k > 0$. Lemma A.2 gives the virtual reference (x^0_k, g^0_k).

Corollary B.13 (Virtual-launch envelope)**.** *Let the reference have at least $m+1$ atoms. Run full-memory XRFC for m iterations with a positive C^2 trial rule, and set $\hat{x}_m = x_m + S_m w_m$ and $g_{\mathrm{out}} = g_m + BS_m w_m$ for its unevaluated output. For a function realization of the segment operators, $g_{\mathrm{out}} = \nabla f(\hat{x}_m)$. Then*

$$\|g_{\mathrm{out}}\|_{A^{-1}} = \|\hat{e}_m\| + \frac{\langle \hat{e}_m,\ A^{-1/2}\sum_{j=1}^m \varphi_{m,j}(A)\hat{\zeta}_j + \sum_{l=0}^{m-1} a^{(m)}_l \widehat{E}_l\hat{s}_l\rangle}{\|\hat{e}_m\|} + O(\omega^2).$$

Here $\hat{\zeta}_j = \zeta_j + \Delta^{\mathrm{tr}}_j - \Delta^{\mathrm{tr}}_{j-1}$, with Δ^{tr}_j given by (37)*. Exact trials remove all trial terms, recovering Theorem B.11.*

Proof of Corollary B.13. With actual steps and coefficients, secant telescoping gives exactly $\hat{g}_k = A(\hat{x}_k - x_*) + \hat{J}^{\mathrm{ex}}_k$, $g_{\mathrm{out}} = \hat{g}_m + \sum_{i<m} w_{m,i}(B - A_i)s_i$. The quadratic step matrix is bidiagonal in CG directions. Back substitution gives

$$\sum_{i\le l} w^0_{k,i}\bar{s}_i = (-1)^{k-1-l}\rho_l \prod_{l<j<k}\epsilon'_j\, d^0_l, \qquad w^0_{k,k-1} = \epsilon'_{k-1}. \tag{36}$$

The filtration proof applies to effective gradients: differentiating $\hat{g}_{k+1} = g_{k+1} + Y_{k+1}w_{k+1}$ solves for δg_{k+1} because $1 + w^0_{k+1,k} = \alpha_k/\tau_k \neq 0$; coefficient-derivative terms lie in $\mathcal{K}_{k+2}$. Thus $\delta\hat{x}_m - \hat{\xi}_m \in \mathcal{K}_m$ and $A\hat{\xi}_m + \hat{J}_m = \sum_j \varphi_{m,j}(A)\hat{\zeta}_j$. The output term telescopes by (36) to $\sum_l a^{(m)}_l \widehat{E}_l\hat{s}_l$. Project onto $\hat{e}_m$ to obtain the derivative of the inverse-reference norm.

Finally $\sum_i(1 + w^0_{k,i})\bar{s}_i = \sum_i s^0_i$; summation by parts and (36) give

$$A^{-1/2}\Delta^{\mathrm{tr}}_k = \sum_{l\le k-2}\epsilon_l\sqrt{c_l}\Big(1 - (-1)^{k-1-l}\prod_{l<i<k}\epsilon'_i\Big)\widehat{E}_l\hat{s}_l. \tag{37}$$

Consecutive differences give $\hat{\zeta}_j$. Exact trials remove these terms, and Lemma B.1 gives the uniform remainder. □

Corollary B.14 (Realization dependence before and near exhaustion)**.** *For two methods satisfying (H3) on a reference with $N > m$ atoms, put $a = \|\hat{e}_m + \iota_m\|$. For $a > 0$, their inverse-reference gradient norms $\|g_m\|_{A^{-1}}$ differ by at most*

$$\min\Big\{\frac{\|\nu_m\|^2 + \|\nu'_m\|^2}{2a},\ \max(\|\nu_m\|, \|\nu'_m\|)\Big\} + O(\omega^2)\|e_0\|_A.$$

At $a = 0$, the bound with $\max(\|\nu_m\|, \|\nu'_m\|)$ still applies. (a) Before exhaustion ($\|\hat{e}_m\| \gg \omega\|e_0\|_A$) this is of second order: for any two methods satisfying (H3), realization dependence is invisible at first order; this includes FC and RFC. (b) Near exhaustion the difference is of first order in general, e.g. $\nu_m^{\mathrm{FC}} - \nu_m^{\mathrm{RFC}} = \sum_l P_l \widehat{E}_l u_l$, $u_l = \sum_{j=l+1}^{m-1} \sqrt{c_j}\hat{s}_j$.

Proof of Corollary B.14. Apply B.11 and rationalize the difference of the two square roots. The two elementary bounds displayed in the statement follow. Away from exhaustion $\nu, \nu' = O(\omega\|e_0\|_A)$ and $a \geq \|\hat{e}_m\|/2$; near exhaustion use (34) for FC/RFC. □

Let $T_m = LDL^\top$ be the Jacobi matrix of the reference run, $D = \mathrm{diag}(1/\alpha_k)$, L unit lower bidiagonal with subdiagonal $-\sqrt{\beta_k}$, with eigenvalues (Ritz values) $\vartheta_1^{(m)} < \cdots < \vartheta_m^{(m)}$.

Proposition B.15 (Continuation determinants)**.** *Let $S_{m,j} = T_m/T_j$ be the Schur complement of the leading $j \times j$ block and $I_{\geq j} = \mathrm{diag}(0_j, I_{m-j})$. Then*

$$\varphi_{m,j}(\lambda) = \det(I - \lambda S_{m,j}^{-1}) = \frac{\det(T_m - \lambda I_{\geq j})}{\det T_m}.$$

If ν has more than m atoms, the zeros are real, simple and lie in $[\vartheta_1^{(m)}, \vartheta_m^{(m)}] \subset (\mu, L)$. On $(-\infty, \vartheta_1^{(m)})$ the polynomial is positive and nonincreasing (strictly decreasing for $j < m$), with $\varphi_{m,j}(0) = 1$.

Proof. The insertion identity (22) and the block determinant formula give both determinant expressions. Since $S_{m,j}^{-1}$ is a principal block of T_m^{-1}, interlacing bounds its eigenvalues by the Ritz interval. The nonzero off-diagonal entries give simple zeros, and $\varphi_{m,j} = \prod_i (1 - \lambda/\sigma_i)$ proves the sign and monotonicity assertions.

□

B.4. Minimax selection

Fix $m \geq 2$, $N \geq m + 1$. Let $\mathcal{B}(\omega)$ contain scenarios with $\|\widehat{E}_l\| \leq \omega_l$ for $l \leq m - 2$ and every correction perturbation bounded by $\bar{\omega} = \max_l \omega_l$. Let $\mathfrak{M}$ be the methods satisfying (H3), $\mathfrak{M}_1$ those with at most one correction, and $\mathfrak{M}_0$ the correction-free ones; fixed reference-dependent parameters are allowed, but CG-consistency is required on all quadratic data. Write $\nu_m^{\mathfrak{m}}$ for the linear Krylov response. Its absorbed shadow replaces $\widehat{E}_l$ by $P_l^\perp \widehat{E}_l P_l^\perp$, $P_l^\perp = I - P_l$. For $q(\lambda) = \sum_{j<m} q_j \lambda^j$, set

$$v_q = A^{1/2} q(A) g_0, \quad \Lambda_q = \sum_{j=1}^{m-1} q_j \sum_{r<j} \mathrm{sym}(A^{r+1}\hat{e}_m \otimes A^{j-r}\hat{e}_0), \quad \mathcal{S}_m = \{q : \|v_q\| = 1\}.$$

This is the symmetrized Daleckii–Krein derivative [57].

Theorem B.16 (Minimax transverse variation)**.** *For $N \geq m + 1$ and every budget profile,*

$$\min_{\mathfrak{m}\in\mathfrak{M}} \max_{\Theta\in\mathcal{B}(\omega)} \|\nu_m^{\mathfrak{m}}(\Theta)\| = \min_{\mathfrak{m}\in\mathfrak{M}_1} \max_{\Theta\in\mathcal{B}(\omega)} \|\nu_m^{\mathfrak{m}}(\Theta)\| = V_m(\omega) = \max_{q\in\mathcal{S}_m} \sum_{l=0}^{m-3} \omega_l \|P_l^\perp \Lambda_q P_l^\perp\|_*.$$

RFC attains the value with $\nu_m^{\mathrm{RFC}} = -\varkappa_m$. Every method of $\mathfrak{M}$ followed by one latest-information correction at iteration $m - 1$ also attains it. The bound is forced by the absorbed scenarios: $\nu_m^{\mathfrak{m}}(\Theta^\flat) = -\varkappa_m(\Theta)$ for all $\mathfrak{m} \in \mathfrak{M}$.

Write δ for the derivative along $t\Theta$, $\mathcal{Z} = \{\Theta = \Theta^\flat\}$, $Q_l = P_{m-2} - P_l$, $\mathcal{Z}^\sharp = \{\Theta : P_l \widehat{E}_l Q_l = 0\} \supset \mathcal{Z}$, $h = \hat{s}_{m-1}$, $\hat{f}_m = \sum_{l \le m-2} P_l \widehat{E}_l h$ and $\bar{E}_p = \sum_{l<p} \widehat{E}_l$. Polynomial degree and CG orthogonality give, for $r + j \le N - 1$,

$$A^j \hat{s}_r \in \operatorname{ran} P_{r+j}, \qquad \hat{e}_k \perp \operatorname{ran} P_{k-1}. \tag{38}$$

For $N \ge m + 1$, taking inner products with the orthonormal steps gives

$$A\hat{e}_m = \|g_m^0\|^2 \Big(\frac{\hat{s}_{m-1}}{\sqrt{c_{m-1}}} - \frac{\hat{s}_m}{\sqrt{c_m}} \Big), \qquad \|A\hat{e}_m\|^2 = \|g_m^0\|^4 \Big(\frac{1}{c_{m-1}} + \frac{1}{c_m} \Big). \tag{39}$$

The observed residual ϖ_m is defined by (34).

Lemma B.17 (The information fibre and its residual)**.** *Let $N \ge m$ and $\mathfrak{m} \in \mathfrak{M}$.*

(a) $\nu_m^{\mathfrak{m}} = \varpi_m^{\mathfrak{m}} - \varkappa_m$, where $\hat{s}_i^\top \varkappa_m = \langle \hat{e}_m, A^{1/2} \delta s_i \rangle / \sqrt{c_i}$ is independent of the method. A final Galerkin correction with $H - U^\top B U = O(\bar{\omega})$ makes $\varpi_m = 0$; in particular RFC does so.

(b) The common tilt is

$$\varkappa_m = \sum_{p=1}^{m-2} \sum_{i=p+1}^{m-1} \alpha_i \sqrt{\frac{c_p}{c_i}} \, \langle \psi_{i,p+1}(A) A \hat{e}_m, \bar{E}_p \hat{s}_p \rangle \, \hat{s}_i.$$

Both $\varkappa_m$ and ι_m depend only on $\Theta^\flat$; $\varkappa_m$ is independent of $\widehat{E}_{m-2}$ and $\varkappa_2 = 0$.

(c) On $\mathcal{Z}$, every answer and step agrees to first order with CG for $A(t) = A + tA^{1/2} Z A^{1/2}$, $Z = \sum_l \widehat{E}_l$, and fixed g_0. Thus δg_m is common to all methods and $\varpi_m = 0$, $\nu_m = -\varkappa_m$.

(d) On $\mathcal{Z}^\sharp$, every correction-free method has $\varpi_m = \sqrt{c_{m-1}} \hat{f}_m$.

(e) For every polynomial q of degree at most $m - 1$,

$$\langle v_q, \varkappa_m \rangle = \sum_{l=0}^{m-2} \langle \widehat{E}_l, P_l^\perp \Lambda_q P_l^\perp \rangle, \qquad P_{m-2}^\perp \Lambda_q P_{m-2}^\perp = 0.$$

Proof. (a) Differentiate $s_i^\top g_m$ and use (34). A final correction has $w = O(\bar{\omega})$ and model error $O(\bar{\omega})$, so (29) makes its projected residual quadratic.

(b) The frozen recurrence gives $\xi_{i+1} - \xi_i = \alpha_i \sum_{j \le i} \psi_{i,j}(A) \zeta_j$. Substitute $\zeta_{p+1} = \sqrt{c_p} A^{1/2} \bar{E}_p \hat{s}_p$ into the tilt. Polynomial degree and (38) imply $\psi_{i,p+1}(A) A \hat{e}_m \perp \operatorname{ran} P_p$ and $(I - P_{m-1}) \varphi_{m,i+1}(A) P_l = 0$ for $i > l$. Thus both tilt and exterior response depend only on absorbed blocks; the last increment gives an empty sum.

(c) On absorbed scenarios, the derivative of the difference between the scenario probe and the constant-$A(t)$ probe is proportional to $A^{1/2} \sum_{l \ge k} \widehat{E}_l \hat{s}_k = 0$. Corrections have identical derivative $AU(0)\dot{w}$, since $w(0) = 0$. Oracle induction and CG-consistency identify the runs to first order; quadratic orthogonality proves (c).

(d) On $\mathcal{Z}^\sharp$ use instead $Z = \sum_l P_l^\perp \widehat{E}_l P_l^\perp$. Before the last probe the difference is $\sum_{l<k} P_l \widehat{E}_l Q_l \hat{s}_k = 0$, so old conjugacy products have zero derivative. In the exact transfer identity

$$s_i^\top g_m = \sum_{j=i+1}^{m-1} (\mathsf{D}_{ij} + s_j^\top y_i), \quad \delta \mathsf{D}_{ij} = \sqrt{c_i c_j} \sum_{l=i}^{j-1} \hat{s}_i^\top \widehat{E}_l \hat{s}_j, \tag{40}$$

only $j = m - 1$ survives, giving the asserted ϖ_m.

(e) On the absorbed shadow differentiate $\langle q(A(t)) g_0, g_m(t) \rangle = 0$. The product rule $\delta A^j = \sum_{r<j} A^r (\delta A) A^{j-r-1}$ gives Λ_q, while (c) gives $-\varkappa_m$. Each summand has a factor in $\operatorname{ran} P_{m-2}$, proving the final vanishing identity. □

Proof of Theorem B.16. The shadow preserves budgets and forces $\nu_m = -\varkappa_m$ for every method; RFC and every final Galerkin correction have that value on all scenarios. Thus both minima equal $\max_{\mathcal{B}(\omega)} \|\varkappa_m\|$. The map $q \mapsto v_q$ covers the unit sphere of $\operatorname{ran} P_{m-1}$; Lemma B.17 and nuclear duality give the stated formula, attained by $\widehat{E}_l = \omega_l \operatorname{sign}(P_l^\perp \Lambda_{q^*} P_l^\perp)$.

In particular V_m is convex, homogeneous, monotone and independent of ω_{m-2}, with $V_2 = 0$ and $\max_l \omega_l v_l^{\mathrm{sel}} \le V_m \le \sum_l \omega_l v_l^{\mathrm{sel}}$, where $v_l^{\mathrm{sel}} = V_m(\mathbf{u}_l)$. For the last nontrivial budget the tilt reduces to $-\alpha_{m-1}\sqrt{c_{m-2}/c_{m-1}}\,\langle A\hat{e}_m, \widehat{E}_{m-3}\hat{s}_{m-2}\rangle \hat{s}_{m-1}$. The paired vectors are orthogonal by (39); a rank-two increment attains their norm product. Hence $v_{m-3}^{\mathrm{sel}} = \beta_{m-1}\sqrt{c_{m-2}(c_{m-1}+c_m)/c_m} > 0$. □

Proposition B.18 (CG-consistency is the operative hypothesis)**.** *If CG-consistency is required only on the reference data, the infimum in Theorem B.16 is* 0*: a smooth method of $\mathcal{M}$ with one correction, at iteration $m-1$, that is CG on the reference data has $\nu_m \equiv 0$.*

Proof of Proposition B.18. Run FC and estimate $\bar{E}_p\hat{s}_p$ by $A^{-1/2}(z_p - Ad_p)/\|A^{1/2}d_p\|$, using the fixed reference A. These estimates are exact to first order. Form $\hat{\varkappa}$ from Lemma B.17(b) and at the final iteration solve $Hw = \hat{r} - U^\top g'$, where $\hat{r}_i = \sqrt{c_i}\hat{s}_i^\top\hat{\varkappa}$ $(i < m-1)$, $\hat{r}_{m-1} = \sqrt{c_{m-1}}\hat{s}_{m-1}^\top\hat{\varkappa}/\alpha_{m-1}$. The method is CG on the reference, since $\hat{r} = 0$ there. The correction identity gives $U^\top g_m = \hat{r} + O(\bar{\omega}^2)$; translating its last direction coordinate into the recorded step multiplies it by α_{m-1}. Hence $\varpi_m = \varkappa_m$ and (34) gives $\nu_m = 0$. □

Corollary B.19 (The price without a final correction)**.** *Every correction-free method satisfies*

$$\max_{\mathcal{B}(\omega)} \|\nu_m\| \ge \left(V_m^2 + c_{m-1}\omega_{m-2}^2\right)^{1/2}.$$

Thus it cannot attain V_m when $\omega_{m-2} > 0$.

Proof. Choose a maximizing absorbed scenario for V_m, on which every method has $\nu_m = -\varkappa_m$. Replace its newest increment by $\omega_{m-2}(xh^\top + hx^\top)$, where $h = \hat{s}_{m-1}$ and $x \in \operatorname{ran} P_{m-2}$ is a unit vector orthogonal to $\varkappa_m$; this is possible for $m \ge 3$, and for $m = 2$ the tilt is zero. The new increment leaves $\varkappa_m$ unchanged. The scenario belongs to $\mathcal{Z}^\sharp$, and Lemma B.17(d) gives $\nu_m = \sqrt{c_{m-1}}\omega_{m-2}x - \varkappa_m$. Pythagoras proves the bound. □

Corollary B.20 (Gradient-norm minimax comparison)**.** *For every fixed $\mathfrak{m} \in \mathfrak{M}$, uniformly over budget profiles and references in compact subsets of $\mathcal{U}_m$,*

$$\sup_{\Theta\in\mathcal{B}(\omega)} \|g_m^{\mathrm{RFC}}\|_{A^{-1}} \le \sup_{\Theta\in\mathcal{B}(\omega)} \|g_m^{\mathfrak{m}}\|_{A^{-1}} + C_{\mathfrak{m}}\bar{\omega}^2\|e_0\|_A.$$

RFC minimizes the worst realization term $(a^2 + \|\nu_m\|^2)^{1/2} - a$, $a = \|\hat{e}_m + \iota_m\|$, exactly.

Proof of Corollary B.20. The shadow leaves $a(\Theta) = \|\hat{e}_m + \iota_m(\Theta)\|$ unchanged and forces every method to have $\nu_m = -\varkappa_m$, exactly RFC's value. It therefore forces the same realization term $\sqrt{a^2 + \|\varkappa_m\|^2} - a = \frac{\|\varkappa_m\|^2}{a+\sqrt{a^2+\|\varkappa_m\|^2}} \le \frac{V_m^2}{2a}$. Maximization proves its exact minimax property; Theorem B.11 adds the stated method-dependent quadratic remainder. No uniform remainder over the unrestricted method class is required. A final Galerkin solve removes ϖ_m. □

B.5. Second-order residuals and information regimes

For $W > 0$ the maximizer in (5) is $a = \sqrt{1-c^2}$, $c = (\sqrt{\omega'^2 + 8W^2} - \omega')/(4W)$. Also

$$\max(\tfrac{1}{2}W^2, W\omega') \le K(W,\omega') \le \tfrac{1}{2}W^2 + W\omega', \qquad K(W,\omega') = \tfrac{1}{2}W^2 + \omega' W/\sqrt{2} + O(\omega'^2).$$

These follow by $a\sqrt{1-a^2} \le 1/2$ and the test values $a = 1, 1/\sqrt{2}$. For the maximizer, differentiate $\sqrt{1-c^2}(Wc+\omega')$: its derivative has numerator $W - 2Wc^2 - \omega' c$, with one zero on $[0,1]$. Differentiation at $\omega' = 0$ gives the expansion.

Leading expansion in Theorem 3.3. Put $U = [s_0, \dots, s_{n-2}, d]$ and $D_c = U(0)^\top A U(0)$, which is diagonal. Lemma B.3 gives $A^{1/2} U D_c^{-1/2} = Q + O(\omega)$. By the pair-error estimate and flag identities, congruence by $D_c^{-1/2}$ takes H to $I + O(\omega)$ and $U^\top B U - H$ to $Q^\top \mathcal{L} Q + O(\omega^2)$. For $i < n-1$, conjugacy gives $s_i^\top g' = \alpha s_i^\top (A_{n-1} - A_i) d + O(\omega^2)$, while $d^\top g' = 0$. Hence $D_c^{-1/2} U^\top g' = \sqrt{c_{n-1}} Q^\top \hat{f} + O(\omega^2)$. Substitute in (29), or (15), to obtain $-\sqrt{c_{n-1}} \mathcal{L} \hat{f} + O(\omega^3)$. □

Lemma B.21 (A modified last model)**.** *Replace only the last correction model by $H + \Delta_H$, with $D = D_c^{-1/2} \Delta_H D_c^{-1/2} = O(\omega)$ symmetric. Then*

$$A^{-1/2} g_n = -\|e_{n-1}\|_A (\mathcal{L} - \widehat{\Delta}) \hat{f} + O(\omega^3), \qquad \widehat{\Delta} = Q D Q^\top.$$

For a diagonal change in the old-pair entries, $\widehat{\Delta} = \sum_{a<n-1} c_a^{-1} (\Delta_H)_{aa} \hat{s}_a \hat{s}_a^\top$.

Proof. In the preceding calculation the normalized model-error factor becomes $Q^\top \mathcal{L} Q - D + O(\omega^2)$ and the model inverse remains $I + O(\omega)$. The common correction identity gives the result. □

Sharpness and minimax assertion in Theorem 3.3. Lemma B.2 gives, for $t = \|\hat{f}\| \le W_0$, $\|\mathcal{L}\hat{f}\| \le t(\sqrt{W_0^2 - t^2} + \omega') \le K$. If $W_0 = 0$ the lower bound is immediate. Otherwise put $e = \hat{s}_0$, $h = \hat{s}_{n-1}$, $t = 1 - W_0$, $\gamma = \sqrt{1 - a^2}$ and use, for $\varsigma = \pm 1$,

$$\widehat{A}_0^\varsigma = I + \varsigma t \gamma W_0 h h^\top, \quad \widehat{E}_l^\varsigma = t\omega_l [a(e h^\top + h e^\top) + \varsigma \gamma (e e^\top - h h^\top)], \quad \widehat{E}'^\varsigma = \varsigma t \omega' e e^\top,$$

with $F_k = 0$. Their own-base increment norms are at most $t\omega_l/(1 - t\gamma W_0) \le \omega_l$, and likewise for E'. They are positive for small ω and their base displacement is at most W_0.

Lemma B.4 gives common data, $\rho = t a W_0 \sqrt{c_{n-1}} e$, centre $M = I + t a W_0 (e h^\top + h e^\top)$ and opposite hidden coefficients $\pm t(\gamma W_0 + \omega')$. Its radius is $t^2 a W_0 (\gamma W_0 + \omega') \sqrt{c_{n-1}} (1 + O(\omega^2))$. Both normalization changes are $1 + O(\omega)$; explicitly $\phi(A_0^\varsigma) = (1 + \varsigma t \gamma W_0)^{-1/2} \sqrt{c_{n-1}}$. Maximize over a to get $K - O(\omega^3)$; the centred RFC solve attains $K + O(\omega^3)$. Choose r_0 so the closed base ball lies in $\mathcal{U}_n$. Lemma B.1 makes the upper bound uniform there; for every fixed $r > 0$ the two lower scenarios enter it once $W_0 \le r$. All choices are uniform over profiles, including zero budgets. □

Proposition B.22 (Frobenius budgets)**.** *In the setting of Theorem 3.3, for independent budgets $\|\widehat{E}_l\|_F \le \omega_l$, $\|\widehat{E}'\|_F \le \omega'$, the sharp FC first-order constant is $\frac{1}{\sqrt{2}} \sum_l \omega_l \|e_{l+1}\|_A$, and the RFC second-order minimax constant is $\frac{1}{\sqrt{2}} K(W_0, \omega') \|e_{n-1}\|_A$, for every profile and already with a known base; RFC attains it.*

Proof. For FC, the Frobenius form of (17), $v_l \perp \operatorname{ran} P_l$, gives the factor $1/\sqrt{2}$, attained by aligned rank-two increments. For RFC, (26) and $\|\widehat{E}'\| \le \omega'$ give $\|\mathcal{L}\hat{f}\| \le t(\sqrt{W_0^2 - 2t^2} + \omega') \le \frac{1}{\sqrt{2}} K(W_0, \omega')$, $0 \le t \le W_0/\sqrt{2}$.

For the lower bound keep $A_0 = A$ and apply Lemma B.4 with $x_l = a\omega_l$, $y_l = \varsigma c \omega_l$, $z_l = 0$, $\zeta = \varsigma \omega'$, $2a^2 + c^2 = 1$. The Frobenius budgets hold exactly; its radius is $a W_0 (c W_0 + \omega') \|e_{n-1}\|_A (1 - O(\omega^2))$. Set $a = \hat{a}/\sqrt{2}$, $c = \sqrt{1 - \hat{a}^2}$ and maximize. □

B.6. The information centre

Consider a defined one-correction run with invertible direction matrix $D = [d_0, \dots, d_{n-1}]$. Its flag $\mathcal{V}_l = \operatorname{span}\{d_0, \dots, d_l\}$ and observations define

$$\Pi_l = D I_{\le l} D^{-1}, \quad \bar{\Pi}_l = I - \Pi_l, \quad \Theta_l = 2\Pi_l - I, \quad \overline{T}^{\mathsf{D}}_{jk} = d^\top_{\min(j,k)} z_{\max(j,k)}, \quad J_{\mathsf{D}} = D^{-\top} \overline{T}^{\mathsf{D}} D^{-1}.$$

Write $\mathcal{Y}_k = \sum_{l<k} \Pi_l^\top E_l \Pi_l$ and $\Delta = \sum_{l \le n-2} \Pi_l^\top E_l \Pi_l + E'$. Define the reflection

$$A_0^\sharp = A_0 + 2 \sum_l \bar{\Pi}_l^\top E_l \bar{\Pi}_l, \quad E_l^\sharp = -\Theta_l^\top E_l \Theta_l, \quad E'^\sharp = -E', \quad F_k^\sharp = F_k + 2\mathcal{Y}_k.$$

Its admissibility is controlled by the actual direction Gram matrix:

$$N = \operatorname{diag}(\|d_k\|_{A_0}), \quad \Sigma = A_0^{1/2} D N^{-1}, \quad \varepsilon = \|\Sigma^\top \Sigma - I\|, \quad \kappa_\varepsilon^2 = \frac{1+\varepsilon}{1-\varepsilon}, \quad \varpi = 2\kappa_\varepsilon^2 \sum_l \|E_l\|_{A_0}.$$

For $\Xi \in \mathcal{S}$, $\varepsilon, \varpi < 1$, impose

$$\begin{aligned}
&\text{(C1)} \quad \|E_l\|_{A_0} \le (1-\varpi)\kappa_\varepsilon^{-2}\omega_l,\\
&\text{(C2)} \quad \|E'\|_{A_0} \le (1-\varpi)\omega',\\
&\text{(C3)} \quad \|A^{-1/2}(A_0 - A)A^{-1/2}\| + (1+r)\varpi \le r,\\
&\text{(C4)} \quad \|(A_0^\sharp)^{-1/2}(F_k + 2\mathcal{Y}_k)u_k\| \le \omega \|(A_0^\sharp)^{1/2} u_k\| \quad (k \le n-2).
\end{aligned}$$

Condition (C4) is automatic for $u_k = 0$.

Proposition B.23 (Latest information is the centre)**.** *The reflection is a data-preserving involution:*

$$A_k^\sharp d_k = A_k d_k, \qquad B_k^\sharp|_{\mathcal{V}_k} = B_k|_{\mathcal{V}_k} \ (k \le n-2), \qquad B_{n-1} - J_{\mathsf{D}} = \Delta = -(B_{n-1}^\sharp - J_{\mathsf{D}}).$$

It preserves observed blocks and zero increment budgets. Under (C1)–(C4), $(1-\varpi)A_0 \preceq A_0^\sharp \preceq (1+\varpi)A_0$*, and replacing intermediate correction operators gives an admissible twin in* $\mathcal{S}$ *with identical observations and terminal operator. No replacement is needed if* $\|F_k + 2\mathcal{Y}_k\|_{A_0^\sharp} \le \omega$*; if all* $u_k = 0$*, take* $F_k^\sharp = 0$*.*

If J_{D} *is invertible, put* $u_c = -J_{\mathsf{D}}^{-1} g'$*,* $\widehat{J} = A_0^{-1/2} J_{\mathsf{D}} A_0^{-1/2}$ *and* $\widehat{\Delta} = A_0^{-1/2} \Delta A_0^{-1/2}$*. Every last correction* u *satisfies*

$$\max\{\|g' + B_{n-1}u\|_{A_0^{-1}}, \|g' + B_{n-1}^\sharp u\|_{A_0^{-1}}\} \ge \frac{\|\widehat{\Delta} A_0^{1/2} u_c\|}{\sqrt{1 + \|\widehat{\Delta}\widehat{J}^{-1}\|^2}},$$

whereas u_c *leaves* $\pm\Delta u_c$*. For every seminorm,* J_{D} *is a centre relative to the admissible inner class (C1)–(C4), as in Lemma A.1(d).*

Proof. In direction coordinates $\widetilde{X} = D^\top X D$, write $\bar{I}_l = I - I_{\le l}$. Direct block reflection gives

$$\widetilde{A_k^\sharp - A_k} = 2\sum_{l \ge k} \bar{I}_l \widetilde{E}_l \bar{I}_l - 2\sum_{l<k} I_{\le l} \widetilde{E}_l I_{\le l}. \tag{41}$$

This vanishes on d_k and equals $-2\mathcal{Y}_k$ on $\mathcal{V}_k$. The correction change cancels its action on $u_k \in \mathcal{V}_k$. Induction therefore preserves every answer, direction, search, correction and recorded pair. The flag is unchanged, and applying the reflection twice restores the operators because $\Theta_l^2 = I$. The terminal case of (41), together with (19), proves reflection about J_{D}.

All normalized projectors and reflections have squared norms at most κ_ε^2. Thus the base changes by at most ϖA_0, and relative norms increase by at most $(1-\varpi)^{-1}$. Conditions (C1)–(C3) give the increment budgets and base radius. Condition (C4) is precisely the symmetric image-ball criterion for realizing the required action by an operator of relative norm at most ω; such replacements preserve the run. If $u_k = 0$, its action is vacuous.

Lemma A.1(d), applied with $G = \widehat{J}$, gives the residual bound, while $g' + J_{\mathsf{D}} u_c = 0$ gives the opposite centred residuals. Pairing every admissible inner instance with its reflection proves the claim for every seminorm, including unbounded fibres. □

Let $\mathcal{Q}$ contain scenarios satisfying $A_k \bar{d}_k = \bar{A}\bar{d}_k$ for $k \le n-2$, where $(\bar{d}_k)$ are the CG directions of some n-atom $(\bar{A}, g_0)$. Define $\mathcal{S}^\natural \subset \mathcal{S}$ by

$$\|E_l\|_{A_0} \le (1 - 4nW_0)\omega_l, \quad \|E'\|_{A_0} \le (1-4nW_0)\omega', \quad \|A^{-1/2}(A_0 - A)A^{-1/2}\| \le r - 4(1+r)W_0.$$

Let L_ϕ be a relative Lipschitz constant of the positive last CG error ϕ on the compact base ball, and put $c_1 = n(6 + 4(1+r)L_\phi)$.

Corollary B.24 (Pre-quadratic scenarios: RFC is instance-optimal)**.** *For $r \le r_0$, $W_0 \le 1/(4n)$, $\omega \le 1/(6n)$ and $\Xi \in \mathcal{Q} \cap \mathcal{S}^\natural$, the twin $\Xi^\sharp$ is common to all CG-consistent methods and RFC's last correction is centred. Every CG-consistent $M \in \mathcal{M}$ with at most one correction per iteration satisfies*

$$\max\{\rho_M(\Xi), \rho_M(\Xi^\sharp)\} \ge (1 - c_1\omega)\rho_{\mathrm{RFC}}(\Xi) \ge \rho_{\mathrm{RFC}}(\Xi) - c_2\omega^3,$$

where c_2 depends only on (A, g_0, r). Moreover $\sup_{\mathcal{Q}\cap\mathcal{S}^\natural} \rho_{\mathrm{RFC}} \ge K(W_0, \omega') - O(\omega^3)$ for every budget profile, including degenerate ones.

Proof. The preterminal probes in $\mathcal{Q}$ coincide with those of one quadratic. CG-consistency therefore fixes all preceding directions and zero corrections, giving a common flag. Its relative Gram matrix has off-diagonal entries bounded by W_0, so $\varepsilon \le (n-1)W_0 \le \frac{1}{4}$, $\kappa_\varepsilon^2 \le 1+4\varepsilon \le 2$, $\varpi \le 4W_0$, $(1-\varpi)\kappa_\varepsilon^{-2} \ge 1-4nW_0$. Hence (C1)–(C3) hold, (C4) is vacuous, and the twin remains in $\mathcal{Q}\cap\mathcal{S}$. RFC has $H = U^\top J_{\mathsf{D}} U$, so its correction is u_c and its residuals are $\pm\Delta u_c$.

The estimates $\|\widehat{\Delta}\| \le 2n\omega$ and $\widehat{J} \succeq (1-3n\omega)I \succeq I/2$ give a factor $(1+4n\omega)^{-1}$ in the preceding proposition. The twin's metric costs at most $1-2W_0$, and its normalization changes by at most $1+4(1+r)L_\phi W_0$. Multiplication, $W_0 \le n\omega$ and the uniform bound $\rho_{\mathrm{RFC}} = O(\omega^2)$ prove both comparisons. Finally the signed scenarios proving Theorem 3.3, with budgets scaled by $1-4nW_0$, belong to $\mathcal{Q}\cap\mathcal{S}^\natural$ for small ω and yield $K + O(\omega^3)$. □

B.7. Uniform stability and the singular-model certificate

$$\|e_{n-2}\|_A \;\ge\; M_0\,\omega\,\|e_0\|_A \tag{N}$$

requires that the penultimate CG error dominate the perturbation scale; the last energy is unrestricted.

Theorem B.25 (Uniformity and the spectral gate)**.** *Fix $n \ge 2$, $\kappa \ge 1$. Uniformly over n-atom references in $[\mu, L]$, $L/\mu \le \kappa$, ordinary RFC has constants M_0, ω_0, C depending only on (n, κ) such that $\omega \le \omega_0$ and (N) imply gate acceptance at every iteration and normalized models $I + O(C\omega)$. If its last FC direction is nonzero,*

$$\|A^{-1/2}g_n + \|e_{n-1}\|_A \mathcal{L}\hat{f}\| \le C\omega^3\|e_0\|_A, \qquad \|g_n\|_{A^{-1}} \le K(W_0, \omega')\|e_{n-1}\|_A + C\omega^3\|e_0\|_A.$$

For $n \le 3$, omit (N).

For fixed $\gamma_{\mathrm{g}} \in (0,1)$, every defined gated RFC cycle on the full spectral class satisfies, for sufficiently small ω,

$$\|g_n\|_{A^{-1}} \le C\omega^2\|g_0\|_{A^{-1}}, \qquad \|g_n\|_{A^{-1}} \le C(\omega^2\|g_{n-1}\|_{A^{-1}} + \omega^3\|g_0\|_{A^{-1}})$$

where the second bound requires the last correction to be made. Constants depend only on $(n, \kappa, \gamma_{\mathrm{g}})$. On n-atom references satisfying (N), its gate never acts and the ordinary expansion applies.

For the proof normalize $\|\hat{e}_0\| = \mu = 1$ and suppress canonical hats where unambiguous. At iteration k let σ_k^ℓ be the line step, χ_k the correction, $\sigma_k = \sigma_k^\ell + \chi_k$, $\tau_k = \sigma_k/\|\sigma_k\|$ and u_k the unit line direction. Normalize $U_k = [S_k, d_k]$ by its canonical column lengths D_k, and put $\widetilde{H}_k = D_k^{-1}H_kD_k^{-1}$ and use ρ_k, N_k as in Lemma B.3. With $T_k^+ = [\tau_0, \dots, \tau_{k-1}, u_k]$, $\tilde{w}_k = D_k w_k = -\widetilde{H}_k^{-1}\rho_k$, one has

$$T_k^{+\top}\hat{g}_{k+1} = N_k\tilde{w}_k. \tag{42}$$

The pairs obey $\hat{y}_i = \widehat{A}_i\sigma_i + \xi_i$, $\xi_i = (\widehat{B}_i - \widehat{A}_i)\chi_i$, and conjugacy becomes

$$\tau_i^\top\widehat{A}_i u_k = -\xi_i^\top u_k/\|\sigma_i\|. \tag{43}$$

All operators are within $(n+1)\omega$ of I; hence the normalized entry/residual bounds are precisely (27) with $\varepsilon = C\omega$.

Ordinary RFC in Theorem B.25. Normalize $\|\hat{e}_0\| = \mu = 1$. For $k \le n-2$ inductively maintain $\sigma_k^\ell - \sigma_k^0, \sigma_k - \sigma_k^0 = O(\omega)$, $\chi_k = O(\omega^2 + \omega\|\sigma_k^\ell\|)$, $\xi_k = O(\omega^2\|\sigma_k\|)$, with angles $O(\omega)$, projected residuals $O(\omega^2)$ and $\widetilde{H}_k = I + O(\omega)$, $N_k = O(\omega)$. Nonstagnation and (N) give $\sqrt{c_k} \ge M_0\omega/\sqrt{\kappa}$, so recorded steps remain comparable to reference steps for large M_0. The perturbed-step lemma supplies each next line step; (27) gives $\rho_k = O(\omega^2 + \omega\|\sigma_k^\ell\|)$, then the correction and next residual. Dividing by the reference step lower bound closes the angle and pair-error induction.

At the last step no lower energy bound is used. With $T = T_{n-1}^+$, the exact identity (42) and the flag formulas give

$$\hat{g}_n = -T^{-\top} N \widetilde{H}^{-1}\rho = -\mathcal{L}^\tau f^\tau + O(\omega^3), \quad \mathcal{L}^\tau = \sum_l P_l^\tau \widehat{E}_l P_l^\tau + \widehat{E}', \quad f^\tau = \sum_l P_l^\tau \widehat{E}_l \sigma_{n-1}^\ell.$$

Here P_l^τ projects onto the first normalized recorded columns. Weighted estimate (32) replaces the last product by $\sqrt{c_{n-1}}\mathcal{L}\hat{f}$ with cubic error. For $n = 2$ there is no intermediate small step. For $n = 3$, $d_1 = d_1^0$ and $\sigma_1^\ell = \sigma_1^0/(1 + \hat{s}_1^\top \widehat{E}_0 \hat{s}_1)$, so the angle and correction bounds hold relative to $\sqrt{c_1}$ without (N). □

Gated RFC in Theorem B.25. Normalize $\|\hat{e}_0\| = 1$. At a passed gate the normalized model and Gram inverse are bounded. Since $\rho = (\rho^{\mathrm{o}}, 0)$, its correction coefficients obey $\|\tilde{w}^{\mathrm{o}}\| \ge \|\rho^{\mathrm{o}}\|/(n+1)$, $|\tilde{w}^{\mathrm{l}}| \le 2\|\rho^{\mathrm{o}}\|/\gamma_{\mathrm{g}}$. The line changes only the last coefficient and cannot cancel the old ones. Induction gives $\|\chi_k\| \le C_*\|\sigma_k\|$, with $C_* = 2\sqrt{n}(1 + 2(n+1)/\gamma_{\mathrm{g}})/\sqrt{\gamma_{\mathrm{g}}}$, and consequently pair errors $O(\omega)\|\sigma_k\|$ and bounded one-step gradient growth.

Until the first line step below $M\omega^2$, maintain angles at most $(1-\gamma_{\mathrm{g}})/(4n)$ and old residual at most $C_s\omega^2$. These imply gate acceptance. Equation (27) propagates the residual, while its correction estimate bounds the next angle by $C\omega + CC_s/M$; choose C_s, then M, then ω_0. If a small step occurs, absorb the pair errors into symmetric $O(\omega)$ secant perturbations using the image-ball identity; Lemma E.4(a),(b) gives $\hat{g}_k = O(\omega^2)$. All later gradients remain that small, and a final passed gate improves this to $O(\omega^3)$. Otherwise (27) gives directly $\|\hat{g}_n\| \le C(\omega^3 + \omega^2\|\hat{g}_{n-1}\|)$. These prove the two bounds; under (N), the ordinary proof gives models $I + O(\omega)$, so every gate passes. Restore scaling. □

Proposition B.26 (An unbounded RFC family; certified)**.** *For $n = 4$, $A = \mathrm{diag}(1, 4, 9, 16)$, $\omega = 10^{-3}$ and $g_0 = A^{1/2}(1, 1, \omega^4, \omega^4)^\top$, there is an affine family of perturbations with all canonical increment and correction budgets less than ω on which plain RFC passes every canonical-defect gate, but its terminal residual is unbounded. Thus a small defect alone does not control a nearly singular correction model.*

Proof. For a row (a, b, c, d, e, f) let $[a, b, c; d, e, f] = \mathrm{diag}(\left(\begin{smallmatrix} a & b \\ b & c \end{smallmatrix}\right), \left(\begin{smallmatrix} d & e \\ e & f \end{smallmatrix}\right))/32$. Use

	a	b	c	d	e	f
X_0	−1	6	−5	−7	2	−2
X_1	0	−2	8	−7	2	6
X_2	−8	−7	−6	4	6	−8
X_3	−6	−7	5	7	2	−4
Y_1	7	−1	6	1	2	2
Y_2	−8	2	2	−4	6	2
X'	−3	0	−6	3	−1	−1

and set $\widehat{E}_0(s) = \omega((1-s)X_0 + sX_1)$, $\widehat{E}_1 = \omega X_2$, $\widehat{E}_2 = \omega X_3$, $\widehat{F}_1 = \omega Y_1$, $\widehat{F}_2 = \omega Y_2$, $F_0 = 0$, $\widehat{E}' = \omega X'$. Their squared Frobenius norms are at most $175/512 < 1$; convexity gives the budget bounds for every $s \in [0, 1]$, and all operators remain positive definite.

Exact dyadic endpoints and rational interval witnesses certify the construction. Each RFC update uses rational field operations; outward interval evaluation with nonzero reciprocal domains certifies the following strict inequalities. The arithmetic expressions are decoded as the actual observed RFC run, so

the certificate concerns its genuine directions, pairs, models and gates. Take $I = [q_* 2^{-420}, (q_*+1)2^{-420}]$, where the exact integer is

$$q_* = 26464549281804814561712511468479180561216769836755377736434086 7\,10^{64} + 43686391600749724320415731437602501545883676531629073943623914 85.$$

On this interval, near $s = 0.97738647059296281323\ldots$, it verifies nonzero preterminal denominators, positive line-search lengths, $\|\delta\|_F^2 < 10^{-6}$ at all four iterations, positive preterminal models and positive first three leading minors of the last model H. At the two endpoints, $\det H(s_-) > 2.6 \cdot 10^{-167}$, $\det H(s_+) < -7.5 \cdot 10^{-167}$. Uniformly on I, with $v = B_3 U \operatorname{adj}(H) U^\top g'$, $\|v\|^2 > 10^{-107}$, $\|g'\|^2 < 4 \cdot 10^{-23}$, $\det U \neq 0$. These strict interval comparisons include all gate inequalities. Continuity gives a zero of $\det H$ approached by positive-definite models. RFC takes the last correction there, and $\|g_4\| = \|g' - v/\det H\| \geq \|v\|/|\det H| - \|g'\| \longrightarrow \infty$. □

C. One-gradient information and curvature completion

The final point is unevaluated, and its last segment carries an independent symmetric operator B. Its output vector is $g_{\text{out}} = g_n + Bu_n$ for output correction u_n; under a function realization of the segment operators this vector is the gradient at the output. An evaluated quadratic prefix can therefore determine the output point exactly while leaving g_{out} uncertain.

C.1. One cycle of XRFC

Use the perturbation convention and reference CG quantities fixed in Section 3. Full-memory XRFC has virtual points $\hat{x}_k = x_k^0$ on quadratic data for every positive trial rule; the evaluated point is $x_{k+1} = x_k^0 + \tau_k d_k^0$ (Lemma A.2). For the reference trials put

$$\varepsilon_k = 1 - \frac{\tau_k}{\alpha_k}, \qquad \vartheta_k = \frac{\varepsilon_k}{1-\varepsilon_k} = \frac{\alpha_k}{\tau_k} - 1, \qquad a_l = (-1)^{n-1-l}\,\varepsilon_l \Big(\prod_{i=l+1}^{n-1} \vartheta_i \Big) \sqrt{c_l}. \tag{44}$$

Thus ε_k is the relative trial error and ϑ_k the relative virtual correction. The reference steps $\bar{s}_j = \varepsilon_{j-1}\alpha_{j-1} d_{j-1}^0 + \tau_j d_j^0$ ($\varepsilon_{-1} := 0$) are not the CG steps; with $\Sigma_0 = A^{1/2}[\bar{s}_0, \ldots, \bar{s}_{n-1}]$, $\Pi_l = \Sigma_0 I_{\leq l} \Sigma_0^{-1}$ is the projector onto $\operatorname{span}\{\hat{s}_0, \ldots, \hat{s}_l\}$ along the later columns. For the unevaluated output segment write $\widehat{E}_{n-1} := \widehat{E}'$, $\omega_{n-1} := \omega'$, $\Pi_{n-1} := I$.

Harmonic-mean trials minimize the last factor $|\varepsilon|$; arithmetic-mean trials minimize later $|\vartheta|$ factors (Lemma C.3).

Theorem C.1 (XRFC variation)**.** *Run full-memory XRFC for n iterations with a positive C^2 trial rule near the reference. At its virtual output $\hat{x}_n = x_n + S_n w_n$, put $g_{\text{out}} = g_n + BS_n w_n$. Then*

$$A^{-1/2} g_{\text{out}} = \sum_{l=0}^{n-1} a_l \Pi_l^\top \widehat{E}_l \hat{s}_l + O(\omega^2).$$

The sharp first-order bound over $\|\widehat{E}_l\| \leq \omega_l$ is

$$C_{\text{S}} = \max_{\|u\|=1} \sum_l \omega_l |a_l| \|\Pi_l u\|.$$

If the reference trials are exact, $a_l = 0$ and the output error is $O(\omega^2)$.

Fix an n-atom reference and use the notation and canonical coordinates of Appendix A; put $\widehat{E}_{n-1} = \widehat{E}'$ for the final output segment.

C.2. Quadratic trajectories and extrapolation errors

On a quadratic the evaluated steps and virtual corrections satisfy

$$\bar{s}_j = \varepsilon_{j-1}\alpha_{j-1} d_{j-1}^0 + \tau_j d_j^0, \qquad \bar{S}_k w_k^0 = \varepsilon_{k-1}\alpha_{k-1} d_{k-1}^0,$$

with the first term omitted for $j = 0$; hence $\operatorname{span} \bar{S}_k = \mathcal{K}_k$.

Lemma C.2 (Structure of the XRFC steps)**.** $A^{1/2}\bar{s}_j = \varepsilon_{j-1}\sqrt{c_{j-1}}\,\hat{s}_{j-1} + (1-\varepsilon_j)\sqrt{c_j}\,\hat{s}_j$ *($\varepsilon_{-1} := 0$), so Σ_0 is upper bidiagonal in the basis $\{\hat{s}_j\}$ and* $\operatorname{span}\{A^{1/2}\bar{s}_0, \dots, A^{1/2}\bar{s}_l\} = \operatorname{span}\{\hat{s}_0, \dots, \hat{s}_l\}$*. Moreover $\Pi_l \hat{s}_k = \hat{s}_k$ for $k \le l$ and $\Pi_l \hat{s}_k = r_{l,k}\hat{s}_l$ for $k > l$, with*

$$r_{l,k} = \prod_{j=l+1}^{k} \Big(- \frac{\varepsilon_{j-1}\sqrt{c_{j-1}}}{(1-\varepsilon_j)\sqrt{c_j}} \Big),$$

so $\Pi_l = \sum_{k<l} \hat{s}_k \hat{s}_k^\top + \hat{s}_l r_l^\top$ with $r_l = \sum_{k \ge l} r_{l,k}\hat{s}_k$, $r_{l,l} = 1$, $\|\Pi_l\| = \|r_l\|$, and $a_{n-1} r_{l,n-1} = a_l$.

Proof. The trajectory formula follows from Lemma A.2; its positive diagonal entries give the flag. Since $\Pi_l A^{1/2}\bar{s}_k = 0$ for $k > l$, back substitution gives $\Pi_l \hat{s}_k = -\varepsilon_{k-1}\sqrt{c_{k-1}}\Pi_l \hat{s}_{k-1}/((1-\varepsilon_k)\sqrt{c_k})$. Iteration gives $r_{l,k}$. The orthogonal block form of Π_l gives $\|\Pi_l\| = \|r_l\|$; telescoping the square roots and $\varepsilon_i\alpha_i/\tau_i = \vartheta_i$ gives $a_{n-1} r_{l,n-1} = a_l$. □

Lemma C.3 (Trial errors)**.** *(a) $|a_l| = |\varepsilon_l| \prod_{i>l} |\vartheta_i| \sqrt{c_l}$; an overshoot ($\varepsilon_i \le 0$) has $|\vartheta_i| < 1$, an undershoot by more than a factor 2 has $|\vartheta_i| > 1$. (b) Let $\alpha \in [\alpha_{\mathrm{lo}}, \alpha_{\mathrm{hi}}]$, $r = \alpha_{\mathrm{hi}}/\alpha_{\mathrm{lo}} > 1$. Then $\min_\tau \max_\alpha |1 - \tau/\alpha| = \frac{r-1}{r+1}$, attained only at the harmonic mean of $\alpha_{\mathrm{lo}}, \alpha_{\mathrm{hi}}$, and $\min_\tau \max_\alpha |\alpha/\tau - 1| = \frac{r-1}{r+1}$, attained only at their arithmetic mean. At the harmonic mean $|\vartheta| \le \frac{r-1}{2}$; at the arithmetic mean $|\varepsilon| \le \frac{r-1}{2}$.*

Proof of Lemma C.3. Part (a) follows from $\vartheta = \varepsilon/(1-\varepsilon)$ and $\varepsilon < 1$. For (b), the two endpoint errors $1 - \tau/\alpha$ equioscillate only at the harmonic mean; the errors $\alpha/\tau - 1$ do so only at the arithmetic mean. The common minimax value is $(r-1)/(r+1)$. Substitution gives, respectively, $\vartheta \in [-(r-1)/(2r), (r-1)/2]$ and $\varepsilon \in [-(r-1)/2, (r-1)/(2r)]$. □

Proposition C.4 (Compressions times extrapolation distance)**.** *Let points $x_0, \dots, x_n$ with $g_{j+1} = g_j + A_j s_j$ depend in a C^1 way on the increments, with linearly independent reference steps $\bar{s}_j$ defining Π_l. Let $\hat{x} = x_n + Sw$, $w = -\overline{T}^{-1} S^\top g_n$, assign operator B to the output segment, and put $g_{\mathrm{out}} = g_n + BSw$ and $\hat{e} = A^{1/2}(\bar{x}_n - x_*)$. Then*

$$A^{-1/2} g_{\mathrm{out}} = -\Big(\sum_{l=0}^{n-1} \Pi_l^\top \widehat{E}_l \Pi_l \Big) \hat{e} + O(\omega^2), \qquad \omega = \max(\|\widehat{E}\|, \|\widehat{E}'\|).$$

For XRFC, $\hat{e} = -\varepsilon_{n-1}\|e_{n-1}\|_A \hat{s}_{n-1}$, and the first-order term is $\sum_l a_l \Pi_l^\top \widehat{E}_l \hat{s}_l$.

Proof. Lemma A.3 gives the model-error factor $\sum_l \Pi_l(S)^\top \widehat{E}_l \Pi_l(S)$. At the reference $\widehat{J} = I$ and $\hat{g}_n = \hat{e}$, so (15) gives the expansion with the reference flag. Substitute $\hat{e} = -a_{n-1}\hat{s}_{n-1}$ and $a_{n-1}\Pi_l \hat{s}_{n-1} = a_l \hat{s}_l$ from Lemma C.2. □

Proof of Theorem C.1. The output identity is Lemma A.1(a). Proposition C.4 gives the expansion; regularity follows from Lemma B.1. For the sharp bound, apply (17) with $p = \infty$, $v_l = \hat{s}_l$ and $L_l = a_l \Pi_l^\top$.

Exact reference trials make every a_l zero.

□

C.3. The missing pivot and moment

Use the Jacobi notation $J_0, z, b_A, \sigma_J, A_b$ of Section 4, and write $\mathsf{b} = J_{n-2,n-1}$.

Definition C.5 (Krylov information and scenarios)**.** The information $\mathcal{I} = (g_0, A|_{\mathcal{K}_{n-1}})$ has fibre $\{A' \succ 0 : \mathcal{I}(A', g_0) = \mathcal{I}\}$. For an admissible interval $[b_{\mathrm{lo}}, b_{\mathrm{hi}}]$, the step endpoints, ratio r, scenario (b, B) and normalized output error are those preceding Theorem 4.1. We use $\mathcal{A}[\mu, L] = \{A' : \mu I \preceq A' \preceq LI\}$ and $\mathcal{B}_\rho(A_c) = \{A' : \|A' - A_c\|_{A_c} \le \rho\}$, so $\mathcal{A}[\mu, L] = \mathcal{B}_{\rho_\kappa}((L+\mu)I/2)$. For relative balls assume $A_c \succ 0$ and $0 \le \rho < 1$.

Lemma C.6 (The fibre is a Gauss–Radau segment)**.** *Let (A, g_0) have $n \geq 2$ atoms.*

(a) Its fibre is $\{A_b : b > \sigma_J\}$, with $Z^\top A_b Z = J(b)$. Every completion has n atoms and the same $x^0_{n-1}, g^0_{n-1}, d^0_{n-1}$. Only the final $LDL^\top$ pivot varies: $\alpha(b) = (b - \sigma_J)^{-1}$.

(b) If it meets $\mathcal{A}[\mu, L]$, then $\operatorname{spec} J_0 \subset (\mu, L)$ and the admissible interval is $[b_-, b_+]$, where

$$b_- = \mu + \mathsf{b}^2 \mathbf{u}_{n-2}^\top (J_0 - \mu I)^{-1} \mathbf{u}_{n-2}, \qquad b_+ = L - \mathsf{b}^2 \mathbf{u}_{n-2}^\top (LI - J_0)^{-1} \mathbf{u}_{n-2}.$$

The endpoints are the n-point Gauss–Radau matrices with prescribed node μ and L [25, 58, 59]. Their step ratio is $r = (b_+ - \sigma_J)/(b_- - \sigma_J) < \kappa$, with $r \to \kappa$ as $\mathsf{b} \to 0$.

(c) If it meets $\mathcal{B}_\rho(A_c)$, its admissible interval is compact and $r \leq (1 + \rho)/(1 - \rho)$. On the central fibre its endpoints are $b_{A_c} \pm \rho/\alpha_{n-1}(A_c)$ and equality holds.

Proof of Lemma C.6. A symmetric operator annihilating $\mathcal{K}_{n-1} = z^\perp$ is a multiple of $zz^\top$. The last Schur complement of $J(b)$ is $b - \sigma_J$, giving positivity and $\alpha(b) = (b - \sigma_J)^{-1}$. The irreducible prefix fixes all preceding CG quantities and preserves n atoms. If $g^0_{n-1} = \gamma z$, conjugacy gives $d^\top A_b d = \gamma^2 (b - \sigma_J)$, hence $x_*(b) = x^0_{n-1} + \alpha(b) d$.

Strict interlacing and the Schur criteria for $J(b) - \mu I$ and $LI - J(b)$ give the endpoints and their Gauss–Radau interpretation [58, 59]. Resolvent subtraction gives $b_- - \sigma_J = \mu[1 + \mathsf{b}^2 e^\top J_0^{-1}(J_0 - \mu I)^{-1} e] > \mu$, $\quad b_+ - \sigma_J = L[1 - \mathsf{b}^2 e^\top J_0^{-1}(LI - J_0)^{-1} e] < L$, where $e = \mathbf{u}_{n-2}$, proving $r < \kappa$ and its limiting supremum. A relative ball gives a compact interval; inverse monotonicity bounds its step ratio. On the central fibre the relative displacement norm is exactly $|b - b_{A_c}| z^\top A_c^{-1} z = |b - b_{A_c}| \alpha_{n-1}(A_c)$, giving equality and the claimed endpoints. □

Lemma C.7 (What a one-gradient method knows before its last evaluation)**.** *Let a method of $\mathcal{M}_1$ run on constant operators $A_k \equiv A'$, $k \leq n - 1$, with A' in the fibre of $\mathcal{I}$. Then $g_0, \ldots, g_{n-1}$, $s_0, \ldots, s_{n-1}$ and x_n are functions of $\mathcal{I}$, and $x_n - x^0_{n-1} = v + t\, d$ with $v \in \mathcal{K}_{n-1}$ and v, t independent of A'.*

Proof. By induction, $g_j \in \mathcal{K}_{j+1}$ and $s_j \in \operatorname{span}\{g_0, \ldots, g_j, s_0, \ldots, s_{j-1}\} \subset \mathcal{K}_{j+1}$. For $j \leq n-2$, $s_j \in \mathcal{K}_{n-1}$, so $g_{j+1} = g_j + A' s_j = g_j + A s_j$ is determined by $\mathcal{I}$; s_{n-1} is a function of the data $g_0, \ldots, g_{n-1}, s_0, \ldots, s_{n-2}$. Finally $x_n - x_0 \in \mathcal{K}_n = \mathcal{K}_{n-1} \oplus \operatorname{span}\{d\}$ and $x^0_{n-1} - x_0 \in \mathcal{K}_{n-1}$. □

Proof of Theorem 4.1. Lemma C.7 gives a common $x_n - x^0_{n-1} = v + td$ on the endpoint completions. Put $\rho_b = A_b^{1/2}(x_n - x_*(b))$. For every output correction, symmetric image-ball duality and $\omega' \leq 1$ give worst residual at least $\omega' \|\rho_b\|$. Since $v \perp_{A_b} d$, $\frac{\|\rho_b\|^2}{\|e_{n-1}(b)\|^2_{A_b}} = \frac{\|v\|^2_{A_b}}{\alpha(b)^2 \|d\|^2_{A_b}} + |t/\alpha(b) - 1|^2$. The two endpoints and Lemma C.3 force the lower bound. XRFC's full final model yields $u_n = -A_b^{-1} g_n$, so its exact worst residual is $\omega' |1 - \tau_{n-1}/\alpha(b)| \|e_{n-1}(b)\|_{A_b}$. The observed prefix determines the fibre; its harmonic-mean rule attains the endpoint minimum for arbitrary positive earlier trials. □

Corollary C.8 (Spectral and local quantification)**.** *For constant prefixes, $n \geq 2$ and $0 \leq \omega' < 1$, the uniform minimax value with known (μ, L, n) is $\rho_\kappa \omega'$, approached as $\mathsf{b} \to 0$ but not attained by a reference. On a known relative base ball $\mathcal{B}_\rho(A)$ it is $\rho\omega'$, attained at $A_{b_A \pm \rho/\alpha_{n-1}(A)}$. XRFC attains both minimax values by the harmonic-mean trial on the observed fibre; at the centre data of the ball this trial is $\alpha_{n-1}(A)$. The lower bounds persist in larger scenario classes; earlier variation adds Theorem C.1 with compact-uniform remainders.*

Proof of Corollary C.8. Theorem 4.1 gives minimax value $\omega' \sup_{\mathcal{I}} (r(\mathcal{I}) - 1)/(r(\mathcal{I}) + 1)$, with the same fibrewise harmonic-mean rule attaining it. Lemma C.6(b) gives $\sup r = \kappa$; part (c) gives $\sup r = (1 + \rho)/(1 - \rho)$ in the relative ball and harmonic mean $\alpha_{n-1}(A)$ on its central fibre. Constant prefixes form a subclass of every enlarged scenario class; Theorem C.1 and Lemma B.1 give the uniform upper expansion for varying prefixes. □

Corollary C.9 (The undetermined moment)**.** *Let $m_k = \int t^k \, d\nu_g$ be the moments of the spectral measure ν_g of $g_0/\|g_0\|$, and let the fibre meet $\mathcal{A}[\mu, L]$. (a) $\mathcal{I}$ determines $m_0, \dots, m_{2n-2}$, and on the fibre $m_{2n-1} = (\mathsf{b}_0 \cdots \mathsf{b}_{n-2})^2 b + c_{\mathcal{I}}$. (b) Over all probability measures on $[\mu, L]$ with the moments $m_0, \dots, m_{2n-2}$, the values of $\int t^{2n-1} d\nu'$ form the image of $[b_-, b_+]$; the ends are the two Gauss–Radau measures, the principal representations of the moment sequence [60, 61]. When $b_- < b_+$, the canonical moment $p_{2n-1} = \frac{b-b_-}{b_+-b_-} \in [0,1]$ is free; when $b_- = b_+$, the missing moment is fixed. (c) On the scenarios of Theorem 4.1, FC and RFC have error exactly* 0*.*

Proof of Corollary C.9. A path in $(J^k)_{00}$ reaches the last site and returns only after $2n-2$ edges; using its diagonal takes one more step. This gives the affine moment formula. Any measure sharing those moments has the same orthogonal polynomials through degree $n-1$; compressing multiplication by t gives $J(b')$ with spectrum in $[\mu, L]$, hence $b' \in [b_-, b_+]$. Conversely every such matrix realizes the moments. The endpoints are the Gauss–Radau principal representations [60, 61]. FC/RFC terminate on the constant prefix, proving (c). □

Remark C.10 (The free moment is a terminal spring)*.* The reconstruction of the returned static solution in Remark 4.2 is energy telescoping: $\|g_j^0\| h_j^+ = \|e_j\|_A^2$ and the decrements c_i give

$$\|g_j^0\| h_j^+ = \sum_{i=j}^{n-2} c_i + \|g_{n-1}^0\| h_{n-1}^+, \qquad 0 \le j \le n-1.$$

The observed prefix has its own last bead weight

$$(J_0^{-1})_{n-2,n-2} = \frac{\sigma_J}{\mathsf{b}_{n-2}^2},$$

by the definition of σ_J; the Schur complement of J_0 in $J(b)$ is the free last pivot $b - \sigma_J$.

The principal representations in (b) belong to Krein's theory of the Markov moment problem. There, truncated Stieltjes moment data are carried by beaded strings [60], [62, Suppl. II]. Apply Lemma D.15(b) to ν_g in the coordinate $\zeta = \lambda/\mu - 1$. The moments $m_0, \dots, m_{2n-2}$ fix the orthonormal polynomials $P_0, \dots, P_{n-1}$ and therefore all n bead masses and $n-1$ gaps of the Stieltjes string; write m_j for its bead masses. Only the spring $\varrho = -P_{n-1}(0)\tilde{P}_n(0)$ at the far end remains free. It is affine in b with slope $\mathsf{m}_{n-1}/\mu > 0$. By Lemma D.15(b), $\varrho = 0$ exactly when μ is an atom, which is the completion b_-. By Proposition D.17, the stiffest spring compatible with $\|H\| \le \kappa - 1$ is the completion b_+. When $b_- < b_+$, the free canonical moment is the relative stiffness of the terminal spring, $p_{2n-1} = \varrho/\varrho(b_+)$. Seen from the extrapolation point, the same parameter is the terminal compliance $\mathsf{d}_{n-1} = \alpha(b)$ of Remark 4.2. The Lanczos data fix the beads; the last unobserved moment fixes how the string is held at its far end.

Proof of Theorem 4.3. For $Z = [S, d]$ the prescribed compression is $\Gamma = \operatorname{diag}(\overline{T}, c)$. Spectral feasibility is exactly $\mu Z^\top Z \preceq \Gamma \preceq L Z^\top Z$. The generalized Schur criterion [29] gives the stated ranges and $\kappa_{\mathrm{lo}}\|d\|^2 \le c \le \kappa_{\mathrm{hi}}\|d\|^2$. These suffice: extend the compressed operator by μI on $\operatorname{ran} Z^\perp$, $A' = Z(Z^\top Z)^{-1}\Gamma(Z^\top Z)^{-1}Z^\top + \mu[I - Z(Z^\top Z)^{-1}Z^\top]$. To preserve full action $A'S = Y$, use orthogonal blocks on $\operatorname{ran} S \oplus \operatorname{ran} S^\perp$. The free block ranges over $\mu I + A_{WU}(A_{UU} - \mu I)^+ A_{UW} \preceq K \preceq LI - A_{WU}(LI - A_{UU})^+ A_{UW}$. The endpoint quadratic forms along d are precisely $\kappa_{\mathrm{lo}}\|d\|^2$, $\kappa_{\mathrm{hi}}\|d\|^2$, by their zero Schur complements; interpolation fills the interval. Consistent boundary data satisfy the range conditions because each null direction of $\overline{T} - \mu G$ gives a μ-eigenvector orthogonal to d, and similarly at L. Strict margins make the interval nondegenerate. Inconsistent data can have an empty interval: $S = \mathbf{u}_0$, $\overline{T} = 3$, $d = (0.9, 0.1)^\top$, $[\mu, L] = [1, 10]$ gives $\kappa_{\mathrm{lo}} > 1.49$, $\kappa_{\mathrm{hi}} < -4.1$.

At the final Krylov step $d^\top A_b d = \gamma^2(b - \sigma_J)$, identifying this interval with the Jacobi fibre. Endpoint equioscillation (Lemma C.3) proves the unique midpoint rule and its harmonic-mean interpretation. Clipping a positive estimate decreases both $|1 - c'/\hat{c}|$ and $|\hat{c}/c' - 1|$ for every consistent c', hence every product $|a_l|$. Valid estimated spectral bounds give $\hat{r} \le \hat{L}/\hat{\mu}$; Lemma A.2 makes the virtual reference and its intervals trial-independent. □

C.4. Information separation

Write $\mathcal{U}(\mu, L)$ for the references with spectrum in $[\mu, L]$ and n atoms, and $\mathcal{U}^\gamma$ for those with $c_k \geq \gamma \|e_0\|_A^2$, $k \leq n-2$; the last energy is unrestricted. For a known reference (A, g_0) let $\mathcal{B}_r(A)$ be its relative base ball, and use $E = \|g_{\mathrm{out}}\|_{A_0^{-1}} / \|e_{n-1}^{A_0}\|_{A_0}$ when A_0 varies in that ball. Budgets and the output convention are those of Definition 2.1.

Proposition C.11 (Uniform information-class separation)**.** *Let $n \geq 2$. For sufficiently small $\gamma > 0$ there is $c(n, \kappa, \gamma) > 0$ such that every $M \in \mathcal{M}_1$ satisfies*

$$\sup_{\substack{(A_0, g_0) \in \mathcal{U}^\gamma \\ A_k \equiv A_0,\ \|\widehat{E}'\| \leq \omega'}} \frac{\|g_{\mathrm{out}}\|_{A_0^{-1}}}{\|e_0\|_{A_0}} \geq c(n, \kappa, \gamma)\omega', \qquad 0 \leq \omega' \leq 1.$$

On the full spectral class, defined gated RFC cycles instead satisfy $\|g_n\|_{A_0^{-1}} / \|e_0\|_{A_0} \leq C\omega^2$ for sufficiently small ω, with constants depending only on $(n, \kappa, \gamma_{\mathrm{g}})$. On a sufficiently small known base ball $\mathcal{B}_r(A)$, XRFC with trials $\tau_k = \alpha_k(A)$ satisfies

$$E \leq C_1 r\omega + C_2\omega^2.$$

Thus the first- versus second-order separation for fixed $r > 0$ and $\omega' \asymp \omega$ need not persist at $r = 0$.

Proof of Proposition C.11. Over same-information endpoint pairs in $\mathcal{U}^\gamma$, consider the supremum of

$$\frac{r-1}{r+1} \min_{\pm} \frac{\|e_{n-1}^{\pm}\|_{A^\pm}}{\|e_0^{\pm}\|_{A^\pm}}.$$

An interior reference has distinct positive final lengths and errors, with common positive preterminal energies; this supremum is positive for small γ.

Theorem 4.1 gives the lower bound; Theorem B.25 gives the gated upper bound. For ordinary RFC combine B.25 with $K \leq [\frac{1}{2}(n-1)^2 + n - 1]\omega^2$ and $\|e_{n-1}\|_A \leq \operatorname{sech}((n-1)\theta)\|e_0\|_A$. On a small compact base ball, Lemma B.1 gives a uniform quadratic remainder and Theorem C.1 has $\varepsilon_k = O(r)$ with bounded projectors, giving $C_1 r\omega + C_2\omega^2$. The constant-prefix lower bound is Corollary C.8. □

D. Extrapolation designs and extremal strings

This appendix proves the extrapolation criterion and the extremal-string results of Section 5. Throughout, n denotes cyclic spectral dimension, $0 < \mu < L$, $\kappa = L/\mu$ and $\eta = \frac{\kappa+1}{\kappa-1} = \cosh\theta$, so that

$$\sinh\tfrac{\theta}{2} = (\kappa-1)^{-1/2}, \qquad \tanh\tfrac{\theta}{2} = \kappa^{-1/2}, \qquad 2\kappa^{-1/2} \leq \theta \leq 2(\kappa-1)^{-1/2}. \tag{45}$$

Let $\mathcal{P}$ be the compact convex set of probability measures on $[\mu, L]$ with its weak-$*$ topology. We use the Christoffel criteria of Section 5 and the string coordinate $\zeta = \lambda/\mu - 1 \in [0, \kappa - 1]$.

D.1. The worst spectrum: an extrapolation design problem

For probability measures on $[\mu, L]$ define

$$\Phi_{w,\gamma}(\nu) = \sum_{m=1}^{M} w_m \lambda_m(0;\nu)^\gamma, \qquad \Gamma^*_{w,\gamma} = \sup_\nu \Phi_{w,\gamma}(\nu), \qquad 0 < \gamma \leq 1.$$

Write $\Gamma^*_{M,\gamma}$ for $w \equiv 1$, and $\Gamma^*_{\infty,\gamma}$ for the infinite sum; omit γ when it is $1/2$, and put $\Gamma^* = \Gamma^*_{\infty,1/2}$. Lemma D.4 allows exactly n atoms in the finite problem, so Theorem 3.1 gives $\Gamma_\omega = \Gamma^*_{w,1/2}$, $w_m = \omega_{m-1}$, and $\Gamma_\omega = \omega\Gamma^*_{n-1}$ for uniform budgets. The case $\gamma = 1$ controls total Hilbert–Schmidt budgets. This is a mixture of nested polynomial c-information criteria [63].

Theorem D.1 (Mixed extrapolation criterion)**.** *Let $w_m \ge 0$, $w_M > 0$ and $0 < \gamma \le 1$. The maximum of $\Phi_{w,\gamma}$ is attained by a measure with at most $M+1$ atoms; if $\gamma < 1$, every maximizer has at least $M+1$ support points. For a measure with at least $M+1$ support points, let*

$$\psi_\nu = \sum_{m=1}^{M} w_m \lambda_m(0;\nu)^{\gamma-1} p_m^2,$$

where p_m minimizes the Christoffel criterion. Such a measure ν° is optimal if and only if

$$\max_{[\mu,L]} \psi_{\nu^\circ} \le \Phi_{w,\gamma}(\nu^\circ),$$

with equality on its support. Every such ν gives the certificate

$$\Gamma^*_{w,\gamma} \le \Phi_{w,\gamma}(\nu) + \gamma\big(\max_{[\mu,L]} \psi_\nu - \Phi_{w,\gamma}(\nu)\big).$$

The equivalence criterion specializes the general design theorem [63–65]. The sharp single-degree design gives

$$\max_{M' \le M} \operatorname{sech}(M'\theta) \sum_{m \le M'} w_m \le \Gamma^*_{w,1/2} \le \sum_{m=1}^{M} w_m \operatorname{sech}(m\theta).$$

For $w \equiv 1$, the corresponding limiting constants range from $\max_{y \ge 0} y \operatorname{sech} y = 0.66274\ldots$ to $\pi/2$.

Lemma D.2 (Chebyshev extrapolation design; [66, 67])**.** *For every $m \ge 1$,*

$$\sup_\nu \lambda_m(0;\nu) = T_m(\eta)^{-2} = \operatorname{sech}^2(m\theta).$$

An attaining measure has nodes $\lambda_j = (L+\mu)/2 - (L-\mu)\cos(j\pi/m)/2$ and masses proportional to $|\ell_j(0)|$, where ℓ_j are their Lagrange polynomials.

Proof. Let p_* be the normalized Chebyshev polynomial and $c = 1/T_m(\eta)$. Its alternating values at the Lobatto nodes have the signs of $\ell_j(0)$. Thus, for $\nu_j \propto |\ell_j(0)|$ and $q(0) = 0$, $\deg q \le m$, $\sum \nu_j p_*(\lambda_j) q(\lambda_j) = C \sum \ell_j(0) q(\lambda_j) = 0$. These are the minimizing normal equations, with value c^2. Conversely $\|p_*\|_\infty = c$ bounds the criterion for every measure. This is the Hoel–Levine extrapolation design [67], equivalently Greenbaum's sharp CG extremum [66]. □

In the Green coordinate of Remark 3.2, the bound is $\mathsf{X}_m \ge \log T_m(\eta) = \log\cosh(m\theta)$. It bounds the accumulated advance; individual bond lengths may vary.

Remark D.3 (Fixed nodes: the persymmetric adversary)*.* The same computation answers the isospectral question. Fix nodes $\lambda_1 < \cdots < \lambda_n$ in $[\mu, L]$ with Lagrange polynomials ℓ_k, and put $\omega(\lambda) = \prod_k (\lambda - \lambda_k)$. A polynomial of degree at most $n-1$ is determined by its nodal values y_k, and $p(0) = \sum_k \ell_k(0) y_k$. Hence, for strictly positive weights $\nu_k = \nu(\{\lambda_k\})$,

$$\lambda_{n-1}(0;\nu) = \Big(\sum_k \frac{\ell_k(0)^2}{\nu_k}\Big)^{-1} \le \Big(\sum_k |\ell_k(0)|\Big)^{-2},$$

with equality exactly for $\nu_k \propto |\ell_k(0)|$. If a weight vanishes, $\lambda_{n-1} = 0$. This is the fixed-support case of Elfving's c-optimality [63, 68]. The corresponding gradient weights $\propto \lambda_k \nu_k$ are proportional to $|\omega'(\lambda_k)|^{-1}$. These are the reciprocal-derivative weights of the unique persymmetric Jacobi matrix with this spectrum [69], the canonical realization of [21, (1.3)]. In the coordinates of Remark 3.2, the canonical chain is the realization of shortest Green length, $\mathsf{X}_{n-1} = \log \sum_k |\ell_k(0)|$. Lemma D.2 is the case of Chebyshev–Lobatto nodes.

For the zeros of $P_n^{(\alpha,\beta)}$, $\alpha,\beta > -1$, under $\lambda = 1-\zeta$, the sum $\sum_k |\ell_k(0)|$ is the Lebesgue function of the zeros at the hard edge. For each fixed k, the Mehler–Heine formula and its derivative [70, 18.11.5] give

$$|\ell_k(0)| \longrightarrow \frac{2^{1-\alpha}\, j_{\alpha,k}^{\alpha-1}}{\Gamma(\alpha+1)\,|J'_\alpha(j_{\alpha,k})|},$$

which is of order $k^{\alpha-1/2}$ [70, §10.21]. The limiting edge series therefore converges exactly when $\alpha < -1/2$. This is exactly the case in which the canonical density of [21, Sec. 2.4, Prop. C.4] has an end at finite Green distance.

Proof of Theorem D.1. With $f_m = (1, t, \dots, t^m)^\top$, $\mathbf{M}_m = \int f_m f_m^\top d\nu$ and $c = f_m(0)$,

$$\lambda_m = \inf_{c^\top \alpha = 1} \alpha^\top \mathbf{M}_m \alpha = (c^\top \mathbf{M}_m^{-1} c)^{-1} \quad (\mathbf{M}_m \succ 0).$$

Hence each λ_m, and therefore $\Phi_{w,\gamma}$, is concave and upper semicontinuous. Compactness gives a maximum; an $(M+1)$-point Gauss rule preserves the required moments and gives the support reduction [25, 71].

For $\#\operatorname{supp}\nu \geq M+1$, $p_m = \lambda_m c^\top \mathbf{M}_m^{-1} f_m$ and differentiation along $(1-s)\nu + s\nu'$ gives

$$\partial\Phi_{w,\gamma}(\nu; \nu' - \nu) = \gamma\Big(\int \psi_\nu \, d\nu' - \Phi_{w,\gamma}(\nu)\Big). \tag{46}$$

The concave tangent inequality gives both optimality and the upper certificate by testing point masses; $\int \psi_\nu d\nu = \Phi_{w,\gamma}(\nu)$ gives equality on the support. This is the general equivalence argument [63–65].

If $\#\operatorname{supp}\nu = k \leq M$ and $\gamma < 1$, mix it with a fixed $(M+1)$-point measure of mass s. Concavity gives $\lambda_M \geq cs$ with $c > 0$. Terms with $m < k$ change by $O(s)$, while the other terms were zero and remain nonnegative. Thus the objective increases by at least $w_M c^\gamma s^\gamma - O(s) > 0$, excluding such a maximizer. Trailing zero weights may be discarded; the all-zero objective is trivial. □

Lemma D.4 (Measures with a prescribed number of atoms)**.** *For every $N \geq M+1$, the supremum of $\Phi_{w,\gamma}$ over the measures in $\mathcal{P}$ with exactly N atoms equals $\Gamma^*_{w,\gamma}$. In particular, $\Gamma^*_{n-1}(\kappa)$ is the supremum of $\sum_{m=1}^{n-1} \lambda_m(0;\nu)^{1/2}$ over spectral measures ν with n atoms in $[\mu, L]$, which by Theorem 3.1 is the worst first-order response of FC over initial gradients, for the budgets $\omega_l \equiv 1$.*

Proof. Add $N - \#\operatorname{supp}\nu^*$ distinct atoms of total mass ε to a maximizer ν^* with at most $M+1$ atoms. Concavity gives $\Phi_{w,\gamma}((1-\varepsilon)\nu^* + \varepsilon\nu'') \geq (1-\varepsilon)^\gamma \Gamma^*_{w,\gamma}$; let $\varepsilon \downarrow 0$. □

Lemma D.5 (Poisson summation for the Chebyshev majorants)**.** *For $\theta > 0$,*

$$\sum_{m\geq 1} \operatorname{sech}(m\theta) = \frac{\pi}{2\theta} - \frac{1}{2} + \frac{\pi}{\theta} \sum_{k\geq 1} \operatorname{sech}\frac{\pi^2 k}{\theta}, \qquad \sum_{m\geq 1} \operatorname{sech}^2(m\theta) = \frac{1}{\theta} - \frac{1}{2} + \frac{2\pi^2}{\theta^2} \sum_{k\geq 1} \frac{k}{\sinh(\pi^2 k/\theta)}.$$

Equivalently $\sum_{m\geq 1} \operatorname{sech}(m\theta) = K(k)/\pi - 1/2$, where $e^{-\theta}$ is the elliptic nome and $K(k)$ the complete elliptic integral [72]. The remainders are positive and equal $\frac{2\pi}{\theta} e^{-\pi^2/\theta}(1+o(1))$ and $\frac{4\pi^2}{\theta^2} e^{-\pi^2/\theta}(1+o(1))$ as $\theta \to 0$.

Proof. In the convention $\hat{f}(\xi) = \int f(y) e^{-i\xi y} dy$, the transforms are $\widehat{\operatorname{sech}}(\xi) = \pi \operatorname{sech}(\pi\xi/2)$ and $\widehat{\operatorname{sech}^2}(\xi) = \pi\xi/\sinh(\pi\xi/2)$. Poisson summation gives both identities and their leading dual terms. The elliptic form is Jacobi's identity $\vartheta_3(q)^2 = 1 + 4\sum_{m\geq 1} q^m/(1+q^{2m}) = 2K(k)/\pi$, $q = e^{-\theta}$ [72]. □

Proof of the Chebyshev majorants. The upper estimate is termwise Lemma D.2; the lower uses its degree-M' design and $\lambda_m \geq \lambda_{M'}$ for $m \leq M'$. The preceding Poisson identity gives the upper limiting constant. For the lower one, the mesh $m\theta$ approaches the maximum of $y \operatorname{sech} y$, characterized by $y \tanh y = 1$. □

Definition D.6 (Strings and their reward)**.** A string is a pair (ℓ, M), $\ell \in (0,\infty]$, with $M : [0,\ell) \to [1,\infty)$ nondecreasing, right-continuous, $M(0) = 1$, $M(0-) = 0$ (a unit mass at the left end). Write $dM = \varsigma^2 dx + dM_{\mathrm{s}}$ (Lebesgue decomposition) and $y(x) = \int_0^x \varsigma$ (the type). The state (φ, p, K) solves $\varphi(x) = 1 + \int_0^x p$, $p(x) = 1 + \int_{(0,x]} \varphi\, dM$, $K(x) = 1 + \int_{(0,x]} \varphi^2\, dM$ (the string equation $d\varphi' = -\zeta\varphi\, dM$ at $\zeta = -1$). The reward is $R_\gamma(M) = \int_{[0,\ell)} K^{-\gamma} dy$, and $a_{\infty,\gamma} = \sup_M R_\gamma(M)$ over all strings.

Remark D.7 (Where the strings come from)*.* Two classical constructions turn a positive Jacobi chain into a Krein string. The inverse is a single-pair matrix; up to diagonal congruence, such matrices are the influence matrices of beaded strings [26, §3, 4°], [62]. This is the oscillation-theoretic construction of [21, Rem. 2.3, 3.2]: the Green coordinate [73] and the Liouville map $\xi = \tanh x$ [74, §3] give a string fixed at both ends, whose static self-influences determine its energy. The moment construction instead reads masses and gaps from the Stieltjes continued fraction of the first-vertex measure [75], [62, Suppl. II]; its launched end is free. Krein's spectral theory joins these constructions through the characteristic function and its spectral measure [76].

For one CG reference, write ν_λ for its energy measure on $[\mu, L]$ and ν_g for its gradient measure. Three strings occur:

	construction and static point	exact quantity
(S1)	ν_g, at $\lambda = 0$	influence matrix $\lvert\mathcal{G}\rvert$
(S2)	ν_λ, at $\lambda = \mu$	kernel $K_m = \sum_{k\le m} m_k \varphi_k(-1)^2$
(S3)	(S2) rebased at $\lambda = 0$	accumulated mass K_m

Lemma D.15 constructs (S2) in the coordinate $\zeta = \lambda/\mu - 1$, with extrapolation at $\zeta = -1$. Remark D.16 constructs (S1); the string of [21, Rem. 3.2] has the same influence matrix in its two-fixed-end parametrization. The Christoffel transform $\nu_g(d\lambda) \propto \lambda\nu_\lambda(d\lambda)$ changes (S3) to (S1), whereas a change of static frame relates (S2) and (S3). Their shared Green coordinate is identified in Remark D.16.

For the change of frame, if $f > 0$ solves $df' = -cf\, dM$, then

$$d\tilde{x} = \frac{dx}{f^2}, \qquad d\tilde{M} = f^2\, dM, \qquad \tilde{u} = \frac{u}{f}$$

carry the equation at ζ to that at $\zeta - c$; the identity $d(fu' - f'u) = f\, du' - u\, df'$ also covers atoms. Take $f = \varphi$, $c = -1$. The rebased string (S3) keeps the unit left atom and

$$\tilde{M}(\tilde{x}(x)) = 1 + \int_{(0,x]} \varphi^2\, dM = K(x), \qquad \lambda_m(0;\nu_\lambda) = K_m^{-1}.$$

Thus CG energy is reciprocal accumulated mass in the extrapolation frame. Krein's type is invariant, since $\sqrt{d\tilde{M}\, d\tilde{x}} = \sqrt{dM\, dx}$, as it is under the map of [21, Rem. 2.3]. The two spectral ends record mass at zero frequency [76, 77] and type at high frequency [78, 79]. In both papers the continuum problem preserves the extrapolation normalization and the step count through these string quantities.

The value is determined by a planar Lotka–Volterra system; on $\{s > 0,\ U > 0\}$ let

$$(LV_\gamma) \qquad \dot{s} = 2s\,(s - 1 - \gamma U), \qquad \dot{U} = U\,(s + 1 - U). \tag{47}$$

For $\gamma < 1$, let $s = \sigma_\gamma(U)$ be the stable manifold of the saddle $F_\gamma = ((1+\gamma)/(1-\gamma), 2/(1-\gamma))$. For $\gamma = 1$, let it be the separating orbit on $U \ge 1$ characterized by $U + 1 < \sigma_1(U) \le 4U$ and $U \uparrow \infty$. Their real-analytic graphs and unique crossings $s_\gamma = \sigma_\gamma(1)$ are established below. Put $s_0 = s_{1/2}$, $s_* = s_1$ and $t_* = \sqrt{\sigma_\gamma}$.

Theorem D.8 (Optimal continuum string)**.** *For $0 < \gamma \le 1$ there is a unique maximizing string M^*, with positive real-analytic density, and*

$$R_\gamma(M) \le R_\gamma(M^*) = a_{\infty,\gamma} = \frac{\sqrt{s_\gamma}}{2\gamma}.$$

In particular,

$$a = a_{\infty,1/2} \approx 1.3287, \qquad a_{\infty,1} \approx 0.81343 \quad \textit{(five significant digits).}$$

For $\gamma = 1/2$,

$$M^*(\infty) = a^2, \qquad M^{*\prime}(0+) = a^{-2}, \qquad \nu^*(\{0\}) = a^{-2},$$

where ν^* is the principal probability spectral measure of the unit left atom in the canonical energy realization, with free left end [76]; the precise principal convention is specified in Appendix D.4. Proposition D.14 certifies these roundings. In $\zeta = \lambda/\mu - 1$, the unique extremal measure satisfies

$$\nu^*((\zeta, \infty)) \sim \frac{2}{\pi a}\zeta^{-1/2}, \qquad \zeta \to \infty.$$

Theorem D.9 (Asymptotics of the worst constant)**.** *Let $0 < \gamma \le 1$, $B_\gamma = \gamma$ for $\gamma < 1$ and $B_1 = 1/2$. For every $\kappa > 1$,*

$$\Gamma^*_{\infty,\gamma}(\kappa) \le \frac{a_{\infty,\gamma}}{2}\sqrt{\kappa - 1} + \frac{1 + B_\gamma}{4}\log\kappa + \frac{1}{2} + \frac{1 + 2B_\gamma}{4\gamma},$$

and

$$\lim_{\kappa\to\infty} \theta\Gamma^*_{\infty,\gamma}(\kappa) = a_{\infty,\gamma}, \qquad \theta\Gamma^*_{\infty,\gamma} \le a_{\infty,\gamma} + O(\theta\log(1/\theta)).$$

The bounds include the empty $n = 1$ criterion 0; the same limits hold for $\Gamma^*_{n-1,\gamma}$ when $n\theta \to \infty$. In particular,

$$\Gamma^*(\kappa) \sim \tfrac{a}{2}\sqrt{\kappa}, \qquad \Gamma^*(\kappa) \le \tfrac{a}{2}\sqrt{\kappa - 1} + \tfrac{3}{8}\log\kappa + \tfrac{3}{2},$$

and $\Gamma^*_{\infty,1} \le \frac{1}{2}a_{\infty,1}\sqrt{\kappa - 1} + \frac{3}{8}\log\kappa + 1$. The displayed $\gamma = 1/2$ upper bound improves on $\sum_{m\ge1}\operatorname{sech}(m\theta)$ for $\kappa \ge 1600$.

The discrete-to-continuum proof uses the string value function and consecutive-step cancellation under the spectral constraint. For total Hilbert–Schmidt budgets the resulting response is of order $\kappa^{1/4}$; see Proposition B.9.

For $n \ge 2$ define $\omega_{\mathrm{crit}} = 1/\Gamma^*_{n-1}(\kappa)$; Theorem 3.1 gives a worst FC cycle ratio at most $\omega/\omega_{\mathrm{crit}} + C(n,\kappa)\omega^2$. Thus its first-order contraction threshold satisfies

$$\omega_{\mathrm{crit}} \ge \left(\tfrac{a}{2}\sqrt{\kappa - 1} + \tfrac{3}{8}\log\kappa + \tfrac{3}{2}\right)^{-1}, \qquad \omega_{\mathrm{crit}} \sim \frac{\theta}{a} \sim \frac{2}{a}\kappa^{-1/2} \tag{48}$$

as $\kappa \to \infty$, $n\theta \to \infty$; $2/a \approx 1.5053$ to five significant digits. Uniform first-order contraction over all references is equivalent to $\omega < \omega_{\mathrm{crit}}$.

D.2. The optimal string

Use the strings of Definition D.6, with $dM = \varsigma^2 dx + dM_{\mathrm{s}}$ and state

$$\varphi = 1 + \int_0^x p, \qquad p = 1 + \int_{(0,x]} \varphi\, dM, \qquad K = 1 + \int_{(0,x]} \varphi^2\, dM. \tag{49}$$

Picard iteration gives a unique local state; φ is absolutely continuous, $\varphi' = p$ a.e., and p, K are right-continuous and nondecreasing. An atom m sends (φ, p, K) to $(\varphi, p + m\varphi, K + m\varphi^2)$; a mass-free interval changes only φ. Neither collects type.

Along every string $\varphi, p, K \ge 1$ and $r := \varphi p/K \ge 1$, because $d(\varphi p - K) = p^2 dx \ge 0$ and $\varphi p - K$ starts at zero; impulses leave it unchanged since φ is continuous.

For an absolutely continuous string with $\varsigma > 0$, use the type y as time and $t := p/(\varsigma\varphi)$ as control. Since $\frac{d}{dy} = \varsigma^{-1}\frac{d}{dx}$,

$$\frac{d\log\varphi}{dy} = t, \qquad \frac{d\log p}{dy} = \frac{1}{t}, \qquad \frac{d\log K}{dy} = \frac{r}{t}, \qquad \frac{dr}{dy} = \frac{r}{t}\,(t^2 + 1 - r). \tag{50}$$

The state homogeneities reduce the value function to $K^{-\gamma}f(r)$. With $s = t^2$, $U = r$ and $d\tau = dy/t$, the feedback equation and the reduced r-equation form (LV_γ) of (47). We next identify its relevant orbit.

Remark D.10 (Green profiles against the type). For the Stieltjes strings of Lemma D.15, Remark 3.2 gives $K_m = e^{2\mathsf{X}_m}$, where X is the Green coordinate of the corresponding gradient chain. On strings we accordingly write $\mathsf{X} = \frac{1}{2}\log K$. In (50), $d\mathsf{X}/dy = r/(2t)$ is the Green speed, and the reward is $R_\gamma = \int e^{-2\gamma \mathsf{X}} dy$. The reduction to $K^{-\gamma} f(r)$ is the translation covariance of X; [21, Rem. 2.3] records the same translations as automorphisms of $(-1, 1)$. Eliminating the control $t = r/(2\mathsf{X}')$ from (50) gives

$$r' = \frac{r^2}{2\mathsf{X}'} - 2\mathsf{X}'(r-1), \qquad r(0) = 1.$$

Along absolutely continuous strings, the lower spectral bound thus becomes a local Riccati constraint on the Green profile $\mathsf{X}(y)$.

The type is also the clock of [21]. There, packing gives every vertex type $1/n$ [21, Prop. 3.1, Rem. 3.2]. The canonical profile obeys $dx/dt = 1/\phi_\alpha(t)$, with t the step index divided by n [21, Prop. 2.2]. Take the CG reference with eigenvalues $1 - \zeta_{n,k}$ and persymmetric gradient weights. Remark 3.2 and [21, Thm. 2.1, Prop. 2.2] give

$$\log \frac{\|e_{\lfloor ns \rfloor}\|_A}{\|e_{\lfloor nt \rfloor}\|_A} \longrightarrow \int_s^t \frac{d\tau}{\phi_\alpha(\tau)}, \qquad 0 < s < t < 1.$$

So ϕ_α is the reciprocal logarithmic CG rate per unit clock, and $c_*(\alpha) = \int_0^1 (dx/dt)^{-2} dt$. Up to the normalization of the clock, both limiting problems are variational problems for a Green profile against the type. The companion paper [21] minimizes $\int \mathsf{X}'^{-2}$ when every eigenvalue is prescribed. Theorem D.8 maximizes $\int e^{-2\gamma \mathsf{X}}$ under a one-sided bound, which the Riccati equation makes local.

Put $\mathcal{Q} = \{(s, U) : s > 0,\ U > 0\}$ and $s_{\mathrm{F}} = \frac{1+\gamma}{1-\gamma}$, $U_{\mathrm{F}} = \frac{2}{1-\gamma} = s_{\mathrm{F}} + 1$ ($(3, 4)$ for $\gamma = \frac{1}{2}$), $F = (s_{\mathrm{F}}, U_{\mathrm{F}})$,

$$D_\gamma = \{0 < s < s_{\mathrm{F}},\ 0 < U < s + 1\}, \qquad R_\gamma = \{s \geq s_{\mathrm{F}},\ U \geq (s-1)/\gamma\}.$$

Proposition D.11 (Separatrix, $\gamma \in (0, 1)$)**.** *The only equilibrium in $\mathcal{Q}$ is the hyperbolic saddle F. Its eigenvalues $\lambda_\pm$ solve*

$$\lambda^2 - (2s_{\mathrm{F}} - U_{\mathrm{F}})\lambda - 2(1-\gamma) s_{\mathrm{F}} U_{\mathrm{F}} = 0$$

($1 \pm \sqrt{13}$ for $\gamma = 1/2$). Boundary equilibria are $(0, 0)$, $(1, 0)$ and the hyperbolic sink $(0, 1)$. The regions D_γ, R_γ are backward invariant, with $\dot{U} > 0$ on D_γ and $\dot{U} < 0$ on $R_\gamma \setminus \{F\}$.

The stable manifold is the analytic graph $s = \sigma_\gamma(U)$, $U > 0$, with $\sigma_\gamma(U_{\mathrm{F}}) = s_{\mathrm{F}}$. Its two branches lie in D_γ and R_γ, with U ranging monotonically from 0 and $+\infty$, respectively, to U_{F}; $U - 1 < \sigma_\gamma(U) < s_{\mathrm{F}}$ below U_{F}, and $\sigma_\gamma(U) \geq s_{\mathrm{F}}$ above it. It crosses $U = 1$ once, at $s_\gamma \in (0, s_{\mathrm{F}})$. For $0 < s < s_{\mathrm{F}}$, the forward orbit of $(s, 1)$ exits D_γ through $U = s + 1$ for $s < s_\gamma$ and through $s = s_{\mathrm{F}}$ for $s > s_\gamma$; at $s = s_\gamma$ it tends to F.

Proof. The Jacobian at F is $\left(\begin{smallmatrix} 2s_{\mathrm{F}} & -2\gamma s_{\mathrm{F}} \\ U_{\mathrm{F}} & -U_{\mathrm{F}} \end{smallmatrix}\right)$, of determinant $-2(1-\gamma) s_{\mathrm{F}} U_{\mathrm{F}} < 0$; substitution gives the other equilibria. The stable eigenvector $v = (2\gamma s_{\mathrm{F}}, 2s_{\mathrm{F}} - \lambda_-)$ points into R_γ, and $-v$ into D_γ. On the two non-axis edges of D_γ, $(U-s)' = 2s((1+\gamma) - (1-\gamma)s) > 0 \quad (U = s+1)$, $\qquad s' = 2\gamma s_{\mathrm{F}}(U_{\mathrm{F}} - U) > 0 \quad (s = s_{\mathrm{F}})$. The analogous edge signs make R_γ backward invariant. The analytic stable-manifold theorem [80, Thm. 27] gives branches in the stated regions, with strictly monotone U.

A positive backward limit of U on the bounded branch forces its compact limit set onto $U = s + 1$, hence to F or the sink $(0, 1)$. Monotonicity excludes F, and a nonconstant orbit cannot approach a hyperbolic sink backward. Thus $U \downarrow 0$. Bounded U on the other branch would bound $s \leq 1 + \gamma U$ and similarly force limit F, impossible; hence $U \uparrow \infty$. The graph is analytic because $U' \neq 0$ off F and $v_U \neq 0$ at F. Finally the two exit edges, separated by this graph, give the shooting dichotomy: an orbit that never exits has increasing bounded U and must tend to F. □

(LV_1), $\dot{s} = 2s(s - 1 - U)$, $\dot{U} = U(s + 1 - U)$, has no equilibrium in $\mathcal{Q}$. Let $\mathcal{E} = \{U > 0,\ U < s - 1\}$, where $\dot{s} > 0$ and $\dot{U} > 0$, and $\mathcal{P}_1 = \{s > 4,\ U < s/4\} \subset \mathcal{E}$.

Proposition D.12 (Separating orbit, $\gamma = 1$)**.** *Let $\mathcal{A}_-$ be the set of $s > 2$ for which the forward orbit of $(s, 1)$ leaves $\mathcal{E}$, and $\mathcal{A}_+$ the set of $s > 2$ for which it enters $\mathcal{P}_1$. Then $\mathcal{A}_\pm$ are open, disjoint and nonempty, the set $(2, \infty) \setminus (\mathcal{A}_- \cup \mathcal{A}_+)$ is nonempty and contained in $(2, 4]$, and for each of its points s_* the forward orbit of $(s_*, 1)$ stays in $\mathcal{E} \setminus \mathcal{P}_1$ and $U \uparrow \infty$ along it. It is the graph $\{s = \sigma_1(U) : U \geq 1\}$ of a real-analytic function with $U + 1 < \sigma_1(U) \leq 4U$.*

Proof. On $U = s - 1$, $(U - s)' = 2U > 0$; on $U = s/4$, $(U - s/4)' = (3s/4)(1 - s/4) < 0$; and on $s = 4$, $s' > 0$ for $U \leq 1$. Thus $\mathcal{E}$ has a strict exit and $\mathcal{P}_1$ is forward invariant. The exit and entrance sets are open and disjoint. They contain $2+\varepsilon$ for small $\varepsilon > 0$ and $(4, \infty)$, respectively, so connectedness leaves a separating point in $(2, 4]$. Along its orbit s, U increase. Bounded U would either permit $s \to \infty$, forcing entry into $\mathcal{P}_1$, or yield an equilibrium in $\overline{\mathcal{E}} \cap \{U \geq 1\}$; neither is possible. Hence $U \uparrow \infty$, providing the analytic graph with $U + 1 < s \leq 4U$. □

By Theorem D.8 (see the proof below), $\mathcal{A}_- \cup \mathcal{A}_+$ misses exactly one point s_*, so $\mathcal{A}_- = (2, s_*)$ and $\mathcal{A}_+ = (s_*, \infty)$.

For $\gamma \in (0, 1)$ use σ_γ (Proposition D.11), defined on $(0, \infty)$. For $\gamma = 1$ use σ_1; since $\dot{U} > 0$ at $(s_*, 1)$, the orbit is a graph also slightly before $\tau = 0$, and σ_1 is real-analytic on $(1 - \varepsilon_1, \infty)$ for some $\varepsilon_1 > 0$, with $\sigma_1 > 0$ and $\sigma_1 + 1 - U > 0$ there. Put $t_* = \sqrt{\sigma_\gamma}$ and

$$f_\gamma(r) = \frac{r + \sigma_\gamma(r) - 1}{2\gamma\, r\, t_*(r)}, \qquad W(\varphi, p, K) = K^{-\gamma} f_\gamma\Big(\frac{\varphi p}{K}\Big). \tag{51}$$

Lemma D.13 (Fundamental identity)**.** *For every string (ℓ, M) and $0 < X < \ell$, with $z = (\varphi, p, K)$, $r = \varphi p/K$ and $t_* = t_*(r)$,*

$$\begin{aligned} W(z(0)) - W(z(X)) = {} & \int_0^X K^{-\gamma}\, dy + \frac{1}{2}\int_0^X K^{-\gamma}\Big(\sqrt{\tfrac{p}{\varphi t_*}} - \varsigma\sqrt{\tfrac{\varphi t_*}{p}}\Big)^2 dx \\ & + \frac{1}{2}\int_{(0,X]} K^{-\gamma}\frac{\varphi t_*}{p}\, dM^c_{\mathrm{s}} + \sum_{0<\xi\leq X,\ \Delta M(\xi)>0} J_\xi, \end{aligned} \tag{52}$$

where, for an atom of mass m at ξ, $J_\xi = \frac{1}{2}\int_0^m K_u^{-\gamma}\varphi\, t_*(r_u)/p_u\, du > 0$, $p_u = p(\xi-) + u\varphi$, $K_u = K(\xi-) + u\varphi^2$, $r_u = \varphi p_u/K_u$, $\varphi = \varphi(\xi)$.

Proof. The graph equation gives $f'_\gamma = -1/(2rt_*)$ and $(1 - r)f'_\gamma - \gamma f_\gamma = -t_*/(2r)$, including the saddle by analyticity. Consequently

$$W_\varphi p = -\tfrac{1}{2}K^{-\gamma}p/(\varphi t_*), \qquad W_p\varphi + W_K\varphi^2 = -\tfrac{1}{2}K^{-\gamma}\varphi t_*/p.$$

Apply the BV chain rule [81] to $dz = (p, 0, 0)dx + (0, \varphi, \varphi^2)dM$. Completing the square in its absolutely continuous part yields the first two integrals; the continuous singular part gives the third. At each atom, integrate the same derivative along its straight impulse. Positivity and local bounded variation justify the convergent jump sum. □

Proof of Theorem D.8. Value and feedback. The nonnegative terms of (52) and $W > 0$ give $R_\gamma \leq f_\gamma(1) = \sqrt{s_\gamma}/(2\gamma)$. To attain this, solve

$$r_y = \mathcal{F}(r) := \frac{r}{t_*(r)}(\sigma_\gamma(r) + 1 - r), \quad r(0) = 1, \qquad (\log K)_y = r/t_*, \quad (\log\varphi)_y = t_*,$$

and set $p = rK/\varphi$, $\varsigma = p/(\varphi t_*)$, $x_y = \varsigma^{-1}$. Then $(\log p)_y = 1/t_*$ and (49) holds for $M^* = 1 + \int \varsigma^2 dx$. All three loss terms in the fundamental identity vanish. For $\gamma < 1$, $r \uparrow U_{\mathrm{F}}$, $K \to \infty$ and f_γ stays bounded; for $\gamma = 1$, $r \uparrow \infty$ and $f_1(r) \leq 5/(2\sqrt{r+1})$. Thus $W \to 0$ and the upper bound is attained. Applying this argument to either separating orbit for $\gamma = 1$ gives the same value, hence the same s_*; connectedness then identifies $\mathcal{A}_\pm$.

Uniqueness. Equality forces every loss term in (52) to vanish. The mass is absolutely continuous and $\varsigma = p/(\varphi t_*)$ a.e.; the zero-density set is null because the square is strictly positive there. The resulting closed-loop system $\varphi' = p, \qquad p' = p^2/(\varphi\sigma_\gamma(r)), \qquad K' = p^2/\sigma_\gamma(r)$ is locally Lipschitz and has the prescribed initial state. Thus it is M^* up to its maximal length. A shorter string leaves a positive terminal value W and cannot attain the bound.

Shape and bottom atom. For $\gamma < 1$, $\mathcal{F}'(U_\mathrm{F}) < 0$ since $\sigma'_\gamma(U_\mathrm{F}) = 2\gamma s_\mathrm{F}/(2s_\mathrm{F} - \lambda_-) < 1$. Hence $r \to U_\mathrm{F}$ exponentially, and $(\log K)_y \to U_\mathrm{F}/\sqrt{s_\mathrm{F}}, \qquad (\log\varsigma)_y \to -(s_\mathrm{F}-1)/\sqrt{s_\mathrm{F}}$, with integrable errors. Therefore the mass is finite, the length infinite, and $\varsigma(0) = s_\gamma^{-1/2}$. For $\gamma = 1/2$, both $K^{-1/2}$ and ς are comparable to $e^{-2y/\sqrt{3}}$.

Put $H = (\sigma_\gamma(r) - 1)K/\varphi^2$. Direct differentiation with the reduced equations gives

$$H_y = \frac{Kr}{\varphi^2 t_*}((1-2\gamma)\sigma_\gamma - 1).$$

At $\gamma = 1/2$ this is $-\varsigma$; since $H(0) = s_0 - 1$ and $H(\infty) = 0$, the total mass is $s_0 = a^2$ and $M^{*\prime}(0+) = a^{-2}$. The atom of the principal spectral measure at zero is the reciprocal total mass [77, (1.14)]; hence $\nu^*(\{0\}) = a^{-2}$. □

The qualitative results above are analytic. The following finite rational certificate gives their numerical values.

Proposition D.14 (Certified numerical values)**.** *The separating crossings satisfy*

$$1.765356 < s_0 < 1.765358, \qquad 2.64667 < s_* < 2.64668.$$

The orbits from the lower and upper half-gamma endpoints exit $D_{1/2}$ through $U = s + 1$ and $s = 3$, respectively. At $\gamma = 1$, the lower orbit leaves $\mathcal{E}$, and the upper orbit enters $\mathcal{P}_1$. The following roundings have error strictly smaller than half a unit in the final place:

quantity	s_0	s_*	a	a^{-2}	$2/a$	$a_{\infty,1}$	$\sqrt{a_{\infty,1}/4}$
value	1.7654	2.6467	1.3287	0.56646	1.5053	0.81343	0.45095.

Proof of Proposition D.14. The exact rational polynomials are specified in Table 4. For $\gamma = 1/2$, set $v = 4 - U$ and use the rational polynomial barriers

$$b_\varepsilon(U) = 3 - v(P(v) - \varepsilon 10^{-7}), \qquad \varepsilon \in \{-1, 1\}, \qquad \deg P = 16.$$

Their common saddle value is 3. The actual saddle slope r is the root in $(0, 1/2)$ of $4r^2 - 10r + 3 = 0$; rational root isolation gives $0.34861217 < r < 0.34861220$. Direct rational substitution gives

$$\varepsilon(r - b'_\varepsilon(4)) > 0, \qquad \varepsilon\mathcal{R}(b_\varepsilon)(U) < 0 \quad (1 \le U < 4), \qquad \mathcal{R}(b) = Ub'(b + 1 - U) - 2b(b - 1 - U/2).$$

The first inequality puts $b_{-1} < \sigma_{1/2} < b_1$ just left of the saddle. At a first contact, the residual inequality points strictly into the same ordering when U decreases. Thus this ordering holds throughout $[1, 4)$, and evaluation at $U = 1$ gives the stated half-gamma bracket.

For the endpoint put $e(U) = \sigma_1(U) - U$. It is the unique bounded solution in $1 < e < 3$ of $e' = F(U, e) := \frac{e-3+2e(e-1)/U}{e+1}$. With rational polynomials P_1, Q_1 of degree seven, use $b_\pm(U) = P_1(1/U)/Q_1(1/U) \pm 1/500000$. Here $Q_1 > 0$, $1 \le b_\pm \le 4$ and $F(U, b_-) \le b'_-$, $b'_+ \le F(U, b_+)$. The comparison runs from infinity: for $1 \le f < e \le 4$, $F(U, e) - F(U, f) \ge (e - f)/7$. A positive violation of either desired bound would therefore grow at least exponentially and contradict boundedness. Hence $b_- \le e \le b_+$; their rational values at $U = 1$ give the endpoint bracket.

These inequalities are finite polynomial certificates. For the half-gamma barriers, $\varepsilon\mathcal{R}(b_\varepsilon) = -(4 - U)Q_\varepsilon(4 - U)$, where Q_ε has strictly positive coefficients in the basis $v^j(3 - v)^{33-j}$. For the endpoint, clearing the positive factor $(b_\pm + 1)Q_1(1/U)^3$ gives residuals with the required signs in the basis $x^j(1 - x)^{22-j}$, $x = 1/U$. The coefficient identities reduce the verification to rational polynomial arithmetic,

coefficient signs, quadratic root isolation and endpoint evaluation. The shooting dichotomies give the four exits. Monotonicity of square roots and reciprocals then gives the displayed roundings for the same unique optimal strings. □

Table 4: Exact integer coefficients of the rational barriers in the proof of Proposition D.14: $P(v) = 10^{-20}\sum_{j=0}^{16} p_j v^j$, $P_1(x) = 10^{-10}\sum_{j=0}^{7} a_j x^j$ and $Q_1(x) = 10^{-10}\sum_{j=0}^{7} b_j x^j$.

j	p_j	a_j	b_j
0	34861218113400267672	29999996905	10000000000
1	1238001689541169632	1940342662386	686779738455
2	144294136819071755	39086986865755	15216433459508
3	21422240779829582	301521325207097	134389954874331
4	4336056614927339	918646380355570	486228095462338
5	-1319363149807767	1058544265711540	683147284250093
6	4122546028703320	404422468164464	321811746781579
7	-5815880735813218	35959006309097	34706268176245
8	6348535864521766		
9	-5180145255453823		
10	3194588843389863		
11	-1479053840683332		
12	506612317833225		
13	-124585370355620		
14	20829059284726		
15	-2123338103989		
16	99989154827		

D.3. The discrete-to-continuum limit

Let $N \geq 2$, masses $m_0 = 1$, $m_1, \dots, m_{N-1} > 0$, gaps $l_0, \dots, l_{N-2} > 0$ and $\varrho \geq 0$. Let G be the weighted path Laplacian, $\mathsf{G}_{kk} = \frac{1}{l_{k-1}} + \frac{1}{l_k}$, $\mathsf{G}_{k,k+1} = \mathsf{G}_{k+1,k} = -\frac{1}{l_k}$ (with $1/l_{-1} = 1/l_{N-1} = 0$), with $\mathsf{G}_{N-1,N-1}$ increased by ϱ; let $\mathsf{M} = \mathrm{diag}(m_0, \dots, m_{N-1})$ and $H = \mathsf{M}^{-1}\mathsf{G}$, self-adjoint and positive semidefinite for $\langle u, v\rangle_{\mathsf{M}} = \sum_k m_k u_k v_k$. Define polynomials by $\varphi_0 = 1$, $p_{-1} = 0$ and

$$p_k = p_{k-1} - \zeta m_k \varphi_k, \qquad \varphi_{k+1} = \varphi_k + l_k p_k \qquad (0 \leq k \leq N-2). \tag{53}$$

Lemma D.15 (Stieltjes strings and Christoffel functions)**.** *(a) The spectral measure ν of H at e_0 is a probability measure with exactly N atoms in $[0, \|H\|]$. $\deg \varphi_k = k$, $\int \varphi_k \varphi_{k'} d\nu = \delta_{kk'}/m_k$, and for $m \leq N-1$*

$$\min\Big\{ \int |P|^2 d\nu : \ \deg P \leq m, \ P(-1) = 1 \Big\} = \frac{1}{K_m}, \qquad K_m = \sum_{k=0}^{m} m_k \varphi_k(-1)^2.$$

(b) Conversely, every probability measure ν with N atoms in $[0, \infty)$ is obtained in this way. With P_k its orthonormal polynomials (positive leading coefficients), $\zeta P_k = \beta_{k+1} P_{k+1} + \alpha_k P_k + \beta_k P_{k-1}$, and $\tilde{P}_N = (\zeta - \alpha_{N-1}) P_{N-1} - \beta_{N-1} P_{N-2}$, one has $m_k = P_k(0)^2$, $\varphi_k = P_k/P_k(0)$,

$$l_k = \frac{-1}{\beta_{k+1} P_k(0) P_{k+1}(0)} > 0, \qquad \varrho = -P_{N-1}(0)\tilde{P}_N(0) \geq 0,$$

and $\varrho = 0$ iff $0 \in \operatorname{supp} \nu$. Moreover $\|H\| = \max \operatorname{supp} \nu$.

(c) (Gershgorin.) $\|H\| \leq \max_k \big[\frac{1}{m_k}\big(\frac{1}{l_{k-1}} + \frac{1}{l_k} + \varrho\delta_{k,N-1}\big) + \frac{1}{l_{k-1}\sqrt{m_{k-1}m_k}} + \frac{1}{l_k\sqrt{m_k m_{k+1}}}\big]$.

Proof. The recurrence is the generalized eigen-equation $\mathsf{G}v = \zeta \mathsf{M} v$ in its first $N-1$ rows. Irreducibility gives simple eigenvalues and $v(0) \neq 0$; completeness gives the displayed orthogonality. Since $\deg \varphi_k = k$, the Christoffel–Darboux minimization is K_m^{-1} [25].

Conversely, sign alternation gives $(-1)^k P_k(0) > 0$ for $k < N$. The Christoffel–Darboux identity $p_k = -\zeta \sum_{j\le k} P_j(0)P_j$ gives the gap and mass recurrences; at the last row it gives $p_{N-1} = -\varrho\varphi_{N-1}$ on the nodes. The associated weight is $[\sum m_k\varphi_k^2]^{-1} = [\sum P_k^2]^{-1}$, exactly the prescribed weight. The sign of the terminal polynomial gives $\varrho \ge 0$, with equality precisely for a zero node. Finally apply Gershgorin to $\mathsf{M}^{-1/2}\mathsf{G}\mathsf{M}^{-1/2}$ to obtain (c). □

At $\zeta = -1$, (53) reads

$$p_k = p_{k-1} + m_k\varphi_k, \qquad \varphi_{k+1} = \varphi_k + l_k p_k, \qquad K_k = K_{k-1} + m_k\varphi_k^2,$$

with $\varphi_0 = p_0 = K_0 = 1$: a drift of length l_k followed by an impulse of mass m_{k+1}. This is the string with point masses m_k at $x_k = \sum_{i<k} l_i$, and (φ_k, p_k, K_k) is its state after the mass m_k. If ν_λ is the image of ν under $\zeta \mapsto \lambda = \mu(1+\zeta)$, then $\lambda_m(0;\nu_\lambda) = 1/K_m$ for $m \le N-1$ (polynomials in λ with value 1 at $\lambda = 0$ are polynomials in ζ with value 1 at $\zeta = -1$), and $\nu_\lambda \in \mathcal{P}$ iff $\|H\| \le \kappa - 1$.

Remark D.16 (CG as a Stieltjes string at the extrapolation point). First recover the static factors of Remark 3.2 directly. For $i \le j$, $e_i - e_j \in \mathcal{K}_j$ is A-orthogonal to e_j, so

$$(g_i^0)^\top A^{-1} g_j^0 = e_i^\top A e_j = \|e_j\|_A^2.$$

The Lanczos vectors are $z_j = \pm g_j^0/\|g_j^0\|$; hence the two static factors are $h_j^- = 1/\|g_j^0\|$ and $h_j^+ = \|e_j\|_A^2/\|g_j^0\|$. Their product is the bead weight $\mathsf{d}_j = \alpha_j\|e_j\|_A^2/c_j$, and $\sum_j \mathsf{d}_j = \operatorname{Tr} J^{-1}$ is fixed by the spectrum.

Apply Lemma D.15(b) to the gradient measure ν_g in the variable λ itself, that is, with base point $\lambda = 0$. The result is CG. The orthonormal polynomials are normalized CG residual polynomials, so the masses are $\tilde m_k = \|g_0^0\|^2/\|g_k^0\|^2$. Write $J_{[k]}$ for the leading $k \times k$ block. By Lemma B.6 and the Gauss-quadrature representation of the CG error [24, 25, 82], the beads sit at

$$\tilde x_k = \frac{\|e_0\|_A^2 - \|e_k\|_A^2}{\|g_0^0\|^2}, \qquad \tilde l_k = \tilde x_{k+1} - \tilde x_k = \frac{c_k}{\|g_0^0\|^2}.$$

Here $\tilde x_0 = 0$; for $1 \le k < n$, $\tilde x_k = (J_{[k]}^{-1})_{00}$. Since $0 \notin \operatorname{supp}\nu_g$, the far end carries a spring. Equivalently, a massless segment of length $\tilde l_{n-1} = \|e_{n-1}\|_A^2/\|g_0^0\|^2$ ends at a fixed point $\tilde L = \|e_0\|_A^2/\|g_0^0\|^2$. The remaining length is the energy, $\tilde L - \tilde x_k = \|e_k\|_A^2/\|g_0^0\|^2$. The influence matrix of this string, free at the launched end, is the one-sided form [21, (5.9)], [27] of the matrix of Remark 3.2:

$$\sqrt{\tilde m_i \tilde m_j}\,(\tilde L - \tilde x_{\max(i,j)}) = |\mathcal{G}_{ij}|.$$

In this frame the Green coordinate is the logarithmic remaining length, $e^{-2\mathsf{X}} = 1 - \tilde x/\tilde L$. The bead weight is mass times remaining length, $\mathsf{d}_k = \tilde m_k(\tilde L - \tilde x_k)$. CG is the Stieltjes string of its gradient measure at the extrapolation point. Its positions are Gauss quadratures, its gaps are energy decrements, and its remaining length is the error.

This string is (S1) of Remark D.7. The string of Lemma D.15 for ν, made static at $\lambda = 0$, is (S3). Their measures $\nu_g \propto \lambda\nu$ and ν are related by a Christoffel transform; the change of static frame above relates (S2) to (S3). At every bead, however, the mass accumulated by (S3) is reciprocal to the fraction of length remaining in (S1):

$$\tilde M_m^{(3)} \cdot \frac{\tilde L - \tilde x_m}{\tilde L} = 1, \qquad \tilde M_m^{(3)} = K_m = \frac{1}{\lambda_m(0;\nu_\lambda)} = e^{2\mathsf{X}_m},$$

because both factors are expressed through $\lambda_m(0;\nu_\lambda)$. The two strings carry one Green coordinate X: (S3) reads it as accumulated mass, and (S1) reads it as remaining length. This is the exact bridge between the string of [21] and the strings of Theorem D.8. The continuum extremals solve different variational problems. Their explicit correspondences use this coordinate, the common clock, and the persymmetric final step of Remark D.3.

Proposition D.17 (Top solution)**.** *Let $\Lambda > 0$, and define $\mathsf{a}_0 = 1$, $\mathsf{b}_{-1} = 0$ and*

$$\mathsf{b}_k = \Lambda m_k \mathsf{a}_k - \mathsf{b}_{k-1}, \qquad \mathsf{a}_{k+1} = l_k \mathsf{b}_k - \mathsf{a}_k \qquad (0 \le k \le N-2).$$

Then $\|H\| < \Lambda$ iff $\mathsf{a}_k > 0$ $(k \le N-1)$, $\mathsf{b}_k > 0$ $(k \le N-2)$ and $\Lambda m_{N-1}\mathsf{a}_{N-1} > \mathsf{b}_{N-2} + \varrho\,\mathsf{a}_{N-1}$.

Proof. For $Z = \Lambda\mathsf{M} - \mathsf{SGS}$, $\mathsf{S} = \mathrm{diag}((-1)^k)$, let D_k be its leading determinants, $D_0 = 1$. The tridiagonal determinant recurrence gives $\mathsf{a}_k = (\prod_{i<k} l_i) D_k$ and the terminal expression $(\prod_{i<N-1} l_i) D_N$. Sylvester's criterion proves the equivalence; $\mathsf{b}_k = (\mathsf{a}_k + \mathsf{a}_{k+1})/l_k$ gives the remaining signs. □

For $\Lambda > \|H\|$ define the top impedances $c_k = \mathsf{b}_k/(\sqrt{\Lambda}\,\mathsf{a}_k)$ and $c'_{k+1} = \mathsf{b}_k/(\sqrt{\Lambda}\,\mathsf{a}_{k+1})$, $0 \le k \le N-2$.

Lemma D.18 (Top identities)**.** *(a) $\sqrt{\Lambda}\,l_j = \frac{1}{c_j} + \frac{1}{c'_{j+1}}$ for $0 \le j \le N-2$, and $\sqrt{\Lambda}\,m_j = c_j + c'_j$ for $1 \le j \le N-2$.*
(b) $\Lambda\, m_{j+1} l_j > 1$ for $0 \le j \le N-2$.
(c) $\sqrt{\Lambda}\, m_{N-1} > c'_{N-1}$.

Proof. Divide the two top recurrences by $\mathsf{b}_j/\sqrt{\Lambda}$ and $\sqrt{\Lambda}\mathsf{a}_j$, respectively, to obtain (a); divide the terminal inequality by $\sqrt{\Lambda}\mathsf{a}_{N-1}$ for (c). For (b), the Rayleigh quotient at e_{j+1} is at least $(l_j m_{j+1})^{-1}$ and is strictly below Λ. □

Put $B_\gamma = \gamma$ for $\gamma \in (0,1)$ and $B_1 = \frac{1}{2}$.

Lemma D.19 (Monotonicity and log-Lipschitz bound)**.** *For $\gamma \in (0,1]$ and $r \ge 1$,*

$$0 \le \frac{d\log t_*}{d\log r} = \rho_\gamma(r) := \frac{\sigma_\gamma(r) - 1 - \gamma r}{\sigma_\gamma(r) + 1 - r} \le B_\gamma$$

(at $r = U_{\mathrm{F}}$ read $\rho_\gamma = r\sigma'_\gamma/(2\sigma_\gamma)$). In particular t_* is nondecreasing on $[1,\infty)$. For $\gamma = 1$, moreover, $U + 1 < \sigma_1(U) < U + 3$ for $U \ge 1$.

Proof. The graph equation gives $\rho_\gamma = U\sigma'_\gamma/(2\sigma_\gamma)$. Above U_{F}, the bounds $s_{\mathrm{F}} \le \sigma_\gamma \le 1 + \gamma U$ give $0 \le \rho_\gamma \le \gamma$ algebraically. Below U_{F}, the same upper bound follows from $\sigma_\gamma < s_{\mathrm{F}}$; positivity follows by backward invariance of the triangle $1 + \gamma U < s < s_{\mathrm{F}}$, which contains the stable branch. Indeed its oblique edge has $(s - 1 - \gamma U)' < 0$, its vertical edge has $s' > 0$, and the stable tangent enters it. Continuity covers U_{F}.

For $\gamma = 1$, put $e = s - U > 1$. Then $e' = U(e-3) + 2e(e-1)$ and $U' = U(e+1)$. If $e \ge 4$, the ratio $w = e/U$ satisfies $dw/dU \ge (1+w)/(5U)$ and eventually forces $s > 4U$, contradicting separation. Thus $e < 4$. If $e \ge 3$, then $e' \ge 12$ until reaching 4, again impossible. Hence $1 < e < 3$ and $0 < \rho_1 = (e-1)/(e+1) < 1/2$. □

Let ν have $N \ge 2$ atoms in $[0, \kappa - 1]$, with Stieltjes string (m, l, ϱ) from Lemma D.15(b), and states $z_k = (\varphi_k, p_k, K_k)$ at $\zeta = -1$. Put $r_k = \varphi_k p_k / K_k \ge 1$, $g_k = K_{k+1}^{-\gamma} = \lambda_{k+1}(0;\nu_\lambda)^\gamma$ (nonincreasing, $g_0 \le 1$), and, for $0 \le j \le N-2$,

$$\delta_j = \log\frac{\varphi_{j+1}}{\varphi_j}, \qquad q_j = \log\frac{p_{j+1}}{p_j}, \qquad k_j = \log\frac{K_{j+1}}{K_j}, \qquad \Delta_j = W(z_j) - W(z_{j+1}).$$

Lemma D.20 (A priori bounds)**.** *Let $\Lambda > \kappa - 1$. Then (a) $1 \le r_j \le 1 + \Lambda$ for every j; (b) $\sum_j g_j k_j \le \frac{1}{\gamma}$.*

Proof. The update $r_{j+1} = \varphi'(p_j + m\varphi')/(K_j + m\varphi'^2)$ decreases with m, because $\varphi' p_j \ge K_j$. Substituting $m \ge 1/(\Lambda l_j)$ and $l_j p_j \le \varphi'$ gives $r_{j+1} \le 1 + \Lambda$. Also $g_j k_j \le \int_{K_j}^{K_{j+1}} x^{-\gamma-1} dx$; summation gives $1/\gamma$. □

Lemma D.21 (One step, exact averages)**.** *Let* $\tau_j = t_*(r_j)$, $\mathcal{D}_j = \int_0^{l_j} \frac{p_j\,du}{\varphi(u)t_*(r(u))}$ *along the drift* j *(*$\varphi(u) = \varphi_j + up_j$, $r(u) = \varphi(u)p_j/K_j$*),* $\mathcal{I}_j = \int_0^{m_{j+1}} \frac{\varphi_{j+1}t_*(r_u)}{p_u}du$ *along the impulse* j *(as in Lemma D.13),* $A_j^D = \mathcal{D}_j/l_j$ *and* $A_j^I = m_{j+1}/\mathcal{I}_j$*. Let* $E_+(x) = \frac{x}{1-e^{-x}}$, $E_-(x) = \frac{x}{e^x-1}$*, with* $E_+(0) = E_-(0) = 1$ *by continuity, and* $B = B_\gamma$*. Then* $0 \le q_j \le k_j$, $\log\frac{r_{j+1}}{r_j} = \delta_j + q_j - k_j$,

$$\Delta_j \ge \tfrac{1}{2}(K_j^{-\gamma}A_j^D l_j + g_j m_{j+1}/A_j^I),$$

$$\frac{A_j^D}{A_j^I} \ge e^{-B(k_j-q_j)}E_+(\delta_j)E_-(q_j), \qquad \frac{A_j^D}{A_{j-1}^I} \ge e^{-B\delta_j}E_-(\delta_j)E_+(q_{j-1}) \quad (j \ge 1).$$

Proof. The update gives $K_{j+1}/K_j = 1 + (e^{q_j}-1)r_j e^{\delta_j} \ge e^{q_j}$ and the identity for r. Apply Lemma D.13 separately to drift and impulse, using $K \le K_{j+1}$ on the latter. For the ratios, Lemma D.19 bounds t_* above on drift j by $\min(\tau_j e^{B\delta_j}, \tau_{j+1}e^{B(k_j-q_j)})$, and below on impulses $j, j-1$ by τ_{j+1}, τ_j. The remaining integrals are exactly

$$\frac{1}{l_j}\int_0^{l_j} \frac{p_j}{\varphi_j + up_j}\,du = \frac{p_j}{\varphi_j}E_-(\delta_j), \quad \frac{1}{m_{j+1}}\int_0^{m_{j+1}} \frac{\varphi_{j+1}}{p_j + u\varphi_{j+1}}\,du = \frac{\varphi_j}{p_j}e^{\delta_j}E_-(q_j).$$

For impulse $j-1$ the average is $(\varphi_j/p_j)E_+(q_{j-1})$. Multiplication and $e^xE_-(x) = E_+(x)$ prove both ratios. □

Upper bound in Theorem D.9. Gauss quadrature reduces a partial sum to an N-atom measure. Fix $\Lambda > \kappa - 1$ and $S_N = \sum g_j$. The top identities split each drift/impulse loss into adjacent site and bond terms. AM–GM, using $g_{j-1} \ge g_j$ and the terminal top inequality, gives

$$2\sqrt{\Lambda}\sum \Delta_j \ge 2\sum_{j=0}^{N-2} g_j\sqrt{A_j^D/A_j^I} + 2\sum_{j=1}^{N-2} g_j\sqrt{A_j^D/A_{j-1}^I}.$$

Put $\Sigma_z = \sum g_j z_j$. Lemma D.21, $E_+ \ge 1$, $E_- \ge e^{-x}$ and $\sqrt{z} \ge 1 + \frac{1}{2}\log z$ imply

$$4S_N \le 2\sqrt{\Lambda}\sum \Delta_j + 2g_0 + (1+B)\Sigma_\delta + B\Sigma_k + (1-B)\Sigma_q.$$

Abel summation of $\delta_j = \log(r_{j+1}/r_j) + k_j - q_j$ gives $\Sigma_\delta \le \log(1+\Lambda) + \Sigma_k - \Sigma_q$. Now $\sum_j \Delta_j \le a_{\infty,\gamma}$, $g_0 \le 1$, and Lemma D.20 gives $0 \le \Sigma_q \le \Sigma_k \le 1/\gamma$. Substitute, divide by four and let $\Lambda \downarrow \kappa - 1$. □

Lower bound in Theorem D.9. Truncate the positive analytic optimizer at type $Y < y_{\max}$. On the mesh $y_k = kh$, $N = \lfloor Y/h \rfloor + 1$, choose $m_0 = 1$, $m_k = h\varsigma(y_k)$, $l_k = h/\varsigma(y_k + h/2)$, $\varrho = 0$. Writing $L_\varsigma = \max|(\log\varsigma)'|$, Gershgorin gives $\|H\| \le 4e^{L_\varsigma h/2}h^{-2}$ for small h: interior rows have four terms bounded by $e^{L_\varsigma h/2}h^{-2}$, endpoint rows fewer, and the initial-atom terms are $O(h^{-3/2})$. Choose h with this bound equal to $\kappa - 1$; then $\theta/h \to 1$. The drift/impulse recurrence is a consistent first-order discretization of $\varphi_y = p/\varsigma$, $p_y = \varsigma\varphi$ on $[0, Y]$. Its local error is $O(h^2)$ and step Lipschitz constant $1 + O(h)$; discrete Gronwall [83, §II.3] gives uniform $O(h)$ state and energy errors. Therefore

$$\theta\Gamma^*_{\infty,\gamma} \ge (\theta/h)h\sum_{m=1}^{N-1} K_m^{-\gamma} \longrightarrow \int_0^Y K^{-\gamma}dy.$$

Let $Y \uparrow y_{\max}$. When $n\theta \to \infty$, eventually $n \ge N$, giving the same recovery for finite sums. □

Conclusion of Theorem D.9. The finite bounds follow from the upper-bound argument above. Put $x = (\kappa-1)^{-1/2}$, so that $\theta\sqrt{\kappa-1}/2 = \operatorname{arsinh}(x)/x \le 1$. Multiplying by θ and applying the lower-bound argument gives the limit, also for finite sums with $n\theta \to \infty$. Equation (45) gives the $\sqrt{\kappa}$ equivalent.

For the Chebyshev comparison, Lemma D.5 gives $\sum_{m\ge1}\operatorname{sech}(m\theta) > \pi/(2\theta) - 1/2 \ge (\pi/4)\sqrt{\kappa-1} - 1/2$. A rational polynomial barrier for the half-gamma separating graph gives the coarse enclosure

$a \le 333/250$. Hence $\pi > 3.1415$ implies $\pi/4 - a/2 \ge 191/1600$. Put $x = \sqrt{\kappa - 1}$. For $\kappa \ge 1600$, $x > 1999/50$, and the gap is bounded below by $F(x) = \frac{191}{1600}x - \frac{3}{8}\log(x^2+1) - 2$. The elementary bound $\log 1600 \le 739/100$ gives $F(1999/50) \ge 109/80000 > 0$, and $F'(x) > 0$ for $x \ge 1999/50$. This proves the comparison without a high-precision value of a. □

Proof of Theorem 5.1. Theorem D.9 gives the upper bound and the finite/infinite limits, with the value characterized by Theorem D.8. The shooting characterization and numerical brackets are Propositions D.11, D.12 and D.14. Lemma D.4 identifies the finite supremum with actual n-atom CG references. □

D.4. The spectral tail

We use Krein's principal spectral measure, with the natural right-end convention fixed through its characteristic function [77, §1.1, (1.2)–(1.5)]. On a finite endpoint with finite total mass, continue M constantly to infinity; this prescribes a free right end. An infinite-mass endpoint remains fixed. Let $\varphi(\cdot, z), \psi(\cdot, z)$ solve $du' = -zu\, dM$ with

$$\varphi(0, z) = \psi'(0-, z) = 1, \qquad \varphi'(0-, z) = \psi(0, z) = 0.$$

The characteristic function is $W_M(z) = \lim_{x\to L} \psi(x, z)/\varphi(x, z)$ at the resulting right endpoint L. Krein's representation gives its principal measure [76]; the unit left atom gives the normalization

$$h_M(t) := W_M(-t) = \int_{[0,\infty)} \frac{d\nu_M(\xi)}{\xi + t}, \qquad \nu_M([0, \infty)) = 1.$$

The associated Dirichlet principle is

$$h_M(t)^{-1} = \min_{u(0)=1} \left\{ Q_t(u) := \int |u'|^2 dx + t \int |u|^2 dM \right\}, \tag{54}$$

over locally absolutely continuous finite-energy functions, with the same right-end convention and minimum-energy interpolation on mass-free intervals. Indeed the normalized physical solution is $u = \varphi(\cdot, -t) - \psi(\cdot, -t)/h_M(t)$. Green's identity gives $Q_t(u) = t - u'(0+) = 1/h_M(t)$; a trace-zero variation has zero cross term and nonnegative energy. This identifies the characteristic measure with the canonical energy realization. For the optimizer it is the same ν^* as in Theorem D.8, and $\rho_0 = M^{*\prime}(0+) = a^{-2}$.

Proposition D.22 (Tail without rate, general strings)**.** *Let (ℓ, M) be a string with $M(0) = 1$, $\ell + M(\ell-) = \infty$, and $M(x) = 1 + \int_0^x \rho$ near 0 with $\operatorname{ess\,lim}_{x\downarrow 0} \rho = \rho_0 \in (0, \infty)$, and let ν be its principal spectral measure defined above. Then, as $x \to \infty$,*

$$\int_{(0,x]} \xi\, d\nu(\xi) \sim \frac{2\sqrt{\rho_0}}{\pi}\, x^{1/2}, \qquad \nu\big((x, \infty)\big) \sim \frac{2\sqrt{\rho_0}}{\pi}\, x^{-1/2}.$$

Proof. Use the Dirichlet principle (54), with $h = h_M$. Choose δ with $0 < \rho_1 \le \rho \le \rho_2$ a.e. on $(0, \delta)$. Completing the square with the linear weight $\beta(1 - x/\delta)$ gives the lower energy bound. The compact physical trial $u(x) = (e^{-kx} - e^{-k\delta})/(1 - e^{-k\delta})$ on $[0, \delta]$, zero afterward, gives the upper bound. Precisely,

$$\frac{1}{t + k_t/(1 - e^{-k_t\delta})^2} \le h(t) \le \frac{1}{t + \beta_t}, \qquad k_t = \sqrt{t\rho_2}, \quad \beta_t^2 + \beta_t/\delta = t\rho_1.$$

Let $t \to \infty$, then $\rho_1 \uparrow \rho_0$, $\rho_2 \downarrow \rho_0$. Because ν is the principal probability measure, this yields $\int \frac{\xi\, d\nu(\xi)}{\xi + t} = 1 - th(t) \sim \sqrt{\rho_0}\, t^{-1/2}$. Apply Karamata's Stieltjes theorem [84, Thm. 1.7.4] to the nondecreasing $U(x) = \int_{(0,x]} \xi\, d\nu(\xi)$; its integrability condition is $\int (1 + \xi)^{-1} dU \le 1$. The constant is $\sqrt{\rho_0}/(\Gamma(1/2)\Gamma(3/2)) = 2\sqrt{\rho_0}/\pi$. Finally Tonelli gives $\nu((x, \infty)) = \int_x^\infty U(u) u^{-2} du - U(x)/x$, proving the same tail constant. □

For M^*, $\rho_0 = 1/a^2$, so Proposition D.22 gives $\nu^*((\zeta, \infty)) \sim 2(\pi a)^{-1} \zeta^{-1/2}$, proving Theorem D.8.

Remark D.23 (The two ends of the spectrum). The optimizer is read at both ends of its spectrum. At zero frequency, the free right end puts the atom $\nu^*(\{0\}) = 1/M^*(\infty) = a^{-2}$ [77, (1.14)]. At infinite frequency, the tail of ν^* is fixed by the type density at the launched end (Proposition D.22). For comparison, a string on a finite interval with continuous positive density ρ, a free left end and a fixed right end has $\sqrt{z_k} \sim \pi k / \int \sqrt{\rho}$ [78], [79, Thm. 5.1]. This global eigenvalue asymptotic uses total type; Proposition D.22 uses the local density at the launched end and applies to the infinite-length optimizer. The companion paper uses the same two ends. Its trace is a mass, $\int (1-\xi^2)\rho = 2\sum_k 1/z_k$, fixed by the spectrum [21, Rem. 2.3]. Its edge asymptotics $j_{\alpha,k} \sim \pi k$ agree with root mass one, the step count. In both papers the zero-frequency end carries the normalization at the extrapolation point, and the infinite-frequency end carries the number of steps.

E. Function expansions and convergence rates

Here n is the ambient dimension; $r_k = \|x_k - x_*\|_{H_*}$ and r_c denotes a cycle-start error. The vectors e_l inside CG energy norms remain quadratic-reference errors. Use canonical coordinates as in Section 6, so $x_* = 0$, $H_* = I$ in transformed formulas. Unadorned norms of transformed vectors and operators are Euclidean, hence canonical in the original coordinates; κ retains the original Euclidean condition number, and $\theta = \operatorname{arccosh}((\kappa+1)/(\kappa-1))$. Values at $\kappa = 1$ are understood by limits. Here L_H is the canonical Hessian Lipschitz constant. Pair operators $\bar{A}_i$ and radial operators A_k^* are the Hessian averages of Definition 2.1. For a step matrix V, $N_V = \operatorname{diag}(\|v_i\|)$ and $H_*^{1/2} V = \hat{V} N_V$, where $\hat{V}$ has unit columns.

E.1. Local expansions and transfer

Lemma E.1 (Secants near a nondegenerate minimizer). *Let $\widehat{H}(y) = H_*^{-1/2} \nabla^2 f(y) H_*^{-1/2}$. In the self-concordant cases, let the domain $D \ni x_*$ be open and convex; all balls used below are assumed contained in D, and sublevel sets are taken in D.*

(i) If f is standard self-concordant and $\|y - x_*\| = t < 1$, then $(1-t)^2 I \preceq \widehat{H}(y) \preceq (1-t)^{-2} I$ [85, Thm. 5.1.7].

(ii) If f is standard self-concordant and $\|x - x_*\| = t < 1$, the sublevel set $\{f \le f(x)\}$ lies in the ball of radius $\varrho(t)$ about x_*, where $\omega(\varrho(t)) = \omega_*(t)$, $\omega(s) = s - \ln(1+s)$, $\omega_*(t) = -t - \ln(1-t)$. Moreover $\varrho(t) = t + \frac{2}{3}t^2 + O(t^3)$ and $\varrho(t) \le 2t$ for $t \le \frac{1}{2}$.

(iii) Let the Hessian be L_H-Lipschitz on a ball $B(x_*, R)$ in canonical coordinates, and let the segments $[a, a+u]$, $[b, b+v]$ lie in it, with secants A, A'. Then

$$\|\widehat{A} - \widehat{A}'\| \le \tfrac{L_H}{2}(\|a-b\| + \|a+u-b-v\|), \qquad \|\widehat{A} - I\| \le \tfrac{L_H}{2}(\|a - x_*\| + \|a+u-x_*\|).$$

If f is standard self-concordant and $R < 1$, the Hessian is L_H-Lipschitz on $B(x_*, R)$ with $L_H = 2(1-R)^{-3}$.

(iv) If f is C^3 near x_*, the secant of a segment contained in $B(x_*, R)$, with midpoint $\bar{x}$, satisfies $\widehat{A} = I + \mathsf{T}[\bar{x} - x_*] + o(R)$ as $R \to 0$, uniformly over such segments. If f is standard self-concordant, $\|\mathsf{T}\|_\sigma \le 2$.

Proof. Parts (i)–(ii) are the Hessian and function comparisons of [85, Thm. 5.1.7]: invert ω in $\omega(\|y - x_*\|) \le \omega_*(t)$ and expand at zero. The inequality $\varrho(t) \le 2t$ follows from $[\omega(2t) - \omega_*(t)]' \ge 0$ on $[0, 1/2]$. Integrating the Hessian bound along segments proves (iii); its self-concordant constant follows from [85, Lem. 5.1.2]. Averaging the first-order Hessian expansion proves (iv); Banach's identity [86, 87] gives the tensor norm bound. □

Lemma E.2 (The exact line search). *Let the Hessian be L_H-Lipschitz on $B(x_*, r_0)$, and let $r_1 \le \min(r_0, \frac{1}{2L_H})$. Let the line search return the minimizer x' of f on the chord $(x + \mathbb{R}d) \cap \bar{B}(x_*, r_1)$. If $e = \|x - x_*\| < r_1/2$, then x' exists and is unique, $\|x' - x_*\| \le \min(\sqrt{3}, 1 + 2L_H e)\, e$, x' is interior to the ball, $d^\top \nabla f(x') = 0$, and $\|x' - x\| \le 3e$.*

Proof. In canonical coordinates $\nabla^2 f \succeq I/2$ and $|f(y)-f(0)-\|y\|^2/2| \le L_H\|y\|^3/6 \le \|y\|^2/4$. Compactness and strict convexity give a unique chord minimizer, and comparison with $f(x)$ gives $\|x'\| \le \sqrt{3}e < r_1$, hence stationarity and the step bound. The sharper comparison $\|x'\|^2 \le e^2 + L_H(1+3\sqrt{3})e^3/3$ gives $\|x'\| \le (1+2L_He)e$. □

Lemma E.3 (R-order of secant recursions)**.** *Let $r_k \to 0$. (a) If $r_k > 0$ and $r_{k+1} \le K r_{k-p} r_{k-q}$ for all large k, with integers $0 \le q < p$, then the R-order is at least the unique positive root of $t^{p+1} = t^{p-q} + 1$, which exceeds 1 [88]. The cases $(p,q) = (n-1, n-2)$, $(n-1, 0)$, $(n, 0)$ give $\tau_{\mathrm{FC}}(n)$, $\tau_1(n)$, $\tau_1(n+1)$. (b) If $r_k \ge 0$ and $r_{k+1} \le K r_k(r_{k-m+1} + r_{k-m}^2)$ for all large k, with $m \ge 1$, then the R-order is at least $\tau_1(m)$.*

Proof. For (a), put $E_k = -\log(Kr_k)$; then $E_{k+1} \ge E_{k-p} + E_{k-q}$. The characteristic identity propagates $E_j \ge c\tau^j$ from any sufficiently late initial block [55, 88]. For (b), scale $\epsilon_k = \max(1, 2K, \sqrt{2K}) r_k$ and induct on $\epsilon_k \le q^{\tau^k}$. Indeed $\tau^k + \tau^{k-m+1} = \tau^{k+1}$ and $\tau^k + 2\tau^{k-m} \ge \tau^{k+1}$ because $\tau \le 2$. Zero errors obey the same induction. □

The next lemma controls the gradient at a point where the full-conjugacy direction fails to be a descent direction or produces a short step. In canonical coordinates the direction $d = -g + S\beta$ uses the Euclidean identification of gradients with vectors, which is the operator H_*: $H_*^{1/2} d = -H_* \hat{g} + H_*^{1/2} S\beta$. This is where κ enters. Here L and μ are the extreme eigenvalues of H_*.

Lemma E.4 (Gradient at a non-descent direction or a short step)**.** *Let $S = [s_0, \dots, s_{k-1}]$, $k \le n$, have normalized Gram matrix $G = \hat{S}^\top \hat{S} \succeq \frac{3}{4} I$; let $y_i = A_i s_i$ with symmetric A_i, $\|\widehat{A}_i - I\| \le \frac{\varepsilon}{2}$, $\sqrt{n}\,\varepsilon \le \sqrt{3}/(8\kappa)$; let $Y^\top S$ be invertible and $d = -g + S\beta$ with $Y^\top d = 0$. Let P be the orthogonal projector onto* $\operatorname{span}\{\hat{\sigma}_i\}$ *and $a = P\hat{g}$. Then $\|a\| \le \frac{2}{\sqrt{3}} \|\hat{S}^\top \hat{g}\|$, and:*

(a) if $g^\top d \ge 0$, then $\|\hat{g}\| \le 2(1+\sqrt{\kappa})\|a\|$;

(b) if $g^\top d < 0$, $\alpha > 0$ and $-g^\top d = \alpha\, d^\top A d$ for a symmetric A with $\|\widehat{A} - I\| \le \frac{\varepsilon}{2}$, then $\|\hat{g}\| \le 4\kappa(\|a\| + \alpha \|H_*^{1/2} d\|_2)$.

Proof. Write $a = P\hat{g}$, $b = (I-P)\hat{g}$, $\hat{d} = H_*^{1/2} d$, and $B = (b^\top H_* b)^{1/2}$. The Gram bound gives $\|a\| \le 2\|\hat{S}^\top \hat{g}\|/\sqrt{3}$. Approximate conjugacy gives

$$\|P\hat{d}\| \le \epsilon_2 \|\hat{d}\|, \quad \epsilon_2 = \sqrt{k/3}\,\epsilon \le (8\kappa)^{-1}, \quad (I-P)\hat{d} = -(I-P)H_*\hat{g}, \quad \|\hat{d}\| \le 2L\|\hat{g}\|.$$

Consequently

$$B^2 \le -g^\top d + \sqrt{L}\, B\|a\| + \frac{\mu}{4}\|a\|^2 + \frac{\sqrt{\mu}}{4} B\|a\|.$$

If $g^\top d \ge 0$, solving this quadratic yields $B \le (\sqrt{L} + 3\sqrt{\mu}/4)\|a\|$ and proves (a). For (b), put $X = \alpha\|\hat{d}\|$. Then $-g^\top d \le (9/4) L X(\|a\| + B/\sqrt{\mu})$. Substitute in the same quadratic, use $B^2 \le pB + q \Rightarrow B \le p + \sqrt{q}$ and $\sqrt{xy} \le (x+y)/2$; this gives $\|b\| \le 3\kappa X + (7\sqrt{\kappa}/4 + 3/4)\|a\|$, hence (b). □

Lemma E.5 (From a finite cycle to a function)**.** *Consider a fixed finite cycle formed from the smooth homogeneous operations of Lemma B.1 and local exact line searches. Let its normalized quadratic references range over a compact nondegenerate set. Start at $x_* + eu$, $\|u\| = 1$, write $R(u) = (H_*, H_* u)$ for the normalized quadratic reference, and let f have a Lipschitz Hessian near x_*.*

All nodes lie in $B(x_*, Ce)$, the canonical operator increments are $O(e)$, and the nodes differ by $O(e^2)$ from those of the quadratic reference. If $f \in C^3$, an increment between two segment secants has the form

$$\Theta_l = e\, \mathsf{T}[\eta_l(u)] + o(e),$$

where η_l is the difference of their normalized reference midpoints; the same rule includes correction and output segments. Suppose the output gradient vanishes to degree $d-1$ in the increments and has leading homogeneous polynomial P_d with a uniform remainder $O(\|\Theta\|^{d+1})$. Then

$$H_*^{-1/2} g_{\mathrm{out}} = e^{d+1} P_d(R(u); (\mathsf{T}[\eta_l(u)])_l) + o(e^{d+1}).$$

If $D^3 f$ is Lipschitz and the coefficient P_d is locally Lipschitz in the reference, the last remainder is $O(e^{d+2})$.

Proof. Uniformly on bounded normalized sets, $e^{-1}\nabla f(x_* + ez) = H_* z + O(e)$. Induction through the finite operations, with denominators bounded away from zero, keeps the normalized nodes bounded and changes them by $O(e)$; local exact searches obey the same conclusion by strict convexity and the implicit-function theorem. Thus actual midpoint errors are $O(e^2)$. Lemma E.1 gives the increment formula, with $O(e^2)$ error when $D^3 f$ is Lipschitz. The base and normalized initial gradient change by $O(e)$. Substitute in P_d, use its degree and Lemma B.1; continuity gives the little-o statement, and Lipschitz continuity the stated big-O improvement. □

E.2. Local expansions and tensor constraints

Write $m_{\mathsf{T}} = \|\mathsf{T}\|_\sigma/2$; standard self-concordance gives $m_{\mathsf{T}} \le 1$. Exact searches are the chord minimizers of Lemma E.2.

For a cycle from x_0 with $r_c = \|x_0 - x_*\|$, take the first-segment base A_0 and its canonical CG data as in Section 3. Put $\eta = \hat{s}_{n-1}$, $\sigma_l = \sqrt{c_l}\hat{s}_l$ and

$$\xi_l = \bar{x}_{l+1} - \bar{x}_l = \tfrac{1}{2}(\sigma_l + \sigma_{l+1}), \qquad \zeta_l = \tfrac{1}{2}\sigma_l + \sum_{j>l} \sigma_j,$$

where $\bar{x}_l$ are reference segment midpoints.

Assumption E.6 (Uniform nondegeneracy)**.** The normalized pairs $(A_0/\|A_0\|, g_0/\|g_0\|)$ of the cycles considered lie in a fixed compact set of references whose measure has n atoms.

Proposition E.7 (Compatible function increments)**.** *Let $f \in C^3$, $n \ge 2$, and assume E.6. Uniformly over the normalized reference set,*

$$\widehat{E}_l = \mathsf{T}[\xi_l] + o(r_c), \qquad \widehat{E}' = \mathsf{T}[\zeta_{n-1}] + o(r_c).$$

Substitution in Theorems 3.1 and 3.3 gives the FC and RFC leading gradients, respectively of degrees 2 and 3 in r_c. More generally, a homogeneous degree-d operator response gives a degree-$(d+1)$ function response by Lemma E.5.

Put $N_i = \sum_{l=i}^{n-2} \xi_l \otimes v_l$ and define $\Gamma_{\mathsf{T}} = \sup_{\|\mathsf{T}\|_\sigma \le 2} \|\sum_i \hat{s}_i \mathsf{T}(\hat{s}_i, N_i)\|$ and $\Gamma_{\mathrm{op}} = 2\sum_l \|\xi_l\|\|v_l\|$.

Proposition E.8 (Tensor amplification)**.** *The compatible-tensor coefficient satisfies*

$$\Gamma_{\mathsf{T}} = 2 \max_{\|u\|=1} \Big\| \operatorname{Sym} \sum_i u_i \hat{s}_i \otimes \operatorname{sym} N_i \Big\|_*, \qquad \sqrt{c_0}\|e_1\|_A \le \Gamma_{\mathsf{T}} \le \Gamma_{\mathrm{op}} \le \|e_0\|_A \|e_1\|_A + (n-2)\|e_1\|_A^2.$$

Here $u_i = \langle u, \hat{s}_i \rangle$ and $\|\cdot\|_*$ is the nuclear norm of 3-tensors [87].

For flat profiles $\Gamma_{\mathsf{T}} \le 10\, cn \log_2(4n)$ while $\Gamma_{\mathrm{op}} \ge \frac{2\sqrt{2}}{3} c(n-1)^{3/2}$ (Proposition E.12): one tensor cannot align all its slices with the remaining errors, and the operator model exceeds the compatible-tensor coefficient by at least order $\sqrt{n}/\log n$.

Let $\Pi = \operatorname{span}\{\hat{s}_0, \hat{s}_{n-1}\}$, $r_{\mathrm{H}} = \sqrt{c_{n-1}/c_0}$, and let $\mathcal{P}_\delta$ consist of symmetric tensors supported on Π with norm at most $2(1-\delta)$. The planar model uses $A_k = A_0 + \mathsf{T}[\bar{x}_k - \bar{x}_0]$, $B = A_0 + \mathsf{T}[\zeta_0]$, with unknown base $A_0 = I + \beta \hat{s}_{n-1}\hat{s}_{n-1}^\top$, $|\beta| \le r_c$.

Theorem E.9 (Cubic realization of the lower bound)**.** *Let $n \ge 3$, $0 < \delta < 1$, and let u be an H_*-unit vector with an n-atom CG reference. For the harmonic cubic*

$$\mathsf{T}_\delta(h,h,h) = 2(1-\delta)\,\mathrm{Re}\,[e^{-i\pi/4}(\langle \hat{s}_0, h\rangle + i\langle \hat{s}_{n-1}, h\rangle)^3],$$
$$f_\delta(x) = \tfrac{1}{2}\|z\|^2 + \tfrac{1}{6}\mathsf{T}_\delta(z,z,z), \qquad z = H_*^{1/2}(x - x_*),$$

f_δ is self-concordant on the H_*-ball of radius $[1-(1-\delta)^{2/3}]/[2(1-\delta)]$. For starts $x_* + r_c u$, RFC has leading error

$$\tfrac{1}{2}(1-\delta)^2 c_0 \|e_{n-1}\|_A (1 + O(r_{\mathrm{H}})) = K(W, \omega')\|e_{n-1}\|_A(1 + O(r_{\mathrm{H}})).$$

Here $W = (1-\delta)\sqrt{c_0}(1+r_{\mathrm{H}})$ and $\omega' = (1-\delta)\sqrt{c_{n-1}}$ are the leading variation budgets. In the planar model, putting $B_\delta = (1-\delta)^2 c_0 \|e_{n-1}\|_A$,

$$\inf_{M \in \mathcal{M}_{\mathrm{CG}}^{(1)}} \sup_{\mathsf{T} \in \mathcal{P}_\delta,\, |\beta| \le r_c} \|g_n\|_{A^{-1}} \ge \tfrac{1}{2}(1+r_{\mathrm{H}})^{-2} B_\delta (1 - O(r_c)),$$
$$\sup_{\mathsf{T} \in \mathcal{P}_\delta,\, |\beta| \le r_c} \|g_n^{\mathrm{RFC}}\|_{A^{-1}} \le \tfrac{1}{2}(1+r_{\mathrm{H}})(1+3r_{\mathrm{H}}) B_\delta (1 + O(r_c)).$$

Proof of Proposition E.7. Lemma E.5 gives the midpoint increments and, with $d = 1, 2$, the two responses

$$H_*^{-1/2} g_n^{\mathrm{FC}} = \sum_l P_l \mathsf{T}[\xi_l] v_l + o(r_c^2),$$
$$H_*^{-1/2} g_n^{\mathrm{RFC}} = -\|e_{n-1}\|_A\, \mathcal{L}\hat{f} + o(r_c^3),$$
$$\mathcal{L} = \sum_l P_l \mathsf{T}[\xi_l] P_l + \mathsf{T}[\zeta_{n-1}], \qquad \hat{f} = \sum_l P_l \mathsf{T}[\xi_l]\eta.$$

Expand in the orthonormal step basis and use $\sum_{l=m}^{n-2} \xi_l = \zeta_m - \zeta_{n-1}$. The FC response becomes $\sum_i \hat{s}_i \mathsf{T}(\hat{s}_i, N_i)$; the RFC product is

$$\mathcal{L}\hat{f} = \sum_{a=0}^{n-1} \sum_{b=0}^{n-2} \hat{s}_a\, \mathsf{T}(\zeta_{\max(a,b)}, \hat{s}_a, \hat{s}_b) \mathsf{T}(\zeta_b - \zeta_{n-1}, \hat{s}_b, \eta). \tag{55}$$

Here $2\|\xi_l\| = \sqrt{c_l + c_{l+1}} \le \|e_l\|_A$ and $2\|\zeta_{n-1}\| = \|e_{n-1}\|_A$ by orthogonality. □

E.3. Tensor constraints and function-value information

Proposition E.10 (Gradient-indistinguishable cubics)**.** *Let $x_{i+1} = x_i + s_i$ $(0 \le i < p)$, S of full column rank, A symmetric, $b \in \mathbb{R}^n$, $S^\top \phi = \mathbf{u}_0$, and for $c \in \mathbb{R}$ let $f_c(x) = \frac{1}{2}x^\top A x - b^\top x + c(\frac{t^3}{6} - \frac{t^2}{4})$, $t = \phi^\top(x - x_0)$. Then $D^3 f_c \equiv c\,\phi^{\otimes 3}$ and $\nabla f_c(x_i) = \nabla f_0(x_i)$ for $0 \le i \le p$: all f_c produce the data $Y = AS$, $\mathsf{D} = 0$, $\overline{T} = G$ of the quadratic. For every correction $u = Sw$ the secant B_c of $[x_p, x_p + u]$ gives $S^\top B_c S - \overline{T} = \frac{c}{2}(1+w_0)\mathbf{u}_0\mathbf{u}_0^\top$: a rank-one drift in a fixed direction (Lemma E.11). Gradient data carry no information on c; the trapezoid defect of the oldest pair, $f_c(x_1) - f_c(x_0) - \frac{1}{2}(g_0 + g_1)^\top s_0 = -c/12$, determines it.*

Proof of Proposition E.10. For $\psi(t) = t^3/6 - t^2/4$, $\psi'(0) = \psi'(1) = 0$, $\psi''(t) = t - 1/2$. The sampled t-values are $0, 1, \ldots, 1$, hence all gradients agree. Midpoint evaluation of the affine Hessian gives $A_0 = A$, $A_i = A + (c/2)\phi\phi^\top$ for $i \ge 1$, and $B_c = A + (c/2)(1+w_0)\phi\phi^\top$. Compress by S. Direct endpoint evaluation gives $\delta_0^{\mathrm{tr}} = -c/12$. □

Lemma E.11 (The family of Proposition E.10)**.** *In Proposition E.10 the increments are $E_0 = \frac{c}{2}\phi\phi^\top$, $E_l = 0$ $(1 \le l \le p-2)$, $E_{p-1} = \frac{c}{2}w_0\phi\phi^\top$ $(p \ge 2)$: a rank-one drift in a fixed direction, monotone if $c, w_0 \ge 0$. Let moreover the steps be orthogonal and $\phi = s_0/\|s_0\|^2$, and let $\bar{x}_0, \bar{x}_B$ be the midpoints of*

$[x_0, x_1]$ *and* $[x_p, x_p{+}u]$. *For every function with affine Hessian,* $|s_0^\top (B{-}A_0)s_0| \le \|D^3 f\|_\sigma \|s_0\|^2 \|\bar{x}_B{-}\bar{x}_0\|$, *with the injective norm* $\|\cdot\|_\sigma$. *For* f_c *with* $c \neq 0$ *and* $\bar{x}_B \neq \bar{x}_0$,

$$\frac{|(S^\top B_c S - \overline{T})_{00}|}{\|D^3 f_c\|_\sigma \|s_0\|^2 \|\bar{x}_B - \bar{x}_0\|} = \frac{|\phi^\top (\bar{x}_B - \bar{x}_0)|}{\|\phi\| \, \|\bar{x}_B - \bar{x}_0\|},$$

the absolute cosine of the angle between s_0 *and* $\bar{x}_B - \bar{x}_0$.

Proof. Read the increments from the secants in Proposition E.10. For an affine Hessian, $s_0^\top (B{-}A_0)s_0 = D^3 f[\bar{x}_B - \bar{x}_0, s_0, s_0]$; Banach's symmetric tensor identity gives the bound. In the displayed family, $S^\top \phi = \mathbf{u}_0$, $\|D^3 f_c\|_\sigma = |c|\|\phi\|^3$ and $\|\phi\| = \|s_0\|^{-1}$. The numerator is $|c| \, |\phi^\top (\bar{x}_B - \bar{x}_0)|$; division gives the absolute cosine. Finally $\phi^\top (\bar{x}_B - \bar{x}_0) = (1 + w_0)/2$, recovering the same model error. □

Proof of Proposition E.8. Pair the FC response with a unit u and use symmetric spectral/nuclear duality [87]; the self-adjoint symmetrizer gives the displayed equality. Banach's identity gives $\|\mathsf{T}[\xi_l]\| \le 2\|\xi_l\|$; the preceding midpoint bounds give the upper estimates. For the lower bound take $\hat{v}_0 = v_0/\|v_0\|$ and $\mathsf{T}(h, h, h) = 2\,\mathrm{Im}(\langle \hat{s}_0, h\rangle + i\langle \hat{v}_0, h\rangle)^3$. It has norm at most 2. Only ξ_0 has an $\hat{s}_0$ component, equal to $\sqrt{c_0}/2$; the response component along $\hat{s}_0$ is therefore $\sqrt{c_0}\|e_1\|_A$. □

Proposition E.12 (Flat profile)**.** *If* $c_l \equiv c$ *for* $l = 0, \dots, n-1$, $n \ge 2$, *then* $\frac{\sqrt{3}}{2} c\,(n-1) \le \Gamma_{\mathsf{T}} \le 10\, c\, n \log_2(4n)$ *and* $\Gamma_{\mathrm{op}} = \sqrt{2}\, c \sum_{j=1}^{n-1} \sqrt{j} \ge \frac{2\sqrt{2}}{3} c (n-1)^{3/2}$.

Proof. Normalize $c = 1$, $\hat{s}_a = e_a$, $v_a = \mathbf{1}_{>a}$. Then $\mathrm{sym}\, N_a = (v_a v_a^\top + \mathrm{sym}(e_a \otimes v_a))/2$. The cubic $3\sqrt{3} h_0 (\hat{v}_0^\top h)^2$ has norm at most 2 and response component $\sqrt{3}(n-1)/2$ along e_0. For the upper bound, duality gives $\Gamma_{\mathsf{T}} \le \|X_1\|_* + \|X_2\|_*$, where

$$X_1 = \sum_a u_a e_a \otimes v_a \otimes v_a, \quad X_2 = \sum_a u_a e_a \otimes \mathrm{sym}(e_a \otimes v_a), \quad \|u\| = 1.$$

The trace-norm bound gives $\|X_2\|_* \le n/\sqrt{2}$. Let $2^{L-1} < n \le 2^L$ and decompose each prefix v_a into disjoint dyadic intervals. At levels $k \le k'$, the corresponding index sets are disjoint at each level and each smaller set meets at most one larger set. Cauchy–Schwarz bounds their total nuclear contribution by $2^{(L+k')/2}$. Summing the two level orders gives $\|X_1\|_* \le 2L2^L/(\sqrt{2} - 1)$, hence $10 n \log_2(4n)$. Finally $\|\xi_l\| = 1/\sqrt{2}$ and $\|v_l\| = \sqrt{n - 1 - l}$; summation gives Γ_{op}. □

Proof of Theorem E.9. Put $e = \hat{s}_0$, $h = \hat{s}_{n-1}$, $r = r_{\mathrm{H}}$, $k = (1 - \delta)\sqrt{c_0}$. Write $\mathsf{T} = 2(1 - \delta)\tau$, $t_j = \tau(e^{3-j}, h^j)$, where repeated arguments are understood. The planar slices are

$$\mathsf{T}[p] = 2(1 - \delta) \begin{pmatrix} p_0 t_0 + p_1 t_1 & p_0 t_1 + p_1 t_2 \\ p_0 t_1 + p_1 t_2 & p_0 t_2 + p_1 t_3 \end{pmatrix}. \tag{56}$$

The orthogonal middle steps drop out of (55), giving

$$\hat{f} = k(t_1 + r t_2) e, \qquad \mathcal{L}|_\Pi = k \begin{pmatrix} t_0 + 2 r t_1 & r t_2 \\ r t_2 & r t_3 \end{pmatrix}. \tag{57}$$

Bernstein–Szegő's inequality $p'^2 + 9p^2 \le 9\|\mathsf{T}\|_\sigma^2$ for the cubic restriction to a circle [89, 90] implies

$$|\mathsf{T}(e, e, e)\mathsf{T}(e, e, h)| \le \tfrac{1}{2}\|\mathsf{T}\|_\sigma^2. \tag{58}$$

The harmonic tensor $t = (1, 1, -1, -1)/\sqrt{2}$ attains this bound. Its model response has norm $k^2|1 - r|\sqrt{(1 + 2r)^2 + r^2}/2$, with budgets $W = k(1 + r)$, $\omega' = kr$. Its Hessian lower bound $[1 - 2(1 - \delta)\|z\|]I$ proves self-concordance on the stated ball.

For the lower bound, only the last probe is affected:

$$A_{n-1} h = h + k(t_1 + r t_2) e + [\beta + k(t_2 + r t_3)] h. \tag{59}$$

Set $v=(1+r)^{-1}$, $\psi=kv/\sqrt{2}$ and choose $t^{\pm}=v(\pm1,1\pm r,\mp1,-1)/\sqrt{2}$, $\beta_{\pm}=\psi(\pm1+r)$. The harmonic part has norm v, the added term norm at most vr, so $\mathsf{T}^{\pm}\in\mathcal{P}_{\delta}$. Both probes equal $h+\psi e$, and their post-search residual is $\psi\sqrt{c_{n-1}}e$. The final operators are $I+\bar{F}\pm\Delta$, where

$$\bar{F}=\psi\begin{pmatrix}2r & 1\\ 1 & -r\end{pmatrix},\qquad \Delta=\psi\begin{pmatrix}1+2r^2 & -r\\ -r & 0\end{pmatrix}.$$

Now $\|\Delta e\|\ge\psi$; Lemma A.1(d) gives the lower constant $\psi^2\sqrt{c_{n-1}}(1-O(r_c))$. For RFC apply Theorem 3.3, $K\le W^2/2+W\omega'$ and the $1+O(r_c)$ base normalization. □

The corrected model $\overline{T}_F$ and the splits are defined in Section 6. The next result gives its diagonal information, remainder and invariances.

Lemma E.13 (Trapezoid remainder)**.** *Let $b_i=a_i+s_i$ and $w_i=x-m_i-\lambda_i s_i$. If D^3f is L_3-Lipschitz on a convex set containing the segment and anchor, then*

$$\begin{aligned}(\overline{T}_F)_{ii}-s_i^{\top}\nabla^2 f(x)s_i &= -D^3f(b_i)[w_i,s_i,s_i]+R_i,\\ |R_i| &\le L_3\Big[(\tfrac{1}{12}+\tfrac{1}{2}|\lambda_i-\tfrac{1}{2}|)\|s_i\|^4+\tfrac{1}{2}\|x-b_i\|^2\|s_i\|^2\Big].\end{aligned}\tag{60}$$

The same bound holds with M_4 when $|D^4f[h,h,h,h]|\le M_4\|h\|^4$. On cubics $R_i=0$ and

$$\delta_i^{\mathrm{tr}}=-\mathsf{T}(s_i,s_i,s_i)/12,\qquad \overline{T}_{ii}=s_i^{\top}\nabla^2 f(m_i)s_i.$$

Thus the corrected diagonal is exact at x whenever $x-m_i=\lambda_i s_i$.

Proof. For $s=s_i$, $a=a_i$, $b=a+s$ and $T(t)=D^3f(a+ts)[s,s,s]$, two integrations by parts give $\delta_i^{\mathrm{tr}}=-\frac{1}{2}\int_0^1 t(1-t)T(t)\,dt$, $\qquad \overline{T}_{ii}-6\delta_i^{\mathrm{tr}}-s^{\top}\nabla^2f(b)s=-\int_0^1 t^2(1-t)T'(t)\,dt$. The second identity uses the a.e. derivative of the Lipschitz function T. Since $\int_0^1 t^2(1-t)\,dt=1/12$, it has absolute value at most $L_3\|s\|^4/12$. Also

$$\begin{aligned}|\delta_i^{\mathrm{tr}}+T(1)/12| &\le L_3\|s\|^4/24,\\ |s^{\top}(\nabla^2f(x)-\nabla^2f(b))s-D^3f(b)[x-b,s,s]| &\le \tfrac{1}{2}L_3\|x-b\|^2\|s\|^2.\end{aligned}$$

Write $\overline{T}_F=(\overline{T}-6\operatorname{diag}\delta^{\mathrm{tr}})-12\operatorname{diag}((\lambda_i-1/2)\delta_i^{\mathrm{tr}})$ and combine the bounds. The diagonal bound on D^4f gives its multilinear norm by Banach's theorem [86]. On cubics Taylor averaging gives both exact identities and zero remainder; on quadratics $\delta_i^{\mathrm{tr}}=0$. The constant $1/12$ at $x=b$, $\lambda=1/2$ is attained by a univariate quartic. Function values and gradient–step pairings are invariant under linear changes of variables, as are λ_i^Y and the corrected model; λ_i^E is invariant under orthogonal changes. □

E.4. Restarted cycles and sliding windows

With Proposition E.7 and Lemma B.1 every one-cycle theorem becomes a cycle map $r_{c+1}=\Phi(u_c)r_c^p+o(r_c^p)$. Put $S_m=\sum_{l=1}^{m}\operatorname{sech}(l\theta)$.

Theorem E.14 (Restarted cycles)**.** *Let f be C^3 near x_*, let the methods be restarted every $n\ge3$ iterations, let Assumption E.6 hold, and let $r_c\to0$.*

With $\frac{\tanh((n-1)\theta)}{\sinh\theta}\le\frac{\kappa-1}{2\sqrt{\kappa}}$,

$$\begin{aligned}&\textit{FC:}\quad r_{c+1}\le m_{\mathsf{T}}\tfrac{\tanh((n-1)\theta)}{\sinh\theta}r_c^2+o(r_c^2),\\ &\textit{RFC:}\quad r_{c+1}\le m_{\mathsf{T}}^2\big[\tfrac{1}{2}(1+S_{n-2})^2+(1+S_{n-2})\operatorname{sech}((n-1)\theta)\big]\operatorname{sech}((n-1)\theta)r_c^3+o(r_c^3).\end{aligned}$$

Quadratic restarted convergence is classical [31, 32, 91]; imperfect searches give order $2^{1/n}$ [92]. The theorem adds sharp-profile constants and RFC's cubic order. Observable restart rules remove Assumption E.6 in Proposition E.17.

For the function-level local model of XR, consider a full-memory XRFC pass of n iterations from $x^{(c)}$, returning the virtual point $\hat{x}_n$ after $n+1$ gradients. Assume (V1) the trial rule is C^2 and positive

near the reference data, (V2) homogeneous of degree 0 under $(x - x_*, g) \mapsto (\lambda(x - x_*), \lambda g)$, (V3) the directions $u_c = (x^{(c)} - x_*)/r_c$ lie in a compact set $\mathcal{U}$ of unit vectors with $\nu(H_*, H_*u)$ having n atoms. For $u \in \mathcal{U}$ let $c_l, \hat{s}_l$ be the CG data of (H_*, H_*u), ε_l the trial errors of XRFC on this quadratic, $\vartheta_l = \varepsilon_l/(1 - \varepsilon_l)$, a_l, Π_l as in Theorem C.1, and ξ_l the midpoint displacements (61) of the reference evaluated segments.

The reference midpoint displacements are

$$\begin{aligned} \xi_l &= \tfrac{1}{2}\big(\varepsilon_{l-1}\sqrt{c_{l-1}}\,\hat{s}_{l-1} + \sqrt{c_l}\,\hat{s}_l + (1 - \varepsilon_{l+1})\sqrt{c_{l+1}}\,\hat{s}_{l+1}\big) \quad (l \le n-2), \\ \xi_{n-1} &= \tfrac{1}{2}\big(\varepsilon_{n-2}\sqrt{c_{n-2}}\,\hat{s}_{n-2} + \sqrt{c_{n-1}}\,\hat{s}_{n-1}\big), \end{aligned} \tag{61}$$

with $\varepsilon_{-1} := 0$; the last one runs from the midpoint of $[\bar{x}_{n-1}, \bar{x}_n]$ to that of the output segment $[\bar{x}_n, x_*]$.

Theorem E.15 (Virtual-launch expansion)**.** *For $f \in C^3$ and trials satisfying (V1)–(V3), uniformly over $u_c \in \mathcal{U}$,*

$$H_*^{-1/2}\nabla f(\hat{x}_n) = r_c^2\Lambda(u_c) + o(r_c^2), \qquad \Lambda(u) = \sum_{l=0}^{n-1} a_l \Pi_l^\top \mathsf{T}[\xi_l]\hat{s}_l,$$

$$\|\Lambda(u)\| \le \|\mathsf{T}\|_\sigma \sum_l |a_l|\|\Pi_l\|\|\xi_l\|.$$

A Lipschitz $D^3 f$ gives remainder $O(r_c^3)$; if the quadratic-reference trial errors satisfy $\max_l |\varepsilon_l(u_c)| = O(r_c)$, then $r_{c+1} = O(r_c^3)$.

For the function lower bound normalize $\sum_l c_l = 1$.

Theorem E.16 (One-gradient lower bound on cubic functions)**.** *Let $n \ge 2$, $\widehat{M} > 0$, $0 < \delta < 1$. For every $M \in \mathcal{M}_1$ there is a rank-one cubic*

$$f(x) = \tfrac{1}{2}x^\top Jx + h\mathbf{u}_0^\top x + \tfrac{t}{6}(\mathbf{u}_{n-1}^\top x)^3, \qquad J \text{ Jacobi}, \quad h > 0,$$

started at 0, with $\operatorname{spec} H_* \subset [\mu(1-\delta), L(1+\delta)]$, $\|\mathsf{T}\|_\sigma \le (1+\delta)\widehat{M}$, and

$$\|\nabla f(\hat{x}_n)\|_{H_*^{-1}} \ge (1-\delta)\frac{\widehat{M}}{2}\rho_\kappa \|e_{n-1}\|_{H_*}^2.$$

For fixed realizable normalized Krylov information $\mathcal{I}$ in (μ, L), replace ρ_κ by $\rho(\mathcal{I}) = (r(\mathcal{I})-1)/(r(\mathcal{I})+1)$; H_* can lie within relative distance δ of an endpoint completion selected by the method. Here e_{n-1} is the reference CG error for $(H_*, \nabla f(0))$. On this family XRFC with the Gauss–Radau harmonic-mean last trial attains the leading value $\frac{\widehat{M}}{2}|\varepsilon_{n-1}|\|e_{n-1}\|_{H_*}^2$, with $|\varepsilon_{n-1}| \le \rho(\mathcal{I})$.

The adversarial cubic may depend on the starting scale; the lower bound is uniform over functions, rather than an asymptotic claim for one fixed cubic.

Proof of Theorem E.14. The transfer lemma gives $\omega_l \le m_\mathsf{T}\|e_l\|_A + o(r_c)$ and $\omega' \le m_\mathsf{T}\|e_{n-1}\|_A + o(r_c)$. The CG bound at $\theta + O(r_c)$ and the telescoping identity

$$\sum_{l=0}^{n-2} \operatorname{sech}(l\theta)\operatorname{sech}((l+1)\theta) = \tanh((n-1)\theta)/\sinh\theta$$

give FC's coefficient. For RFC use $W \le m_\mathsf{T} r_c(1 + S_{n-2}) + o(r_c)$ and $K(W, \omega') \le W^2/2 + W\omega'$. Lemma E.5 supplies the uniform remainders; local gradient–error equivalence converts both responses to errors. □

Proposition E.17 (Uniform orders with restart rules)**.** *Let f be standard self-concordant on an open convex domain D containing the canonical ball $B(x_*, 1/8)$, and let FC and RFC be restarted every n iterations, and also (a new cycle starting at the current iterate) whenever d_k is undefined or not*

a descent direction. For RFC let $\varpi_k = (w_k^\top H_k w_k/(\alpha_k^2 d_k^\top z_k))^{1/2}$*, the computable ratio of the model energies of the correction and of the line-search step; accept a correction only if it does not increase* f*, and restart also when* $\varpi_k > \bar{\varpi}$*, for a fixed* $\bar{\varpi} \in (0, \frac{1}{16n}]$*. There are* $r > 0$*, depending only on* n*,* κ *and* $\bar{\varpi}$*, and an absolute constant* C_0*, such that* $r_c \le r$ *implies*

$$\textit{FC:}\quad r_{c+1} \le C_0(1+\sqrt{\kappa})\, n^{3/2}\, r_c^2, \qquad \textit{RFC:}\quad r_{c+1} \le C_0\, \kappa\, \bar{\varpi}^{-1} n^{3/2}\, r_c^3.$$

Proof of Proposition E.17. Choose the starting radius with $r = \varrho(r_c) < 1/8$, so $r \le 2r_c$ and all accepted segments stay in this ball. Put $\varepsilon = O(r)$ from Lemma E.1. For FC, approximate conjugacy gives Gram error $O(n\varepsilon)$ and staircase residual $O(n^{3/2}\varepsilon r)$. At completion invert the basis; at a non-descent restart use Lemma E.4(a).

For RFC put $t_k = \alpha_k \|H_*^{1/2} d_k\|$ and $R_k = \|\hat{U}_k^\top \hat{g}_{k+1}\|$. The correction-energy gate bounds its length by $1.4\bar{\varpi} t_k$; induction keeps correlations below $\varepsilon/2 + 2\bar{\varpi}$, Gram matrices above $3I/4$ and model matrices above $11I/16$. The defect gate thus passes. Lemma B.3 yields $\|\rho_k\| \le cR_{k-1} + c\sqrt{n}\varepsilon t_k, \qquad R_k \le cn\varepsilon R_{k-1} + cn^{3/2}\varepsilon^2 r$. Taylor's formula makes every energy-gated correction decreasing. Taking $cn\varepsilon \le 1/2$ gives $R_k \le c_* n^{3/2} r^3$ and closes the correlation induction. If the energy gate fails, the first estimate gives $t_k \le cR_{k-1}/\bar{\varpi}$; Lemma E.4(b) bounds the terminating gradient by $c\kappa R_{k-1}/\bar{\varpi}$. Use part (a) at non-descent restarts and invert the full basis at completion. Gradient–error equivalence proves the result. □

Proof of Theorem E.15. On the quadratic reference, $\bar{x}_{k+1} = x_k^0 + \tau_k d_k^0$ and $\bar{s}_k = \varepsilon_{k-1}\sigma_{k-1} + (1-\varepsilon_k)\sigma_k$. Their midpoint differences, including the segment to the virtual output x_*, are exactly (61). Apply Lemma E.5, $d = 1$, to Theorem C.1. This gives the expansion, including homogeneity and uniformity. The inequality $\|\Pi_l^\top \mathsf{T}[\xi_l]\hat{s}_l\| \le \|\Pi_l\|\|\mathsf{T}\|_\sigma \|\xi_l\|$ gives the norm bound. The Lipschitz version of the transfer lemma gives the cubic remainder; if the quadratic-reference errors satisfy $\max_l |\varepsilon_l(u_c)| = O(r_c)$, all leading coefficients a_l are uniformly $O(r_c)$. □

Proof of Theorem E.16. Fix the Krylov fibre $J(b)$ and set $\pi(x) = \mathbf{u}_{n-1}^\top x$. Use the family $f_{b,t}(x) = \frac{1}{2}x^\top J(b)x + h\mathbf{u}_0^\top x + \frac{t}{6}\pi(x)^3$. The induction of Lemma C.7 applies because the cubic term vanishes on the first $n-1$ Krylov coordinate subspaces. Thus x_n is common to all parameters. Write $p = \pi(x_n)$: its gradient reveals only $bp + tp^2/2$, so every parameter pair with this value gives the same output $\hat{x}$.

Normalize the common last direction by $\pi(d) = 1$. Then $y(b) = \pi(x_*(b))$ is proportional to $\alpha(b)$. For interior endpoints $b_\pm^{\delta'} = b_\pm \mp \delta'$, put $r' = y(b_-^{\delta'})/y(b_+^{\delta'})$, $\rho' = (r'-1)/(r'+1)$. Lemma C.3(b) selects an endpoint b_0 with $|p - y_0| \ge \rho'|y_0|$, $y_0 = y(b_0)$.

Put $J = J(b_0)$, $C_z = \|\mathbf{u}_{n-1}\|_{J^{-1}}$, $M = \widehat{M}/C_z^3$ and $M_p = \min(M, 2\delta'/|p|)$, taking $M_p = M$ if $p = 0$. The scenarios $(b_\pm, t_\pm) = (b_0 \mp M_p p/2,\ \pm M_p)$ stay in the fibre and give $bp + tp^2/2 = b_0 p$. Their output gradients are $v \pm w$, with $v = J(\hat{x} - x_*(b_0)), \qquad w = \frac{1}{2}M_p p_{\mathrm{out}}(p_{\mathrm{out}} - p)\mathbf{u}_{n-1}, \quad p_{\mathrm{out}} = \pi(\hat{x})$. Let $R = \max_\pm \|v \pm w\|_{J^{-1}}$, $A_w = M_p C_z/2$, $T_0 = A_w|y_0||y_0 - p|$. Lemma A.1(d) gives $R \ge \max(\|v\|, \|w\|)$ and $|p_{\mathrm{out}} - y_0| \le R/\sqrt{\mu}$. If $R < T_0$, comparison of the two scalar products gives $R \ge T_0 - \eta_1 R, \qquad \eta_1 = \frac{A_w}{\sqrt{\mu}}(|y_0| + |y_0 - p| + T_0/\sqrt{\mu}) = O(Mh + \delta')$. Thus always $R \ge T_0/(1+\eta_1)$. If $M_p = M$, the endpoint inequality gives $T_0 \ge (\widehat{M}/2)\rho'\|e_{n-1}\|_J^2$, since $d^\top J d = C_z^{-2}$ and $\|e_{n-1}\|_J^2 = y_0^2/C_z^2$. If $M_p < M$, choose h so that $|y_0| \le \delta'/M$. Then $|p| > 2|y_0|$ and $T_0 \ge \delta' C_z|y_0|/2$, which is at least the same target.

At the local minimizers the Hessians are $J(b_0)(I + O(\delta' + Mh))$; the canonical cubic norm is $M_p C_z^3(1 + O(\delta' + Mh)) \le \widehat{M}(1 + O(\delta' + Mh))$. Norms and reference CG errors change by the same relative order. Letting δ', then h, tend to zero proves the fixed-fibre assertion. Lemma C.6(b) supplies fibres with $r \uparrow \kappa$, giving the uniform statement.

For attainment, the canonical cubic direction is the newest CG direction: $\mathsf{T} = \widehat{M}\hat{s}_{n-1}^{\otimes 3}$. In Theorem E.15, every slice $\mathsf{T}[\xi_l]\hat{s}_l$ with $l < n-1$ vanishes, whereas $\langle \hat{s}_{n-1}, \xi_{n-1}\rangle = \sqrt{c_{n-1}}/2$. The residual is therefore $(\widehat{M}/2)|\varepsilon_{n-1}|c_{n-1}$. The common data determine the Gauss–Radau interval and its harmonic-mean trial, so Theorem 4.1 gives $|\varepsilon_{n-1}| \le \rho(\mathcal{I})$. □

Sliding FC/RFC retain the newest $n-1$ pairs in Definition 2.6. RFC's augmented window is $U = [S, d]$, with current probe (d, z). The anchor identity $\alpha d - UH^{-1}U^\top(g + \alpha z) = -UH^{-1}U^\top g$ identifies its full-window step with Burdakov's symmetric secant method [10, eq. (2.7)]. Unlike ordinary unrestarted CG [93, 94], full sliding conjugacy yields the following higher-order recursion.

Put $\hat\sigma_j = H_*^{1/2}s_j/\|s_j\|$. Use (A1) a C^2 function with $H_* \succ 0$ and L_H-Lipschitz Hessian; (A2) defined nonzero-direction iterations $x_k \to x_*$, $x_k \neq x_*$; (A3) probe lengths $\|h_k d_k\| \le c_P r_k$ ($c_P = 3$ for exact searches); and (A4) $\sigma_{\min}([\hat\sigma_{k-n+1}, \dots, \hat\sigma_{k-1}]) \ge \sigma > 0$. Independence is vacuous for RFC when $n = 2$. Under (A1)–(A3) alone, RFC's canonical defects tend to zero and eventually pass every fixed positive defect threshold.

Proposition E.18 (Sliding full conjugacy)**.** *Let $n \ge 2$ and let $\tau_{\mathrm{FC}}(n) > 1$ solve $t^n = t + 1$. Under (A1), (A2), sliding FC satisfies*

$$r_{k+1} \le 4L_H(n-1)^{3/2}(1 + o(1))r_{k-n+1}r_{k-n+2}.$$

If $r_{k+1}/r_k \to 0$, then $\limsup_k r_{k+1}/(r_{k-n+1}r_{k-n+2}) \le L_H/2$. Its R-order is at least $\tau_{\mathrm{FC}}(n) > 2^{1/n}$. Conjugacy makes the normalized steps asymptotically orthonormal; no descent or independence assumption is required.

Powell's exact-search BFGS estimate [33, eq. (1.13)] has the same recursion as Proposition E.18; FC stores $n-1$ pairs and updates no matrix. Schuller's multipoint secant recursion [55, Thms. 3.15, 4.7] has lag n; a current probe shortens the lag to $n-1$ in Theorem E.19.

E.5. Secant rates and function-value improvement

Once the window of XRFC holds n independent steps, its new direction vanishes, the next evaluation is at the virtual point, and XRFC becomes Burdakov's sequential $(n+1)$-point method SS1 [10],

$$x_{k+1} = x_k - S_k\overline{T}_k^{-1}S_k^\top g_k = x_k - J_k^{-1}g_k, \qquad S_k = [s_{k-n}, \dots, s_{k-1}], \quad J_k = S_k^{-\top}\overline{T}_k S_k^{-1}: \tag{62}$$

one gradient per iteration and no line search. XR takes this step when its window holds n independent steps and the step is accepted (Theorem 7.1).

Theorem E.19 (Secant steps and the age of a window)**.** *Let $f \in C^2$ have an L_H-Lipschitz Hessian near x_*, $H_* \succ 0$. Suppose SS1 with n stored pairs, or sliding RFC with $n-1$ stored pairs and a current probe, generates $x_k \to x_*$, $x_k \neq x_*$. For SS1 assume uniform independence of the n normalized steps; for RFC assume (A3), (A4) and defined nonzero directions. Put*

$$h = \begin{cases} n, & \text{SS1}, \\ n-1, & \text{RFC}, \end{cases} \qquad \Delta_k = \operatorname{dist}\left(\hat\sigma_{k-h}, \operatorname{span}\{\hat\sigma_{k-h+1}, \dots, \hat\sigma_{k-1}\}\right) \ge \sigma > 0.$$

RFC's canonical defects tend to zero, so its defect gate of Definition 2.6 eventually passes. Its full model is nonsingular on this tail. Both methods then take $x_{k+1} = x_k - J_k^{-1}g_k$, $r_{k+1} = o(r_k)$, and

$$r_{k+1} \le \frac{L_H}{2\Delta_k^2}(1 + o(1))r_k r_{k-h} \le \frac{L_H}{2\sigma^2}(1 + o(1))r_k r_{k-h}.$$

The R-order is at least $\tau_1(h+1)$ per iteration: $\tau_1(n+1)$ per gradient for SS1 and $\tau_1(n)^{1/2}$ for RFC.

Use canonical coordinates and a late local tail of the convergent iteration; all its segment secants then satisfy $\bar A = I + o(1)$.

Proof of Proposition E.18. Write $o = k - n + 1$. Lemma E.2 gives $r_j \le r_i(1 + o(1))$ for $i \le j \le i + n$. Conjugacy gives $|\hat\sigma_i^\top\hat\delta_j| \le \|\bar A_i - I\| \to 0$ for $o \le i < j \le k$. Thus the normalized basis $\hat{\mathbf{S}}_k = [\hat\sigma_o, \dots, \hat\sigma_{k-1}, \hat\delta_k]$ tends to an orthonormal system, without an independence assumption.

The staircase identity, restricted to this window, reads

$$\hat{\mathbf{S}}_k^\top \hat{g}_{k+1} = (\tilde{r}, 0), \qquad \tilde{r}_i = \sum_{j=i+1}^{k} \frac{s_i^\top (\bar{A}_j - \bar{A}_i) s_j}{\|s_i\|}.$$

The secant estimate gives each summand at most $4L_H r_i r_{i+1}(1+o(1)) \le 4L_H r_o r_{o+1}(1+o(1))$. There are at most $n-1$ summands per component and $n-1$ components. Inverting $\hat{\mathbf{S}}_k$ and the radial secant gives $r_{k+1} \le 4L_H(n-1)^{3/2} r_o r_{o+1}(1+o(1))$.

If $r_{k+1}/r_k \to 0$, only $(i,j) = (o, o+1)$ contributes at this scale: its bound is $L_H(\|s_o\| + \|s_{o+1}\|)\|s_{o+1}\|/2 = (L_H/2) r_o r_{o+1}(1+o(1))$; every other term is $o(r_o r_{o+1})$. Lemma E.3(a) gives $\tau_{\mathrm{FC}}(n)$, and $\tau_{\mathrm{FC}}(n)^n = 1 + \tau_{\mathrm{FC}}(n) > 2$. □

Proof of Theorem E.19. For RFC take $U_k = [s_{k-n+1}, \dots, s_{k-1}, d_k]$, oldest index $o = k-n+1$; for SS1 take $U_k = [s_{k-n}, \dots, s_{k-1}]$, $o = k-n$. The secant errors are $o(1)$. For RFC, with $\epsilon_k = L_H(2 + c_P)\bar{r}_o/2$, conjugacy gives $|\hat{\sigma}_i^\top \hat{\delta}_k| \le \epsilon_k$ and $\sigma_{\min}(\hat{U}_k)^2 \ge \sigma^2 - \sqrt{n-1}\epsilon_k$. Its canonical defects are at most $2\epsilon_k/(1-\epsilon_k)$, so the defect gate passes. Independence and the normalized model $I + o(1)$ give nonsingularity; Lemma A.1(a) identifies the full model step.

Lemma A.8 gives $J_k \to I$, hence superlinearity and eventually decreasing errors. Only the oldest diagonal uses segment o: $|X_{00}| \le \frac{1}{2} L_H r_o(1+o(1)), \qquad \|X - X_{00}\mathbf{e}_0\mathbf{e}_0^\top\| + \|A_k^* - I\| = O(r_{o+1}) = o(r_o)$. The oldest-pair estimate (24) gives $r_{k+1} \le L_H r_k r_o(1+o(1))/(2\widetilde{\Delta}_k^2)$. For SS1, $\widetilde{\Delta}_k = \Delta_k$. For RFC, projecting the oldest step and the probe orthogonally to the newer stored span gives $\widetilde{\Delta}_k^2 = \Delta_k^2 - (r^\top \delta')^2$ and $|r^\top \hat{\delta}_k| \le \epsilon_k(1 + \sqrt{n-2}/\sigma) = o(1)$. Thus $\widetilde{\Delta}_k = \Delta_k + o(1)$. Lemma E.3 gives both orders and the stated gradient counts.

The correction-angle refinement is retained as follows. Put $\rho_k = \|U_k w_k\|/\|s_k\|$ and $\delta = \hat{\delta}_k$. The line-search/model identities give $U_k w_k = -(I - \delta\delta^\top)x_k + O(r_o r_k)$ and $\rho_k = \sin\angle(x_k, \delta) + O(r_o)$. Writing $x_k = \hat{S}_k \xi' + \xi_{n-1}\delta$, independence gives $\|\xi'\| \le c(\rho_k + r_o) r_k$. The newest model column is the probe image, so $\|(J_k - A_k^*)x_k\| \le c\{r_o\|\xi'\| + (r_{o+1} + r_k) r_k\}$. This proves $r_{k+1} \le c r_k((\rho_k + r_o) r_o + r_{o+1} + r_k)$. □

Lemma E.20 (Local SS1 convergence; [10])**.** *Under the regularity of Theorem E.19, for every fixed independence bound $\sigma_0 > 0$ there is $r_0 > 0$ such that defined SS1 iterates starting with $x_0, \dots, x_n \in B(x_*, r_0)$ and satisfying $|\det[s_i/\|s_i\|_2]| \ge \sigma_0$ in every window converge to x_* with R-order at least $\tau_1(n+1)$, or terminate there.*

Proof. This is Burdakov's local argument [10]. The determinant bound gives a uniform canonical singular-value bound $\sigma > 0$ since the columns are normalized and the coordinate change is fixed. In a sufficiently small ball, Lemma A.8 gives $\|J_k - I\| \le n L_H r/\sigma^2$ and $r_{k+1} \le 2L_H r(1 + n/\sigma^2) r_k \le r_k/2$. Induction preserves the ball; Theorem E.19 gives the order. □

For SS1-F, write R_i for the remainder in Lemma E.13, with L_3 in place of M_4; denote that identity by the trapezoid estimate.

With $P = I - s_j y_j^\top/(y_j^\top s_j)$, the model-metric split gives $w_j = P(x_k - m_j) = P(x_k - x_{j+1})$, because P annihilates s_j and $x_k - m_j = (x_k - x_{j+1}) + \frac{1}{2}s_j$. In canonical coordinates $y_j = \bar{A}_j s_j$ with $\bar{A}_j \to I$, so $\|P\| = 1/\cos\angle(s_j, y_j) = 1 + o(1)$. The Euclidean split uses the Euclidean orthogonal projector onto $s_j^\perp$, whose canonical norm is at most $\kappa^{1/2}$.

Proof of Theorem 6.1. Use Lemma A.8 with $D = -12\,\mathrm{diag}(\lambda_i \delta_i^{\mathrm{tr}})$. The normalized off-diagonal entries are unchanged, and Lemma E.13 gives

$$X_{ii}^F = \hat{\sigma}_i^\top (\nabla^2 f(x_k) - I)\hat{\sigma}_i - D^3 f(x_{i+1})[w_i, \hat{\sigma}_i, \hat{\sigma}_i] + R_i/\|s_i\|^2.$$

Condition (S) and $|\lambda_i|\|s_i\| \le \|x_k - m_i\| + \|w_i\|$ imply $X^F \to 0$, hence superlinearity. Thus r_k eventually decreases and the oldest step has length $r_{k-n}(1+o(1))$.

For $n \geq 2$ index the window from zero. Condition (S) gives $\|w_0\| \leq C_w r_{k-n+1}(1+o(1))$ and $|\lambda_0 - 1/2| = O(r_{k-n+1}/r_{k-n})$. Therefore

$$|X_{00}^F| \leq C_w T_3 r_{k-n+1} + L_3 r_{k-n}^2/12 + o(T_3 r_{k-n+1} + L_3 r_{k-n}^2) + O(T_3 r_k + L_3 r_k^2).$$

The sole other leading entries are $01, 10$, of modulus at most $(T_3/2)r_{k-n+1}(1+o(1))$; all remaining entries are $O(T_3 r_{k-n+2} + L_3 r_{k-n+1}^2)$, negligible even if $T_3 = 0$. The leading matrix norm is bounded by $|X_{00}^F| + |X_{01}^F|$. Insert this in the exact model-error identity and use $\|A_k^* - I\| = O(T_3 r_k + L_3 r_k^2)$; Lemma E.3(b) gives the order.

For $n = 1$, (S) forces $w_0 = 0$, $\lambda_0 = 1/2$. Compare the model directly with $f''(x_k)$: $|f''(x_k) - A_k^*| \leq T_3 r_k/2 + O(L_3 r_k^2)$ and $|R_0|/s_{k-1}^2 \leq L_3 r_{k-1}^2(1+o(1))/12$. This gives the stated bound and order two. With merely continuous $D^3 f$, the same Taylor identities give $X_{00}^F = O(r_{k-n+1}) + o(r_{k-n})$, proving the final limit. □

Proof of Corollary 7.4. On a window whose segments lie within ρ of x_*, Lemma E.13 gives $|12\lambda_j \delta_j^{\mathrm{tr}}| \leq C\|s_j\|^2(\|s_j\| + \|x - m_j\|)$, so $D_s^{-1}(\overline{T}_F - \overline{T})D_s^{-1} = O(\rho)$ and Lemma F.8 holds for $\overline{T}_F$ with c_T replaced by $c_T - O(\rho)$. The no-deletion, acceptance and full-window arguments in the proof of Theorem 7.3 apply with $\overline{T}_F$ in place of $\overline{T}$. The independence hypothesis of Theorem 6.1 holds with $\sigma = \sigma_H$, and the splits satisfy (S). □

F. Window tests and safeguarded iterations

Here n is the ambient dimension. Operator and window norms are Euclidean; the errors $r_k = \|x_k - x_*\|_{H_*}$ remain canonical. The pair matrix is $C = S^\top Y$.

F.1. Window geometry and the correction metric

A window is unrefuted at level $\theta_{\mathrm{w}} > 0$ when $\overline{T} \succ 0$ and $\Omega_\star < \theta_{\mathrm{w}}$; otherwise it is refuted.

The defect alone does not certify accuracy (Theorem 2.4); a known budget makes its conditional radius informative (Proposition F.3). The antisymmetric secant matrix was previously used to detect nonquadratic behaviour [53, 95]. Its relative metric below supplies a quantitative refutation threshold.

A correction w moves x to $x + Sw$, with window residual $S^\top g + (S^\top BS)w$. For a metric $\Xi \succ 0$ put $\rho = \Xi^{-1/2} S^\top g$, $b = \Xi^{1/2} w$, $N = \Xi^{-1/2}\overline{T}\Xi^{-1/2}$, $E = \Xi^{-1/2}\mathcal{Z}\Xi^{-1/2}$; the uncertainty set of radius Ω is $-\Omega\Xi \preceq \mathcal{Z} \preceq \Omega\Xi$, and the minimax correction solves

$$\min_b \max_{E=E^\top,\ \|E\| \leq \Omega} \|\rho + (N+E)b\|, \tag{63}$$

for $\Omega \geq 0$, with objective $\|\rho + Nb\| + \Omega\|b\|$ by Lemma A.1. For $\rho \neq 0$, the robust least-squares path [48, Thm. 3.6], [49] has Newton and zero thresholds

$$\Omega_1 = \frac{\|N^{-1}\rho\|}{\|N^{-2}\rho\|} \leq \Omega_2 = \frac{\|N\rho\|}{\|\rho\|},$$

with Tikhonov regularization between them (proof of Theorem F.1).

Theorem F.1 (The relative metric)**.** *Let $\overline{T} \succ 0$ and $\Xi \succ 0$, and put $N = \Xi^{-1/2}\overline{T}\Xi^{-1/2}$. The robust least-squares thresholds $\Omega_1(\rho)$ and $\Omega_2(\rho)$ coincide for every $\rho \neq 0$, or just on a nonempty open set, iff $\Xi = \overline{T}/c$ for some $c > 0$. Their common value is then c. For $\Xi = \overline{T}$, the unique minimax correction is $w = -\overline{T}^{-1} S^\top g$ for $\Omega < 1$ and $w = 0$ for $\Omega > 1$; at $\Omega = 1$ the minimizers form the segment between them. Its value is*

$$\min(\Omega, 1)\|S^\top g\|_{\overline{T}^{-1}}, \qquad (1-\Omega)\overline{T} \preceq S^\top BS \preceq (1+\Omega)\overline{T}.$$

Pair-rescaling invariance alone also permits diagonal metrics; the coincident-threshold property selects the model metric.

Proposition F.2 (Hereditary window test)**.** *For $\theta_{\mathrm{w}} > 0$,*

$$\overline{T} + i\mathsf{D}/\theta_{\mathrm{w}} \succ 0 \quad \Longleftrightarrow \quad \overline{T} \succ 0, \quad \Omega_\star < \theta_{\mathrm{w}}.$$

The property is inherited by every ordered subset. The function UnrefutedSuffix$(T, \mathsf{D}, \theta_{\mathrm{w}})$ *reverses the pair order, factors $T + i\mathsf{D}/\theta_{\mathrm{w}}$ by Cholesky until the first nonpositive pivot, and returns $p - j$, where j pivots were positive. Thus j is the length of the longest admissible suffix and $p - j$ is its deletion count.*

The canonical defect gate and this relative test are quantitatively comparable on well-conditioned normalized windows (Lemma F.5). Independence of S is tested separately before computing generalized Ritz values. Earlier suffix rules [40, 96] and cautious updates [97, 98] use different criteria.

Proposition F.3 (With a known budget the defect sharpens the worst case)**.** *Let $p = 2$, $A_0 = A \succ 0$, let the window be A-conjugate, $Q = A^{1/2} S D_A^{-1/2}$ with $D_A = \operatorname{diag}(s_i^\top A s_i)$, and let $\widehat{E}_0 = A^{-1/2} E_0 A^{-1/2}$ and $\widehat{E}' = A^{-1/2} E_1 A^{-1/2}$ have budgets $0 < \omega_0 < 1$ and ω'. Write $Q^\top \widehat{E}_0 Q = \left(\begin{smallmatrix} y & \delta \\ \delta & k \end{smallmatrix}\right)$. The worst case of $\|D_A^{-1/2} \mathcal{Z} D_A^{-1/2}\|$ over explanations of the compressed observations C is:*

(a) if A is revealed (the data then reveal δ, k and hide $y, \widehat{E}'$): $\omega_0 - \frac{\delta^2}{\omega_0 + |k|} + \omega' \le \sqrt{\omega_0^2 - \delta^2} + \omega'$, with equality iff $\delta = 0$ or $|k| = \sqrt{\omega_0^2 - \delta^2}$;

(b) from the data alone, which reveal $\delta' = C_{01}/\sqrt{C_{00} C_{11}} = \delta/\sqrt{1 + k}$ (so $\Omega_\star = |\delta'|/\sqrt{1 - \delta'^2}$ for $|\delta'| < 1$) and are consistent iff $\delta'^2 \le 2 - 2\sqrt{1 - \omega_0^2}$:

$$U(\delta') = \tfrac{1}{2}\big(\delta'^2 + \sqrt{4\omega_0^2 - 4\delta'^2 + \delta'^4}\big) + \omega'.$$

All three decrease strictly in the modulus of the defect and equal $W = \omega_0 + \omega'$ at zero defect. They remain upper bounds when full vector observations are prescribed and are sharp already for $n = 2$. If $\omega_0 = 0$, the defect vanishes and the radius is ω'.

Proof of Proposition F.3. In the conjugate basis, with $D_A = \operatorname{diag}(d_0, d_1)$,

$$C = \begin{pmatrix} d_0 & \sqrt{d_0 d_1}\,\delta \\ 0 & d_1(1 + k) \end{pmatrix}, \qquad D_A^{-1/2} \mathcal{Z} D_A^{-1/2} = y \mathbf{u}_0 \mathbf{u}_0^\top + Z', \quad \|Z'\| \le \omega'.$$

Compressed increments extend by zero. Thus the radius is $\sup|y| + \omega'$, and the completion formula of Lemma A.1(f), with $\delta^2 + k^2 \le \omega_0^2$, proves (a) and its equality case. For (b) set $t = \delta'^2$, so $\delta^2 = t(1 + k)$ and feasibility is $k^2 + tk + t - \omega_0^2 \le 0$. Its endpoints are $k_\pm = (-t \pm \sqrt{t^2 - 4t + 4\omega_0^2})/2$; they are feasible precisely in the asserted range. Each is realized by $d_1 = C_{11}/(1 + k)$ and an SPD base. The function $\omega_0 - t(1 + k)/(\omega_0 + |k|)$ decreases for $k < 0$ and increases for $k > 0$; at either endpoint $y = -k$. Its maximum is $|k_-|$, yielding U. Differentiation gives strict decrease; the canonical defect identity is Lemma F.5. At zero defect take $k = 0$, $y = \pm\omega_0$. □

Proposition F.4 (Variation certified by refutation)**.** *Let the full-rank window be generated by $A_0, \dots, A_{p-1}, B \succ 0$ and $\|B^{-1/2} E_l B^{-1/2}\| \le \omega_l$. Put*

$$G_B = S^\top B S, \quad W = \sum_{l \le p-1} \omega_l, \quad \kappa_\flat = \max\big(1, \max_l \|G_B^{-1/2} I_{\le l} G_B^{1/2}\|\big).$$

Refutation at level θ_{w} implies

$$W \ge \frac{\theta_{\mathrm{w}}}{(2 + \theta_{\mathrm{w}})\kappa_\flat^2}, \qquad W \ge \frac{\theta_{\mathrm{w}}}{1 + \theta_{\mathrm{w}}} \quad \text{for a } B\text{-conjugate window.}$$

Below the respective thresholds the window is unrefuted and $\|\mathcal{Z}_\star\| < \theta_{\mathrm{w}}/2$ (respectively $< \theta_{\mathrm{w}}$). For $\theta_{\mathrm{w}} \le 1$ the full correction is then minimax in the relative uncertainty set.

Algorithm 3 UnrefutedSuffix: chronological suffix selection

Require: Full-column-rank S, p chronological pairs with $s_i^\top y_i > 0$; $\theta_{\mathrm{w}} > 0$
1: **if** $p = 0$ **then**
2: **return** 0
3: **end if**
4: Form $\mathcal{H} = \overline{T} + i\mathsf{D}/\theta_{\mathrm{w}}$ and reverse its row and column order
5: $j \leftarrow$ number of positive Cholesky pivots preceding the first nonpositive pivot (or p if none)
6: $o \leftarrow \min(p - j, p - 1)$
7: **return** o

Proof of Proposition F.4. Write $W' = \sum_{l \le p-2} \omega_l$ and $\Pi_l = G_B^{-1/2} I_{\le l} G_B^{1/2}$. Put $Q = B^{1/2} S G_B^{-1/2}$ and $Z_l = Q^\top \widehat{E}_l Q$, so $Q^\top Q = I$ and $\|Z_l\| \le \omega_l$. The normalized flag summands are $\Pi_l Z_l \Pi_l^\top$ and $\Pi_l Z_l - Z_l \Pi_l^\top$. Their norms are bounded by $\kappa_\flat^2 \omega_l$ and $2\kappa_\flat^2 \omega_l$, using $\|I - \Pi_l\| = \|\Pi_l\|$ for nontrivial idempotents [99]. For orthogonal projectors the commutator is a skew off-diagonal block and its bound improves to ω_l. All Π_l are orthogonal exactly when G_B is diagonal, proving the characterization of $\kappa_\flat = 1$.

Summation gives $\|G_B^{-1/2} \mathcal{Z} G_B^{-1/2}\| \le \kappa_\flat^2 W$ and $\|G_B^{-1/2} \mathsf{D} G_B^{-1/2}\| \le 2\kappa_\flat^2 W'$. Thus $\overline{T} = G_B - \mathcal{Z} \succeq (1 - \kappa_\flat^2 W) G_B$. Changing to the $\overline{T}$ metric costs at most $(1 - \kappa_\flat^2 W)^{-1}$, giving the model-error and defect estimates; use the improved commutator bound for conjugate windows. The refutation and retention thresholds follow by solving these inequalities for W and applying Theorem F.1 when the resulting radius is below one. □

Proof of Theorem F.1. Lemma A.1 gives $\|\rho + Nb\| + \Omega\|b\|$. The robust least-squares theorem [48, Thm. 3.6], with $A = N$, right-hand side $-\rho$ and no right-hand-side uncertainty, gives the two thresholds and intermediate path $b = -(N^2 + \lambda I)^{-1} N\rho$. Cauchy–Schwarz gives $\|N^{-1}\rho\|^2/\|N^{-2}\rho\|^2 \le \|\rho\|^2/\|N^{-1}\rho\|^2 \le \|N\rho\|^2/\|\rho\|^2$. Equality holds precisely when ρ belongs to one eigenspace. Such equality on an open set forces $N = cI$, equivalently $\Xi = \overline{T}/c$. The scalar case is the Euclidean robust correction of Lemma A.1(b); congruence gives the uncertainty interval. □

Proof of Proposition F.2. Real vectors show $\overline{T} \succ 0$. Congruence reduces the test to $I + iK/\theta_{\mathrm{w}} \succ 0$, $K = \overline{T}^{-1/2} \mathsf{D} \overline{T}^{-1/2}$; skew eigenvalue pairs $\pm i\sigma_j$ make this equivalent to $\|K\| < \theta_{\mathrm{w}}$. Ordered restriction gives principal submatrices. Reverse Cholesky thus finds the longest positive suffix. Diagonal corrections preserved by restriction obey the same argument; pair scaling acts by congruence and coordinate changes preserve C. □

The one-pair suffix is positive because its diagonal is $s_{p-1}^\top y_{p-1} > 0$; hence the clamp preserves a pair and agrees with the longest admissible suffix in exact arithmetic. Rank cleanup precedes this procedure; no generalized eigenvalue is computed from a rank-deficient window.

Lemma F.5 (Canonical defect)**.** *Assume $\overline{T} \succ 0$. With $D_c = \mathrm{diag}(s_i^\top y_i)$, $N_c = D_c^{-1/2} \overline{T} D_c^{-1/2}$, $\delta = D_c^{-1/2} \mathsf{D} D_c^{-1/2}$: $\|\delta\|/\lambda_{\max}(N_c) \le \Omega_\star \le \|\delta\|/\lambda_{\min}(N_c)$, and $\Omega_\star = |\delta_{01}|/\sqrt{\det N_c}$ for $p = 2$. The canonical defect and the relative defect are therefore comparable with factors given by the extreme eigenvalues of N_c. These factors determine the corresponding test thresholds on well-conditioned normalized windows; a small $\lambda_{\min}(N_c)$ permits a much larger relative defect.*

Proof. Scale the pairs by $D_c^{-1/2}$ and use $\Omega_\star = \|N_c^{-1/2} \delta N_c^{-1/2}\|$; the two norm inequalities give the bounds. In dimension two, $X^\top \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right) X = \det(X) \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ with $X = N_c^{-1/2}$ gives the exact formula. □

F.2. Model steps and anchor invariance

Use the model and trial of Section 7, with Gram matrix $\mathrm{diag}(T, \hat{c})$ in $[S, d]$ when $d \neq 0$; if $d = 0$, use the restriction with Gram matrix T in S and set $\tau = 0$.

Lemma F.6 (Model step)**.** *Consider the raw model step before safeguard (G), with full-column-rank S, $T \succ 0$, $w = -T^{-1}S^\top g$ and $\hat{g} = g + Yw$. For a nonzero missing direction, assume $Y^\top S$ is nonsingular, solve $(Y^\top S)\beta = Y^\top \hat{g}$, set $d = -\hat{g} + S\beta$, and take $\hat{c} > 0$. For a full window set $d = 0$ and use $u = Sw$; the same restriction applies when the missing direction vanishes. The unique minimizer of $f(x) + g^\top v + \frac{1}{2}\widehat{\mathsf{H}}(v)$ on $\operatorname{span}[S, d]$ is $u = Sw + \tau d$, with $\tau = -g^\top d/\hat{c} = -\hat{g}^\top d/\hat{c}$ if $d \neq 0$, and $\tau = 0$ otherwise. It satisfies $g^\top u = -\widehat{\mathsf{H}}(u)$ and, for $g \neq 0$ and $f \in C^2$ on the trial segment, the unit-trial curvature-ratio identity of Section 7. On quadratic data with $Y = AS$, $T = \overline{T}$, $d^\top(\nabla f(x + tu) - \hat{g}) = t\tau d^\top Ad$.*

Proof. Conjugacy $Y^\top d = 0$ and invertibility of $Y^\top S$ make $[S, d]$ independent when $d \neq 0$. The quadratic model separates into the SPD blocks $T, \hat{c}$; its normal equations give u and $g^\top u = -\widehat{\mathsf{H}}(u)$. Also $g^\top d = \hat{g}^\top d$ because $\hat{g} - g = Yw$. Taylor's integral formula gives χ; on quadratics $d^\top AS = 0$ gives the final identity. If $d = 0$, retain only the T block. □

Lemma F.7 (Anchor invariance)**.** *Let $T = \overline{T} \succ 0$ and $Y^\top S$ be invertible, and let $(x', g') = (x + s_{p-1}, g + y_{p-1})$. The virtual point, extrapolated gradient, direction, trial length τ (for any $\hat{c}$ that depends only on the window and d) and trial point computed from (x', g') coincide with those computed from (x, g). The full displacement $u = Sw + \tau d$ satisfies $u' = u - s_{p-1}$, so $x' + u' = x + u$.*

Proof of Lemma F.7. The newest column satisfies $\overline{T}\mathbf{u}_{p-1} = S^\top y_{p-1}$, so $w' = w - \mathbf{u}_{p-1}$. Hence $x' + Sw' = x + Sw$ and $g' + Yw' = g + Yw$. The direction and curvature estimate are unchanged, as is the slope because $y_{p-1}^\top d = 0$; therefore $\tau' = \tau$, $u' = S(w - \mathbf{u}_{p-1}) + \tau d = u - s_{p-1}$ and $x' + u' = x + u$. □

For the function-value model, suppress δ_i^{tr} when

$$|\delta_i^{\mathrm{tr}}| \le 100\varepsilon_{\mathrm{mach}}(|f(a_i)| + |f(a_i + s_i)| + |\nabla f(a_i)^\top s_i| + |\nabla f(a_i + s_i)^\top s_i|).$$

Omit a diagonal correction if $|12\lambda_i\delta_i^{\mathrm{tr}}| > C_{ii}/2$, and use $\overline{T}_F^{\mathrm{g}}$ only when its Hermitian test passes. The exact-arithmetic convergence statements set $\varepsilon_{\mathrm{mach}} = 0$ in the noise test. The size and Hermitian tests remain in the iteration; the proof of Corollary 7.4 shows that they are eventually inactive.

F.3. Local transition and global convergence

At a nonstationary anchor on the raw model branch of Lemma F.6, the model step of Section 7 satisfies, at $t = 1$,

$$g^\top d = \hat{g}^\top d, \qquad -g^\top u = \widehat{\mathsf{H}}(u) = w^\top Tw + \tau^2\hat{c}, \qquad \chi = \frac{u^\top \bar{A}u}{\widehat{\mathsf{H}}(u)}. \tag{64}$$

Here $\bar{A} = 2\int_0^1 (1 - s)\nabla^2 f(x + su)\, ds$ and $\tau = 0$ for a full window or $d = 0$. For a trial $x + tu$, the last ratio is $t\, u^\top \bar{A}_t u/\widehat{\mathsf{H}}(u)$, with $\bar{A}_t = 2\int_0^1 (1 - s)\nabla^2 f(x + stu)\, ds$. Operator and window norms are Euclidean, while r_k retains its canonical meaning.

Lemma F.8 (The model step near a nondegenerate minimizer)**.** *Let $1 \le p \le n$ pairs satisfy $y_i = A_i s_i$ with symmetric A_i, $\|A_i - H_*\| \le \epsilon$, and $\sigma_{\min}(\hat{S}) \ge \sigma_\flat$ for $\hat{S} = [s_i/\|s_i\|]$. Put $c_T = \frac{3}{4}\sigma_\flat^2\lambda_{\min}$ and assume $\epsilon \le \epsilon_0 := \sigma_\flat^2\lambda_{\min}/(4n)$ and $\epsilon \le \lambda_{\max}$. Let $D_s = \operatorname{diag}(\|s_i\|)$ and $\hat{T} = D_s^{-1}\overline{T}D_s^{-1}$.*

(a) $\hat{T} \succeq c_T I$, $Y^\top S$ is invertible, and $\Omega_\star(C) \le 2n\epsilon/c_T$.

(b) For every g, and, if $p < n$ and $d \neq 0$, every estimate with $\ell\|d\|^2 \le \hat{c} \le \Lambda\|d\|^2$, the model step satisfies $-g^\top u \ge c_\flat\|g\|^2$ and $\|u\| \le C_\flat\|g\|$.

(c) If $g \neq 0$ and $\bar{A}$ is symmetric with $\|\bar{A} - H_*\| \le \epsilon$, then $\chi \le \max(1, d^\top H_* d/\hat{c}) + C_\chi\epsilon$ for $p < n$ and $d \neq 0$, and $\chi \le 1 + C_\chi\epsilon$ for $p = n$ or $d = 0$.

The constants $c_\flat, C_\flat, C_\chi$ depend only on $n, \sigma_\flat, \lambda_{\min}, \lambda_{\max}, \ell, \Lambda$; for the full-window branch their dependence on ℓ, Λ is omitted.

Proof. Let $G_* = \hat{S}^\top H_* \hat{S} \succeq \sigma_\flat^2 \lambda_{\min} I$. The normalized model and pair matrix differ from G_* by at most $p\epsilon$, while the defect has norm at most $2p\epsilon$. This gives the inverse, positivity and relative-defect bounds.

Put $a^2 = w^\top \overline{T} w$ and $q = \|\hat{g}\|$. Uniform inverse bounds and $S^\top \hat{g} = (C - \overline{T})w$ imply $\|Sw\| + \|g - \hat{g}\| \le ca, \quad a \le c\|g\|, \quad \|d\| \le cq, \quad |g^\top d + q^2| \le c\epsilon a q$. For $d \neq 0$, the slope identity therefore gives $-g^\top u = a^2 + (g^\top d)^2/\hat{c} \ge a^2 + c(q - C\epsilon a)_+^2 \ge c'(a^2 + q^2) \ge c_\flat \|g\|^2$. Also $\|\tau d\| \le q/\ell$, proving the length bound. If $d = 0$, the same estimate gives $q \le C\epsilon a$ and $u = Sw$; for $p = n$, invertibility gives $a \ge c\|g\|$ directly.

Finally conjugacy and the model estimate give $|u^\top \bar{A} u - a^2 - \tau^2 d^\top H_* d| \le C\epsilon(a^2 + \tau^2 \|d\|^2)$. Divide by $\widehat{\mathsf{H}}(u) = a^2 + \tau^2 \hat{c}$ and use $\hat{c} \ge \ell \|d\|^2$. With $\tau = 0$ the same calculation proves the zero-direction and full-window bounds. □

Proposition F.9 (The Gauss–Radau trial is adequate)**.** *Let $\mu < \lambda_{\min}$ and $\lambda_{\max} < L$, and let $\kappa_{\rm lo}, \kappa_{\rm hi}$ be the ends of the consistency interval of Theorem 4.3 with class bounds μ, L. In the setting of Lemma F.8 with $p < n$ and $d \neq 0$ there are $\epsilon_R, C_R > 0$, depending only on $n, \sigma_\flat, \mu, L, \lambda_{\min}, \lambda_{\max}$, such that for $\epsilon \le \epsilon_R$ the radau estimate $\hat{c} = \frac{1}{2}(\kappa_{\rm lo} + \kappa_{\rm hi})\|d\|^2$ is defined and*

$$(\kappa_{\rm lo} - C_R\epsilon)\|d\|^2 \le d^\top H_* d \le (\kappa_{\rm hi} + C_R\epsilon)\|d\|^2, \qquad \frac{d^\top H_* d}{\hat{c}} \le \frac{2r}{1+r} + \frac{2C_R}{\mu}\epsilon \le \frac{2L}{L+\mu} + \frac{2C_R}{\mu}\epsilon,$$

with $r = \kappa_{\rm hi}/\kappa_{\rm lo} \le L/\mu$. Hence (G3) holds for the radau trial with $\ell = \mu$, $\Lambda = L$ for k large whenever $\sigma_{\rm A} < \mu/(L+\mu)$.

Proof. The strict spectral margins give $\hat{T} - \mu\hat{G} \succeq ((\lambda_{\min} - \mu)\sigma_\flat^2 - p\epsilon)I$ and the analogous upper bound, so both Schur inverses are uniformly bounded. The Gram matrix of $LI - H_*$ on $[\hat{S}, d]$ gives $L\|d\|^2 - d^\top H_* d \ge (L\hat{b} - \hat{S}^\top H_* d)^\top (L\hat{G} - \hat{S}^\top H_* \hat{S})^{-1} (L\hat{b} - \hat{S}^\top H_* d)$. Here $\|\hat{S}^\top H_* d\| \le \sqrt{p}\epsilon \|d\|$ and $\|\hat{T} - \hat{S}^\top H_* \hat{S}\| \le p\epsilon$. Comparing inverses gives $d^\top H_* d \le (\kappa_{\rm hi} + C_R\epsilon)\|d\|^2$; use $H_* - \mu I$ for the lower estimate. Since $\mu \le \kappa_{\rm lo} \le \kappa_{\rm hi} \le L$, division by their arithmetic mean gives the stated ratio. Its limiting bound $2L/(L+\mu)$ is strictly below $2(1 - \sigma_{\rm A})$ under the displayed condition. □

Proposition F.10 (Adequacy floor)**.** *Let $L > \lambda_{\max}$, $\sigma_{\rm A} < \frac{1}{2}$, $\eta_0 \in (0, 1 - 2\sigma_{\rm A})$, and replace at every model step with $1 \le p < n$, $d \neq 0$, an estimate $\hat{c} > 0$ by $\hat{c}_{\rm A} = \min\{\max\{\hat{c}, \kappa_{\rm hi}\|d\|^2/(2(1-\sigma_{\rm A}) - \eta_0)\}, \kappa_{\rm hi}\|d\|^2\}$, with $\kappa_{\rm hi}$ the upper end of the interval of Theorem 4.3 computed with L only. Under (G1), (G2), for large k, $\hat{c}_{\rm A}$ satisfies (G3) with $\ell = \lambda_{\min}/(2(2(1-\sigma_{\rm A}) - \eta_0))$, $\Lambda = L$, $\eta = \eta_0/2$; on quadratic data the upper clip never increases a trial error.*

The Gauss–Radau trial satisfies (G3) by itself when its class bounds strictly contain the spectrum of H_* and $\sigma_{\rm A} < \mu/(L+\mu)$ (Proposition F.9).

Proof of Proposition F.10. The preceding upper Schur estimate uses only L and gives $\kappa_{\rm hi} \ge \lambda_{\min} - C\epsilon \ge \lambda_{\min}/2$ eventually. Put $a = 2(1 - \sigma_{\rm A}) - \eta_0 > 1$. The clipped estimate lies between $(\kappa_{\rm hi}/a)\|d\|^2$ and $\kappa_{\rm hi}\|d\|^2$, so it has the stated lower and upper bounds, and $d^\top H_* d/\hat{c}_{\rm A} \le a + C\epsilon/\ell \le 2(1-\sigma_{\rm A}) - \eta_0/2$ for large k. On quadratic data the upper clip projects an estimate down to a consistency bound, so cannot increase its trial error (Theorem 4.3). Raising it to the adequacy floor is a separate acceptance safeguard. □

Proof of Theorem 7.3. Put $\bar{r}_k = \max_{k-n \le j \le k} r_j$. The safeguard puts every evaluated node of iteration j within Cr_j of x_*. Each iteration stores an accepted positive-curvature pair eventually, so every window comes from the last n iterations. Its secants and trial Hessian averages differ from H_* by $\epsilon_k = O(\bar{r}_k) \to 0$.

Lemma F.8 gives $\overline{T} \succ 0$, $\Omega_\star = O(\bar{r}_k)$ and the strict safeguard bounds. Thus the window and rank tests drop no pair. If $m_0 = n$, the accepted pairs fill the window within at most n further iterations; apply the full-window estimate below thereafter. If $m_0 < n$, trial adequacy gives $\chi \le$

$\max(1, 2(1-\sigma_{\mathrm{A}})-\eta)+O(\epsilon_k) < 2(1-\sigma_{\mathrm{A}})$; for a full window or $d=0$ use $\chi \le 1+O(\epsilon_k)$. Hence, in the applicable phase, the first trial is accepted and one pair is added.

Once the current cap is filled, q clean iterations increase it. Each increase requires finitely many new pairs; no halving occurs. Thus the target cap is reached in finite time.

At cap n, after at most n further iterations the model is exactly SS1 on the last n accepted pairs. Canonical normalization gives independence $\sigma_H = \sigma_\flat\sqrt{\lambda_{\min}/\lambda_{\max}}$ and Hessian Lipschitz constant at most $L_H^{\mathrm{E}}\lambda_{\min}^{-3/2}$. Apply Theorem E.19. If $m_0 = n$, the window fills within n iterations and only the full-window part of the local lemma is needed, so no trial-adequacy hypothesis is used. □

Lemma F.11 (Local equivalence with XRFC)**.** *Let q be a reference quadratic with n atoms and $m_0 = n$ on whose first n iterations every trial is accepted at $t=1$, the Armijo tests and (G) hold strictly, and the trial rule is a positive C^2 function of the data nearby (strictly decided branches; for the spectral policy, simple extreme Ritz values). Then there are a neighbourhood U of the reference trajectory and $\epsilon_0 > 0$ such that for every $f \in C^2(U)$ with the same value and gradient at x_0 and $\|\nabla^2 f - A\| \le \epsilon_0$ on U, XR performs in these n iterations exactly the iterations of XRFC (F off), resp. XRFC-F (F on, exact f-values), with full memory and the same trial rule; with F off, Theorems C.1 and B.13 hold for XR. In the operator-sequence model, two such trial rules that agree at the reference and differ by $O(\|\widehat{E}\|)$ give outputs that differ by $O(\|\widehat{E}\|^2)$.*

Proof. There are finitely many strict or locally constant decisions. On their common branch all updates are smooth nonsingular solves. A small Hessian perturbation with matched initial value and gradient perturbs the observations uniformly on the finite trajectory, preserving the branch and the XRFC or XRFC-F updates. For the prescribed-trial output $\Psi(\widehat{E}, \tau)$, quadratic trial independence gives $\partial_\tau\Psi(0,\tau) = 0$. Bounded mixed derivatives imply $\partial_\tau\Psi = O(\|\widehat{E}\|)$; integration between trial vectors $O(\|\widehat{E}\|)$ apart proves the second-order comparison. □

Proof of Theorem 7.1. On quadratic data, inductively $Y = AS$, the anchor lies in $x_0+\mathcal{K}_k$, and span $S = \mathcal{K}_k$. The zero defect and Krylov independence keep every pair, and the Galerkin solve gives $(\hat{x}, \hat{g}, d) = (x_k^0, g_k^0, d_k^0)$ (Lemma A.2). Every nonzero trial adds the next Krylov direction, regardless of whether its anchor is retained. At n independent pairs the full model reaches x_*; Armijo accepts it. A smaller cap obeys the same induction until its first deletion.

For global convergence, after at most one probe, backtracking fixes a direction satisfying (G). The descent lemma accepts every $t \le t_* = 2(1-\sigma_{\mathrm{A}})c_1/(Lc_2^2)$. Successive factors lie in $[1/10, 1/2]$, giving $t \ge \min(1, t_*/10) = t_{\min}$. Armijo then gives the displayed decrease; summation proves square summability and stationarity. Bounded sublevel sets supply limit points. This is the gradient-related Armijo argument [36].

For the spectral trial policy, bounded normalized Gram matrices keep the running Ritz upper bound bounded. A strict eventual margin $\hat{L} > \lambda_{\max}(H_*)$ makes Proposition F.10 uniform and supplies (G3) when $m_0 < n$. For either trial policy, Theorem 7.3 applies under its stated local hypotheses; when $m_0 = n$, trial adequacy is unnecessary. With F on, the normalized correction is $O(\bar{r}_k)$, so its Hermitian and size gates eventually pass; the exact-arithmetic noise gate removes only zero defects. Corollary 7.4 then gives the stated SS1-F phase under its additional hypotheses. □

G. Experimental settings and additional comparisons

All comparisons use the stationarity target and limits of Section 8. The ablations use 64 of the primary CUTEst starts, with the corresponding XR/XR-F runs as their reference observations. The local experiments use the complete dimension/condition-number/seed grid specified with Figure 4.

For the controlled objectives, a Gaussian QR basis supplies the eigenvectors of A and independent normalized Gaussian columns supply the u_j. An independent Gaussian start is normalized to the specified A-energy; all methods share each realized objective and start.

Method	Primary memory setting	Source
L-BFGS	20 pairs	[34, 37, 38]
L-CG_DESCENT	11 directions	[39]
MS-L-BFGS	8 stored pairs, 8 simultaneous secants	[40]
SC-MSS	3 model pairs, $q = 5$ preceding curvatures	[41]
SC-SR1	5 model pairs, $q = 7$ preceding curvatures	[42]
O-ACCEL-B	history 20	[43]
XR / XR-F	initial $\min(10, n)$, cap $\min(100, n)$	Section 7.1

Table 5: Settings shared across test problems. Directions, secant pairs and initialization observations count different stored information.

Performance-profile scales. Ordinary profiles use the scheduled cohort $\mathcal{P}$. For the companion view, let $\mathcal{P}_{\text{any}}$ contain the instances solved by at least one method and $N = |\mathcal{P}_{\text{any}}|$. For a ratio allowance $r \geq 1$, let $k_s(r)$ count its problems that method s solves within r times the cheapest successful cost. The empirical-logit ordinate is

$$z_s(r) = \log \frac{k_s(r) + 1/2}{N - k_s(r) + 1/2}.$$

Ticks show the original proportions k_s/N; the half-count correction makes both endpoints finite. An unsuccessful run has infinite cost ratio.

Implementations and settings. Experiments run on an Apple M4 Max with 14 CPU cores and 36 GB of memory, using one solver process at a time and one BLAS thread. Each method–problem pair contributes one timed run. Solver times include initialization, model updates, searches and oracle calls; problem setup, data warm-up and module loading precede timing. XR/XR-F use C model and control routines with BLAS/LAPACK and incremental QR updates. MS-L-BFGS, SC-MSS, SC-SR1 and O-ACCEL-B use independent C implementations of their published algorithms; the SC and O-ACCEL implementations were checked against the authors' source code. L-CG_DESCENT uses its author C implementation; L-BFGS uses SciPy's compiled core with Python reverse communication. The C ports share compiler and floating-point settings and BLAS/LAPACK kernels with XR. All methods use the same objective and derivative interface. A point-budget unit acquires one record in a four-point cache; function and gradient values at a resident point share that unit. Gradient profiles count actual gradient evaluations.

Table 5 lists the primary settings. The SC methods use shape-changing infinity-norm trust regions; MSS has dense initialization, and SR1 uses maximum-curvature initialization (Init. 2). Their model-pair and initialization-history sizes are distinct parameters: each initialization uses the current $\|y\|^2/(s^\top y)$ and q preceding observations, where s is an observed step and y its gradient difference. SC-MSS starts its complement curvature at one and subsequently uses the latest quotient with $s^\top y > 0$. For SC-SR1, the updated maximum-history scalar γ enters the compact storage test

$$|s^\top (y - \gamma s)| > 10^{-8} \max\{\|s\| \, \|y - \gamma s\|, 10^{-10}\}.$$

Both use the authors' initial Moré–Thuente search settings: initial step one, sufficient-decrease and curvature constants 10^{-4} and 0.9, interval tolerance 10^{-6}, maximum step five and at most 3000 trials. A finite sufficient-decrease point seeds the trust-region model when the initial search reaches its step bound. The compact SR1 subproblem is treated in [100]. O-ACCEL-B uses $\delta = 10^{-4}$ and regularization 10^{-12}, as in [43].

L-CG_DESCENT uses its native default and the best tested memory in [39]. L-BFGS uses our choice of 20 within the authors' recommended range 3–20; its SciPy default is 10. MS-L-BFGS uses the published uniform L8M8 variant, MSS its recommended default, and SR1 its preferred published experimental configuration. The 10–100 XR memory policy is our engineering choice.

XR uses $q = 3$ clean steps per capacity increase. With normalized step columns, rank cleanup uses QR pivot tolerance $32\varepsilon_{\text{mach}} \max(n, p)$ and reciprocal condition threshold $32\sqrt{\varepsilon_{\text{mach}}}$. Each accepted

Method	Setting	Solved	Both solved	$N_g^{\text{variant}}/N_g^{\text{base}}$	Variant only	Base only
XR	m0=5	15/16	15	1.086	0	1
XR	m0=10 (base)	16/16	16	1.000	0	0
XR	m0=20	15/16	15	1.210	0	1
XR-F	m0=5	16/16	15	1.297	1	0
XR-F	m0=10 (base)	15/16	15	1.000	0	0
XR-F	m0=20	14/16	14	0.920	0	1
L-BFGS	m=5	12/16	11	1.115	1	0
L-BFGS	m=10	11/16	11	1.076	0	0
L-BFGS	m=20 (base)	11/16	11	1.000	0	0
L-CG_DESCENT	m=5	16/16	16	1.121	0	0
L-CG_DESCENT	m=11 (base)	16/16	16	1.000	0	0
L-CG_DESCENT	m=20	16/16	16	0.882	0	0
MS-L-BFGS	L8M2	13/16	12	0.910	1	2
MS-L-BFGS	L8M4	15/16	14	0.862	1	0
MS-L-BFGS	L8M8 (base)	14/16	14	1.000	0	0

Table 6: Fixed-parameter sensitivity on 16 CUTEst instances. Ratios are geometric means of $N_g^{\text{variant}}/N_g^{\text{base}}$ on instances solved by both settings. Base settings are marked in each method family. Variant only and Base only count exclusive successes. XR and XR-F use memory cap 100 and vary m_0; MS-L-BFGS stores eight pairs and varies its simultaneous-secant limit.

iteration allows at most 80 trial evaluations. XR-F uses the size and Hermitian gates of Section 7 and the roundoff threshold in Appendix F.2. The midpoint ablation uses Ritz inflation $\varpi = 2$ and floor margin $\eta_0 = 10^{-2}$. Exhausted searches and numerical stops enter the profiles as failures under the common stationarity target.

Ratios and memory sensitivity. For a method or setting v, let $\mathcal{S}_v$ be the set of solved problems and $N_{g,p}(v)$ its gradient count on problem p. Tables compare v with a reference b through

$$R(v,b) = \left(\prod_{p \in \mathcal{S}_v \cap \mathcal{S}_b} \frac{N_{g,p}(v)}{N_{g,p}(b)} \right)^{1/|\mathcal{S}_v \cap \mathcal{S}_b|}.$$

A ratio below one favors v. "Both solved" is the size of the intersection; "variant only" and "base only" count problems solved by exactly that member of the comparison. Success fractions use the complete test set.

The settings study uses the same sixteen CUTEst starts for every configuration, four from each nonlinear group, selected by a fixed ordering of problem-identifier hashes. L-BFGS memories are $5, 10, 20$, L-CG_DESCENT memories $5, 11, 20$, and MS-L-BFGS secant limits $2, 4, 8$ with eight stored pairs. XR/XR-F initial memories are $5, 10, 20$, with cap 100. The primary profiles use Table 5 throughout. Initial capacity affects both the early model and the suffix retained by the window test. In this cohort the base XR initialization combines the broadest coverage with the lowest paired gradient cost. For XR-F, a larger initial window reduces paired work at a cost in coverage; a smaller one broadens coverage with more work. The CG memory changes affect work at unchanged coverage, while the L-BFGS and multisecant settings have their own cost–coverage tradeoffs.

Full-matrix reference and higher accuracy. BFGS-SP uses Shanno–Phua scaling [101] and a Moré–Thuente search [102] on the 149 small CUTEst instances and twelve quadratic controls. Its full matrix gives a reference for the limited-memory models. XR attains comparable coverage with lower gradient cost on their common successes (Table 7).

The sixteen settings-study problems are also solved at relative gradient tolerance 10^{-10}, with the same resource limits. L-CG_DESCENT and O-ACCEL-B most consistently reach this stricter threshold. The increase in cost on common successes and the change in solved fractions distinguish accuracy from terminal reliability (Table 8). These supplementary series compare gradient counts; the time profiles use the primary campaign.

Method	CUTEst	$N_g^{\text{method}}/N_g^{\text{XR}}$	Both solved	XR only	Method only
XR	139/149	1.00	151	0	0
XR-F	137/149	1.07	148	3	1
L-BFGS(20)	121/149	1.29	131	20	2
L-CG_DESCENT(11)	139/149	1.20	147	4	4
MS-L-BFGS(8,8)	130/149	1.16	139	12	3
SC-MSS(3,5)	125/149	1.89	133	18	4
SC-SR1(5,7)	121/149	2.08	130	21	3
O-ACCEL-B	132/149	4.11	141	10	3
BFGS-SP	138/149	1.38	144	7	6

Table 7: Full-memory BFGS-SP reference on CUTEst instances with $n \leq 300$ and twelve quadratics, at relative gradient tolerance 10^{-8}. Gradient ratios are geometric means on instances solved by both methods. Paired counts and ratios pool the 149 CUTEst instances and twelve controls (161 problems). XR only and Method only count exclusive successes. Common to all rows: Quadratics = 12/12.

Method	Solved at 10^{-8}	Solved at 10^{-10}	$N_g(10^{-10})/N_g(10^{-8})$	Both solved
XR	16/16	11/16	1.31	11
XR-F	15/16	12/16	1.30	12
L-BFGS(20)	11/16	10/16	1.61	10
L-CG_DESCENT(11)	16/16	15/16	1.59	15
MS-L-BFGS(8,8)	14/16	10/16	1.65	10
SC-MSS(3,5)	11/16	10/16	1.90	10
SC-SR1(5,7)	12/16	11/16	1.90	11
O-ACCEL-B	15/16	14/16	1.67	14

Table 8: Gradient-tolerance sensitivity on 16 CUTEst instances, four per objective/dimension group, selected by the fixed hash rule. Ratios are geometric means on instances solved at both tolerances. Every run uses the same preset, 100000-point budget and 120-second limit.

Table 9 summarizes the full-memory local study by dimension and condition number. Its error is measured in the analytic stationary-Hessian norm. "First full" records the gradient count at the first accepted step computed with n retained pairs.

Logistic datasets. The logistic series uses the complete training sets listed in Table 10. The a9a and w8a representations follow Platt [103], and the binary Covertype construction follows Collobert et al. [104]. Madelon and gisette originate in the NIPS 2003 feature-selection challenge [105]. We use the normalized epsilon distribution from the PASCAL 2008 challenge described in [106], the RCV1

n	κ	Method	Med. N_g/n	First full N_g/n
5	10	XR	2.2	1.4
5	10	XR-F	2.6	1.4
5	100	XR	2.8	1.8
5	100	XR-F	2.8	1.8
10	10	XR	1.7	1.3
10	10	XR-F	1.7	1.3
10	100	XR	1.9	1.3
10	100	XR-F	1.9	1.3
20	10	XR	1.3	1.2
20	10	XR-F	1.3	1.2
20	100	XR	1.25	1.15
20	100	XR-F	1.45	1.15

Table 9: Full-memory local study over 10 seeds per dimension and condition number. The target relative stationary-Hessian norm is 10^{-12}; $\varepsilon = 0.1$, initial base energy 0.03. Costs are medians of actual gradient evaluations divided by n. First full is the first accepted step whose model uses n retained pairs. All 120 runs reach the target and use a full window.

Dataset	Structure	Observations	Features	Storage
a9a	Moderate dimension	32,561	123	csr
w8a	Moderate dimension	49,749	300	csr
covtype.binary	Many observations	581,012	54	dense
madelon	Dense, high dimension	2,000	500	dense
epsilon	Dense, high dimension	400,000	2,000	dense
gisette	Dense, high dimension	6,000	5,000	dense
real-sim	Sparse, high dimension	72,309	20,958	csr
rcv1.binary	Sparse, high dimension	20,242	47,236	csr

Table 10: Binary classification datasets for regularized logistic regression. Features exclude the appended intercept. Every nonzero feature row has unit Euclidean norm; all coefficients, including the intercept, are regularized. Each dataset uses the same frozen regularization grid and the zero initial point.

λ	Method	N_g/XR	Time/XR
10^{-2}	XR-F	0.9	0.9
10^{-2}	L-BFGS(20)	1.12	1.09
10^{-2}	L-CG_DESCENT(11)	1.18	1.51
10^{-2}	MS-L-BFGS(8,8)	0.86	1.06
10^{-2}	SC-MSS(3,5)	2.71	2.63
10^{-2}	SC-SR1(5,7)	1.66	1.82
10^{-2}	O-ACCEL-B	2.36	2.02
10^{-4}	XR-F	0.99	0.98
10^{-4}	L-BFGS(20)	2.2	2.3
10^{-4}	L-CG_DESCENT(11)	2.17	2.65
10^{-4}	MS-L-BFGS(8,8)	1.88	2.74
10^{-4}	SC-MSS(3,5)	7.86	7.86
10^{-4}	SC-SR1(5,7)	6.58	6.45
10^{-4}	O-ACCEL-B	3.3	2.84
10^{-6}	XR-F	1.01	1.03
10^{-6}	L-BFGS(20)	4.31	4.44
10^{-6}	L-CG_DESCENT(11)	4.37	5.48
10^{-6}	MS-L-BFGS(8,8)	3.61	5.3
10^{-6}	SC-MSS(3,5)	21.78	21.86
10^{-6}	SC-SR1(5,7)	17.53	17.36
10^{-6}	O-ACCEL-B	4.05	3.55

Table 11: Paired comparison against XR by regularization strength. Costs are geometric mean method/XR ratios on jointly solved datasets; values above one favor XR. All methods reach the target on all 8 datasets at each regularization level.

corpus [107], and McCallum's real-sim corpus with SVMlin preprocessing.[5] Dense designs and sparse CSR designs share the same normalized full-batch objective. All arrays are loaded and warmed before timing. Comparisons by regularization are in Table 11; each penalty includes the same eight datasets, so each dataset has equal weight in the pooled profiles.

[5]Original sources: McCallum's data collection and SVMlin preprocessing.

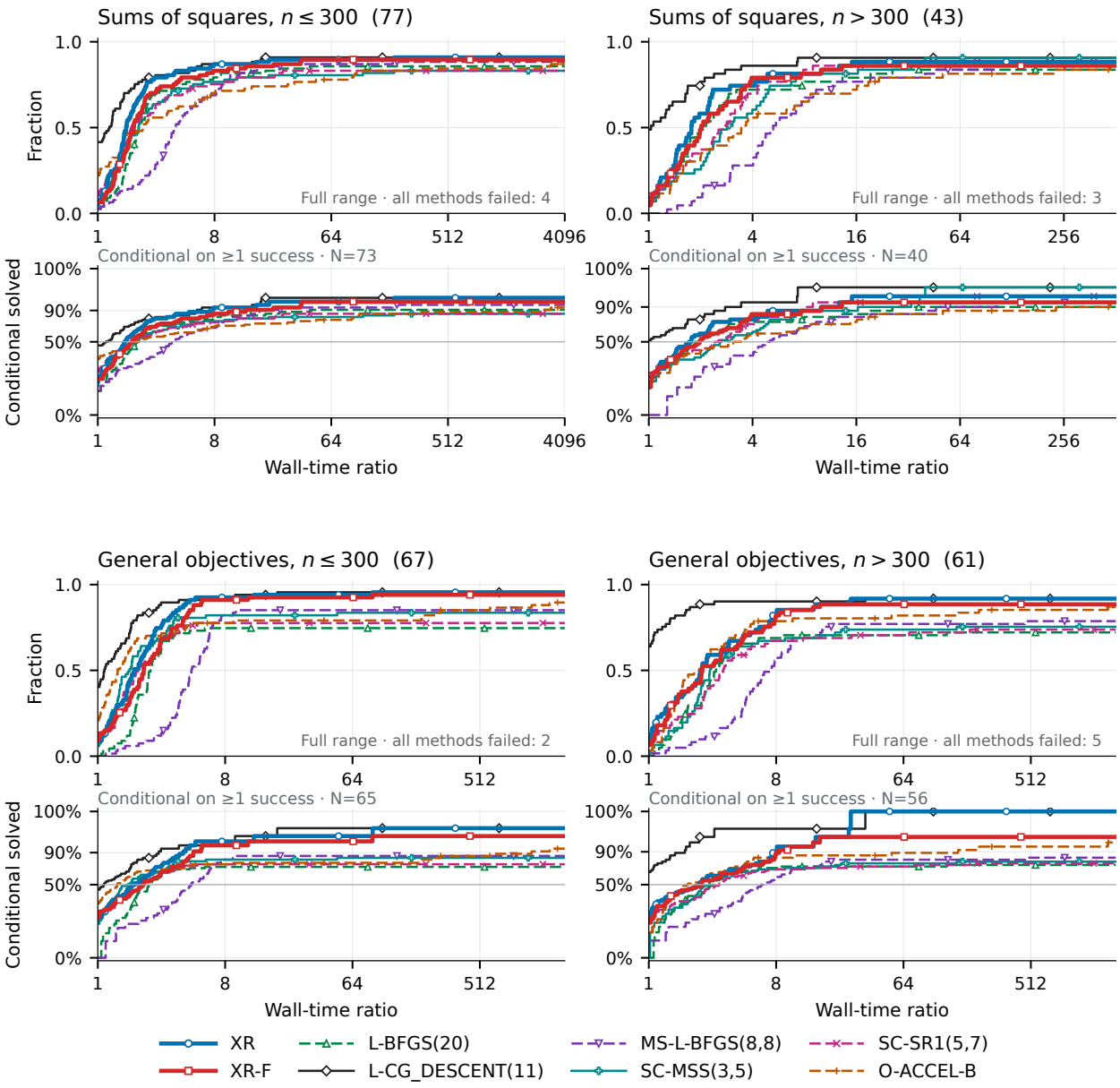


Figure 5: Elapsed-time performance profiles for the same cohorts and targets as Figure 2. Ordinary panels use all scheduled problems; empirical-logit panels use the stated number solved by at least one method. Both views show the complete ratio range.